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Value Creation in Freemium Business Models: A Study of Freemium Business Models in the B2B Market

Master Thesis

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Abstract

The freemium business model, referring to the combination of free and premium offerings, has in later years become increasingly popular as a way to create value for companies operating in virtual markets. Most literature on freemium models has been conducted in the business-to-consumer (B2C) market, as the use of freemium models in business-to-business (B2B) transactions traditionally has been rare. A number of non-academic articles argue that the freemium business model will fail to create value in B2B markets as business customers are less prone to a viral spread and lack trust in the quality and security of free offerings. However, as a result of changes in the software industry, a number of start-ups and established firms operating in the B2B market are today utilizing the freemium business model. Thus, the question of how these offerings survive and create value is becoming increasingly relevant.

As there is limited prior academic research on freemium business models in the B2B enterprise software market, this thesis takes an important step in exploring this research area. A conceptual model for value creation from a business model perspective is used as a baseline for the research. The model is qualitatively tested through interviews with four companies. The findings imply that a freemium business model creates value in the B2B enterprise software market by increasing transaction efficiency, creating lock-in, employing complementarities as well as adding novelty to the offering. This is interesting as it contrasts prior literature on the subject. We find that the freemium business model enables firms to utilize a bottom-up approach to customer acquisition, which serves to greatly eliminate challenges with B2B complexity. Two mechanisms are central in this approach; versioning with zero prices and network effects. Versioning with zero prices reduces adoption barriers as it enables individual employees to start using the offering for free. Network effects serves to effectively spread adoption of the offering and attract complementors.

The findings of this study make a distinct contribution to the extant knowledge of freemium business models in B2B relations. The study is important as an increasing amount of companies compete by structuring their offerings through freemium business models. Focusing academic attention to this field might eventually lead to a better understanding of the different drivers of value, and bear important implications for firms utilizing a freemium business model.

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Abbreviations

B2B	Business to Business
B2C	Business to Consumers
SaaS	Software as a Service
SME	Small and Midsize Enterprises
A&Z	Amit & Zott
cf.	Confer/Compare
PIM	Product Information Management
DAM	Digital Asset Management

Chapter 1 – Introduction

1.1 Background

Changes endured in the enterprise software industry during the last decade have challenged the value creation model of many established firms. Especially prominent is the introduction of the Internet as well as Software as a Service solutions. With the introduction of Software as a Service, the cost of digitalized products and services has decreased, resulting in a dramatic increase in the range of servicized offerings (Anderson, 2009). Anderson (2009) actually finds that in 2009 only 32 of the top 100 companies made a physical product that users can hold. Additionally, it is becoming increasingly popular to employ open source software and open application programming interfaces (Lyman, 2012). As a result of these changes, the question about how firms operating in the software industry create value has become important.

New business models have emerged as a result of the above mentioned changes, and the use of freeware through the freemium business model as a novel way of value creation has been especially prominent. In fact, 98% of AppStore revenue was in 2015 coming from freemium apps (Kim, 2015). Freemium is a combination of 'free' and 'premium' offerings where firms give one part away for free, while other versions are offered for a premium price. Historically, the majority of companies utilizing a freemium business model have targeted the business-to-consumer market (B2C market). Furthermore, this business model has commonly been employed by start-ups for relatively cheap and uncomplex products and services, like Dropbox, Evernote and Spotify (Brandall, 2016). Here, the freemium business model is found to create value through creating and leveraging network effects, lowering customer's barriers to adoption, and the ability to attract a large user base without expending massive resources (Kumar, 2014).

There is however limited academic research on how the freemium business model creates value in B2B markets. Traditionally, there have been distinct differences between B2B and B2C markets (Kotler & Pfoertsch, 2006). B2B markets are understood to have a higher complexity in buying decisions, greater diversity in demand, bigger technical and economic considerations as well as a need for long-lasting relationships with vendors (Kotler & Pfoertsch, 2006). Many non-academic

articles therefore propose it is challenging to utilize a freemium business model in B2B markets, arguing that these markets face higher complexity and more stringent requirements, and that they are less prone to a viral spread than B2C markets. However, as a reaction to the endured market changes, many companies including more established industry players have started utilizing the freemium business model also for complex and traditionally costly offerings in the B2B market (Bakal, 2014). Companies like Microsoft, Google and Salesforce are for example exploring the freemium model for several of their products and services (Capossela, 2015; Schonfeld, 2010). Other freemium offerings like Yammer, Slack and Box are experiencing great success in the B2B enterprise software market (Kim, 2015).

There thus seems to be a knowledge gap in relation to what is believed and known about how firms operating in the B2B market create value through freemium business models. In seeking to understand the value creation potential of this business model, we found that existing literature only partially advance our knowledge of the field, and that there may be a need for alterations or modifications to the existing literature. This is interesting from an academic as well as a managerial point of view, and this thesis therefore aims to add to the existing body of research on value creation in freemium business models by extending it to the B2B enterprise software market.

1.2 Problem Statement

The above discussion leads to the research question that will drive this thesis:

"How does the freemium business model create value for firms operating in the B2B enterprise software market?"

'How' refers to the different ways the freemium business model might create value for the case companies in this research. Furthermore, we put forward the definition of a 'business model' as proposed by Amit & Zott (2009, p.266); "A firm's business model depicts the content, structure and governance of transactions of the firm in relation to the networks in which it is embedded, as well as how it is shared". The business model thus embodies the strategic intent of the firm concerning value creation and appropriation in its exploitation of business opportunities (Zott & Amit, 2009). The freemium business model refers to a business model where some part of an offering is given

away for free, whereas a value added service or enhanced version is offered for a premium price (Wilson, 2006).

'Value' can be studied from a range of different angles. In this thesis we rely on the definition proposed by Amit & Zott (2001, p.515); "Total value created through a business model equals the sum of the values appropriated by all the participants in a business model, over all transactions that the business model enables". Value in this thesis is therefore not limited to monetary benefits for the focal firm, but might also include value through a large user base, brand, reputation and other non-monetary benefits.

A focus on the 'B2B market' refers to markets in which businesses transact with other businesses as opposed to private consumers. Lastly, our research studies a specific industry, namely the enterprise software market. 'Enterprise software' refers to application software used by organizations to conduct their business operations, such as planning, project management, sales and human resources (Enterprise Software Market, 2016).

In answering the research question, this research employs a qualitative case study using four case companies. The research furthermore relies upon a business model perspective on value creation, and more specifically a value creation model offered by Amit & Zott (2001). The model depicts four interconnected sources of value creation, which will be thoroughly discussed in chapter 4. The value creation model facilitates a framework to guide our research, and we find it to be applicable for the purpose of this research in many ways. We do however recognize the need to modify certain aspects of the model to create a better fit with our specific research question. A discussion on exactly how the value creation model of Amit & Zott (2001) is used in this thesis will be provided in chapter 4 and 6.

1.3 Scope and Delimitation

This research focuses on web-based products and services, and is as such limited to information goods. This makes sense in relation to the research topic, the freemium business model, which is enabled by the Internet (Anderson, 2009). The research furthermore examines value creation using a study of four case companies, which leads us to focus on value creation from a firm's point of

view. Even though there might be disadvantages in not highlighting the customer's perception of value, most of the case companies interviewed in this study had already performed customer research regarding this matter, and thus inhibits an understanding of their customer's value perception. The problem statement further demands research in a B2B context, which entails that B2C contexts will be excluded. Lastly, as will be discussed in paragraph 3.6.3, this thesis restricts itself to case studies of a few companies in order to gain a richer understanding of these company's history and choices in relation to the utilization of the freemium business model. As with any qualitative study, the purpose is therefore not to be able to generalize across the total population, but rather to lay the foundation for future research in this area.

1.4 Thesis Structure

The thesis commences with a chapter discussing factors that has enabled the use of freemium business models, as well as literature explaining the freemium business model (cf. chapter 2). Next, the methodology chapter (cf. chapter 3) aims to provide an understanding of how the research was conducted and explain the credibility of the findings. Following, a literature review is provided in chapter 4 in order to highlight the research question and other relevant strands of research. The literature review starts with a discussion of different perspectives on value creation, upon providing an overview of literature on information goods, network theory and B2B transactions. Thereafter, important modifications to the value creation model will be presented. Chapter 5 depicts the four case companies employed in this research. Chapter 6 presents research findings, as well as an interpretation of the case study data with respect to the value creation model. Chapter 7 reviews the most important findings of the research, and explains what implications this bears for literature and managers, as well as perspectivation, possible limitations and suggestions for future research. In chapter 8, the thesis concludes with an overview of every important aspect of this research.

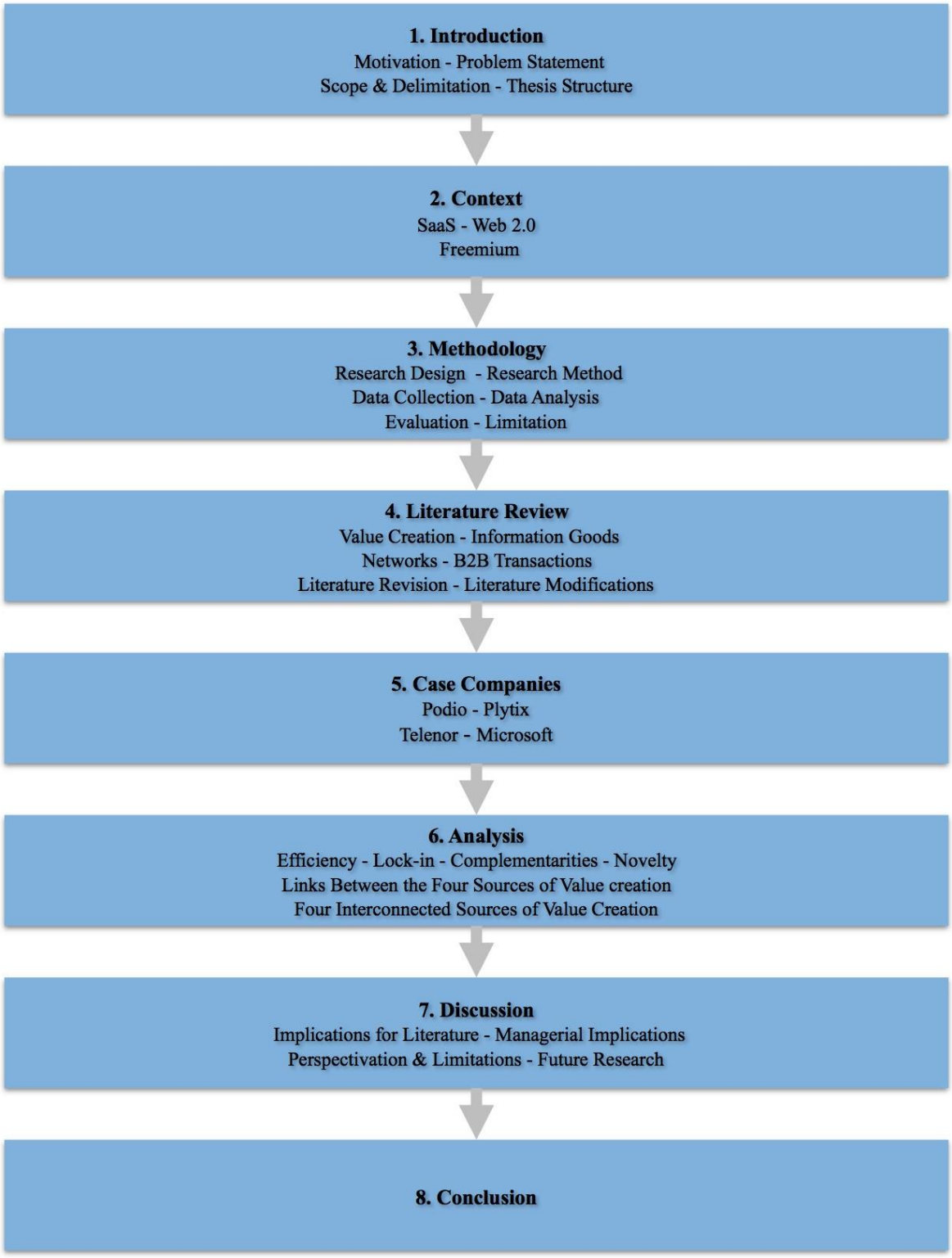

Figure 1. Thesis Overview.

Chapter 2 – Context

To gain an understanding of the freemium business model, this chapter first reviews the enablers of a freemium business model. Specifically, Software as a Service (paragraph 2.1) and Web 2.0 (paragraph 2.2) is discussed. Thereafter, the history and important aspects of the freemium model is reviewed, specifically freeconomics (paragraph 2.3.1), the origin of freemium (paragraph 2.3.2) and different versions of the freemium model (paragraph 2.3.3). Lastly, prior knowledge about value creation in the freemium business model is provided (paragraph 2.4), and specific implications are highlighted (paragraph 2.4.3).

2.1 Software as a Service

The enterprise software industry has over the last few years endured several changes (Ozzie, 2005). Some of the major waves include the introduction of personal computers, service-based architectures and information communication technologies (Gummerus, 2011). The emergence of Software as a Service (SaaS) has offered new opportunities for interaction between firm and customer (Gummerus, 2011). The predominant Software as a Product (SaaP) was characterized by customers paying upfront licenses for on premise software, where add-ons and updates could be purchased separately (Gorokhova, 2014). Beginning in the late 2000s, this dominant design was challenged by a discontinuous technology, namely SaaS (Cusumano, 2008). SaaS seeks to cover the same consumer need, but makes use of cloud computing infrastructure to deliver one application to many users, regardless of their location (Cusumano, Iyer & Venkatraman, 2012). By delivering software over the Internet, SaaS providers have significantly reduced the distribution costs of software.

The revenue model for SaaS has been developed as a subscription model (Covert, 2013; Lardinois, 2012). A subscription model refers to a payment structure that allows an organization or individual customer to subscribe to a vendor's services for a specific time period and at a set price (Whitler, 2016). These developments have lowered the barriers of entering the software enterprise market with new products and services (Lardinois, 2012). With the emergence of SaaS and cloud

computing, established industry leaders like Microsoft, SAP and Oracle have experienced increased competition from players such as Salesforce.com, Google Apps, Amazon, Apple, Podio and many other start-ups and small and medium-enterprises (Bort, 2012).

2.2 Web 2.0

The shift from Web 1.0 to Web 2.0 has been an important enabler of new transaction structures, hereunder the freemium business model. In Web 1.0, information was controlled by a relatively small number of players and distributed to the mass (Lytras, Damiani & de Pablos, 2008). With the development of Web 2.0, user-generated content has become the new norm. Musser & O'Reilly (2007, p.10) define Web 2.0 as "a more mature, distinct medium characterized by user participation, openness, and network effects". With Web 2.0 the Web has thus become a user-centered communications medium (Musser & O'Reilly, 2007), characterized by technology like Open Source, Software as a Service (SaaS) and Cloud Computing. This implies that information is becoming increasingly available and shared, as opposed to being controlled by a few focal firms, which makes it easier for smaller companies to enter the software market.

2.3 Freemium

In this section we review what is called freeconomics (paragraph 2.3.1), the origin of freemium (paragraph 2.3.2) and different types of freemium models (paragraph 2.3.3).

2.3.1 Freeconomics

With the advent of SaaS and Web 2.0, a variety of services offered for free appeared in the enterprise software market (Anderson, 2009), often entailing new business models like freemium and ad-based business models (Lytras, Damiani and Ordónez de Pablos, 2009). Anderson (2009) explains that as the marginal cost for information goods is close to zero, sooner or later, most producers in the digital marketspace will find themselves competing with free products and services. Thus, the underlying technologies of the digital age are drivers of what Anderson (2009) names "freeconomics". There are several reasons for the migration to free economics (Anderson,

2009). First, the supply of content has grown by factors of millions because of the Internet. At the same time, demand has not risen to the same extent, and more competition between suppliers has thus lead to lower price points. The ease of access to products and services has also accelerated due to the progress in the digital industry. Lastly, consumers today expect that a lot more products and services should be offered free, adding pressure to the vendors (Anderson, 2009).

2.3.2 The Origin of Freemium

There is no overall accepted definition of the term ‘freemium’. It was however venture capitalist Fred Wilson who popularized the term ‘freemium’ when he posted about the concept on his blog in 2006 (Wilson, 2006). The business model has since then become increasingly popular, and Anderson (2009) states that it is one of the most relevant and interesting business models in today's software market. In this thesis we will rely upon the definition employed by several authors, like Kumar (2014) and Miller (2009), who depict freemium as a combination of ‘free’ and ‘premium’ subscription models.

As the freemium business model yields revenues only *after* building a customer base (Yogev, 2011), the business model revolves around two distinct user groups; one larger group who consumes the service for free and a small percentage that pays a premium price (Lytras et al. 2009). Even though there might not be a physical revenue stream emulating from the free consumer, the customer pays for the free good with an alternative currency, namely ‘mindshare’ (Pujol, 2010). As seen in Figure 2 (Pujol, 2010 p.4), the free user grants mindshare back to the provider, while the premium customer grants direct revenue. ‘Mindshare’ is the development of awareness for the provider's brand and the consideration for purchase of future commercial services (Pujol, 2010). Thus, what free grants in exchange for forsaking direct revenues are the potential of a huge customer mass (Anderson, 2009).

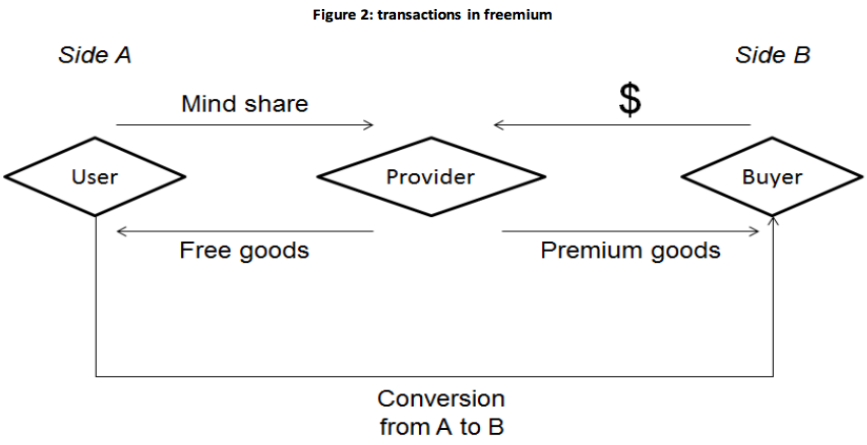

Figure 2. Freemium Business Model (source: Pujol, 2010, p.4).

2.3.3 Different Types of Freemium

How a firm chooses to structure its freemium offerings will impact the value creation potential of the business model. We will in the following summarize different ways of offering a freemium model, namely; time limited freemium models (paragraph 2.3.3.1), feature limited freemium models (paragraph 2.3.3.2), seat, capacity and device limited freemium models (paragraph 2.3.3.3) and customer type limited freemium models (paragraph 2.3.3.4);

2.3.3.1 Time Limited

When utilizing a freemium model, different strategies are available (Pujol, 2010). First, a company might use what is called a *time limited* freemium model (Anderson, 2009), which refers to a time restricted access to a product or service upon having to pay, (Pujol, 2010). Salesforce.com is an example of a firm utilizing a time limited freemium model (Anderson, 2009). Even though this type of freemium model is quite easy to organize for the focal firm, many potential customers might be unwilling to commit when they know they will have to pay at some point (Anderson, 2009). As a time limited freemium model only grants access to a free version for a restricted period of time, we do not recognize it as a proper form of freemium model, but rather what is called a 'trial' model. Thus, in the rest of this paper, time limited freemium model will not be included in our definition of a freemium business model.

2.3.3.2 Feature Limited

Second, a company might utilize a *feature limited* approach, meaning offering a basic version for free and a more sophisticated version for paying customers (Anderson, 2009). Yahoo! and Podio are some of the firms utilizing this freemium model. Even though this strategy maximizes the reach of new customers and attracts loyal and less price sensitive consumers (Anderson, 2009), it requires the firm to create two versions of an offering (Anderson, 2009). Such feature differentiation in business models might be challenging as it requires tradeoffs between growing the free user base and generating revenues (Pujol, 2010).

2.3.3.3 Seat, Capacity and Device Limited

Third, a company might utilize a *seat limited* freemium model. This model refers to a situation where the offering is free for a certain amount of users. After the limit is reached, the customers have to pay (Anderson, 2009). Similarly, businesses may opt for a *capacity limited* freemium model where users have a set capacity of free space, but have to pay a premium price for additional space (Burry, 2014). Familiar examples of this freemium model are Dropbox, Box, iCloud and Gmail (Burry, 2014). Companies may also implement a *device limited* freemium model, in which the offering is free of charge for certain or a certain amount of devices, but where there is a cost for other or additional devices (Burry, 2014). Common for all of these freemium models are that they are easy to implement and understand. However, the focal firm may suffer from cannibalizing potential income from the low end area of the market (Anderson, 2009).

2.3.3.4 Customer Type Limited

Finally, a company might offer what is called *customer type limited* freemium, meaning that certain types of customers get the offering for free, while others have to pay (Anderson, 2009). Microsoft is an excellent example here, as companies less than three years old and under $1 million in revenues get Microsoft's business software for free, while other companies must pay (Alabaster, 2012). The benefit of this model is that it charges customers according to their valuation of the offering and ability to pay, which might attract fast-growing companies in their early phases. The downside is that the process can be challenging to police (Anderson, 2009).

2.4 Knowledge about the Freemium Business Model

As mentioned in chapter 1, it is limited literature available on the use of freemium business models in B2B markets. There are however some insights about the use of the freemium model in B2C markets, and some non-academic articles on how this model might be used in a B2B setting. In this section we therefore first examine some prior knowledge about the use of the freemium model in consumer markets (paragraph 2.4.1), specifically focusing on conversion rates (paragraph 2.4.1.1) and continued innovation (paragraph 2.4.1.2). Then we revise what is currently known about the freemium model in B2B markets (paragraph 2.4.2), with a particular focus on stricter requirements in B2B markets (paragraph 2.4.2.1), viral growth in B2B markets (paragraph 2.4.2.2.) and market maturity (paragraph 2.4.2.3). Lastly we propose implications for the use of freemium models stemming from prior knowledge (paragraph 2.4.3).

2.4.1 Freemium in B2C Markets

The freemium model is found to be compelling in many ways for freemium firms operating in the B2C market (Patel, 2015), with Spotify, Dropbox and Skype as some of the most familiar success stories. First, free features are attractive to customers and thus a potential marketing tool that firms can use without spending resources on costly ads (Kumar, 2014). Second, use of the freemium model gives the consumer the possibility to try the offering upon deciding whether or not to purchase the good (Anderson, 2009), thus lowering the barriers of adoption. Third, social networks are powerful drivers of conversion (Kumar, 2014). If freemium providers run their business in markets inhibiting network effects, they have the opportunity to quickly scale a lot of customers with minimal expenses on customer acquisition (Pujol, 2010). In order to leverage these potential benefits, the freemium model must be structured in a specific way (Kumar, 2014). Two important aspects on the structuring of freemium business models are discussed in the two succumbing paragraphs, namely conversion rates (paragraph 2.3.4.1.1) and continued innovation (paragraph 2.3.4.1.2);

2.4.1.1 Conversion Rates

Freemium firms do not reap any monetary returns from free customers, which make the prediction of conversion rates important for success (Yogev, 2011). A low conversion rate from freemium to

premium users might signal that too much is provided for free, or that customers do not understand the value proposition of the premium offering (Kumar, 2014). At the same time, a very high conversion rate is not necessarily positive as it signals that the free offering is not very compelling, which might limit potential adoption. To be able to continually convert users in a steady stream, it is important to think strategically about what features to put in the different versions of the offering (Kumar, 2014).

2.4.1.2 Continued Innovation

In order to keep increasing upgrades and retain customers, freemium firms need to continue innovating and enhance the value of the offering, especially on the premium service (Kumar, 2014). When firms do not add new and novel content to the premium version(s), the incentives for customers to convert or stay in the premium version decreases (Kumar, 2014). This enhanced value needs to be intuitive for the consumers, as the user's first experience with an offering is found to be highly important (Forbes, 2014).

2.4.2 Freemium in B2B Markets

The limited literature available for value creation through a freemium model in the B2B market makes it necessary to review non-academic articles about this subject. The majority of these articles state that the freemium business model works better in B2C markets than in B2B markets. More specifically, the articles imply that business customers will not appreciate free products as they have stricter requirements to their software than individual consumers (paragraph 2.4.2.1), are less prone to viral growth (paragraph 2.4.2.2) and that the B2B market has not yet matured to embrace freemium offerings (paragraph 2.4.2.3);

2.4.2.1 Stricter Requirements in B2B Markets

Needleman & Loten (2012) find that for companies operating in the B2B market, it rarely makes sense to employ a freemium business model. Enterprise clients typically have budgets for purchasing software and are therefore not as drawn to free products (Needleman & Loten, 2012). B2B enterprise software transactions are traditionally guarded by gatekeepers such as the IT-department. These gatekeepers commonly have requirements in terms of quality and security.

Offerings in B2B markets are thus suggested to meet more stringent requirements than consumer offerings (Stanek, 2013). To secure proprietary data, many firms may actually prohibit the use of non-approved free services with live corporate data, making the freemium model challenging to use in this domain (Murphy, 2010).

2.4.2.2 Viral Growth in B2B Markets

While viral growth and network effects exist in certain enterprise settings, Boris (2014) states that network effects are typically limited in B2B markets. Therefore, customer acquisition proves more costly, straining the already capital-low freemium firms (Boris, 2014). One key element to the viral nature of freemium is that simplicity has a lot to do with virality. Full featured productivity suites are less likely to go viral than single purpose consumer apps (Murphy, 2010). Thus, the "free" strategy of rapid customer growth will not deliver the same payback in B2B settings (Murphy, 2010).

2.4.2.3 Market Maturity

Lastly, Murphy (2010) states that even though consumers in the B2C market have come to understand that offerings like games and music can be obtained for free, business users still associate free as a non-sustainable business model. They will be skeptical of whether the firm will survive in the long term, and thus be reluctant to engage in transactions. Furthermore, especially in mid-market and higher B2B markets, the strategy of putting a free pricing tier to entice usage of a system does not work well. Most business customers still feel that free gives zero value, and are thus not interested in spending their time on services offered through a freemium model (Murphy, 2010).

All in all, the articles on freemium models in B2B and B2C markets differ a great deal. Whereas freemium business models are quite common in B2C markets and have proven to be successful, the articles propose that many consumers are still skeptical to this model in the enterprise market. As B2B firms like Microsoft and Google have started utilizing this model in later years, we set out to review whether and how a freemium business model might create value for firms operating in the B2B market space.

2.4.3 Implications

Based on the previous discussion we in this section propose two implications about the value creation potential of freemium business models.

First, in relation to paragraph 2.4.1.1 and 2.4.1.2, we assume conversion rates and continued innovation will affect the value creation potential of a firm utilizing a freemium business model. We suggest that high innovation skills and attention to strategic conversion analytics might spawn value creation in the businesses. Second, as some non-academic articles suggest there are stricter requirements, less viral growth and a less mature market in relation to freemium models in B2B settings, we assume that it might be more challenging to create value when using a freemium model in B2B markets. These implications will be discussed in greater detail in paragraph 4.6.

Chapter 3 – Research Methodology

This chapter presents the methodology employed to answer the research question of this thesis. The methodological choices have been assessed and determined according to the research question and the purpose of this study. The chapter will first cover the research design (paragraph 3.1), before describing the research method (paragraph 3.2) and data collection techniques employed (paragraph 3.3). Following, the data analysis approach will be covered (paragraph 3.4), before evaluating the validity, reliability and generalizability of this research (paragraph 3.5). Lastly, methodological limitations will be discussed (paragraph 3.6).

3.1 Research Design

Research design relates to the overall choice of strategy when collecting the required data to answer the problem statement (Ghauri & Grønhaug, 2002). The research design includes research philosophy (paragraph 3.1.1), research approach (paragraph 3.1.2), research purpose (paragraph 3.1.3) and research strategy (paragraph 3.1.4).

3.1.1 Research Philosophy

The research philosophy contains important assumptions about the way in which researchers view the world (Saunders, Lewis & Thornhill, 2009). Saunders et al. (2009) differs between *interpretivism*, *positivism*, *pragmatism* and *realism*. In this research we adopt a *pragmatic view*, letting the problem statement guide our research and method. In the pragmatic view, both quantitative and qualitative approaches are appropriate, and it is up to the researcher to decide what yields value in each choice (Saunders et al., 2009). The pragmatic view thus rejects methodological orthodoxy in favor of methodological appropriateness as the primary criterion for methodological quality (Patton, 1980).

3.1.2 Research Approach

Ghauri & Grønhaug (2002) distinguish between two ways of establishing conclusions, namely inductive and deductive research approach. In a deductive approach, researchers make use of existing literature to develop hypotheses to test on data, and draw conclusions based on logical reasoning (Saunders et al., 2009). The purpose is to validate or extend existing literature on a topic. Inductive approaches, on the other hand, explore and analyze this data in order to develop theories (Saunders et al., 2009). An inductive approach is useful when existing theory on a concept is incomplete, and researchers seek to extend the knowledge of this concept (Saunders et al., 2009). Patton (1990, p.46) explains that in practice, these approaches are often combined; "there is often a flow from inductive approaches to find out what the important questions and variables are, to deductive hypothesis testing aimed at confirming exploratory findings, then back again to inductive analysis to look for rival hypotheses and unanticipated unmeasured factors".

This research aims to explain how freemium business models create value in B2B markets through a conceptual framework for value creation. This conceptual framework lays the foundation for data analysis. Taking a starting point in literature, one can thus argue we are taking a deductive research approach. However, as there is little academic research on the freemium model, and particularly in B2B markets, this research also aim to inductively identify new concepts that enhance our understanding of value creation in a freemium model. We are therefore taking a mixed research approach (Patton, 1990). This enables us to rely on existing theory, and at the same time be flexible and open to adaption as new insights are revealed in order to find alternative explanations than those posed by existing literature (Saunders et al., 2009).

3.1.3 Research Purpose

All research has different purposes, and literature distinguishes between three essential ones: exploratory, explanatory and descriptive (Saunders et al., 2009). *Explanatory* or causal research deals with structured problems, but the researcher is at the same time confronted with cause-and-effect issues. The main task is to isolate causes as well as establish whether and to what extent causes results in effects (Ghauri & Grønhaug, 2002).

The research purpose has important implications for the design of the study and how the research question should be answered (Saunders et al., 2009). The purpose of this thesis is to study the freemium business model in the B2B enterprise software market, and find an explanation as to how it creates value. Thus, we seek to find a causal explanation as to how one variable affects, or is responsible for, changes in another (Blumberg, Cooper, & Schindler, 2008) - specifically how changes in the structure of a specific business model affects a firm's value creation potential. Thus, we are performing an explanatory research.

3.1.4 Research Strategy

When aiming to perform an explanatory research, posing “how” or “why” questions, case studies are the preferred strategy (Yin, 2003). Yin (2003) distinguishes between four case study strategies based on two discrete dimensions: single case vs. multiple case & holistic case vs. embedded case. This research will employ a holistic multiple case study strategy to answer the research question. The holistic dimension implies that our research is concerned with the organization as a whole, and not differences between units within the organization. Using a multiple case study further enables us to establish whether the findings in one case occur in other cases (Yin, 2003). The case study strategy is appropriate in examining contemporary events, where the relevant behaviors cannot be manipulated and the researchers have little or no control. Web 2.0 is a central enabler of the freemium model. As Web 2.0 is an ongoing development (Anderson, 2009), we study a contemporary development in which a case study yields the greatest understanding due to its ability to perform direct observations, and its unique strength of dealing with a variety of evidence (Yin, 2003).

3.2 Research Method

This section highlights the research method used in set study, focusing on quantitative and qualitative research methods (paragraph 3.2.1), primary and secondary data sources (paragraph 3.2.2), the study's time horizon (paragraph 3.2.3) and literature (paragraph 3.2.4).

3.2.1 Quantitative and Qualitative Method

Researchers distinguish between quantitative and qualitative methods. The quantitative method refers to a data collection technique that generates or uses numerical data, and is mainly used when the objective is to examine the relationship between variables and present findings in measurable quantities (Saunders et al., 2009). In contrast, qualitative methods predominantly use or generate non-numerical data, and is well suited to study phenomenon's in depth (Saunders et al., 2009). A qualitative method yields a more holistic view as it enables researchers to observe many variables, and adapt their search and observations as they gain new insights (Saunders et al., 2009).

As this research has an explanatory purpose and aims to study a phenomenon in depth in order to add to the existing body of research, a qualitative method seems appropriate. The research will further be using more than one qualitative technique, which will be covered in depth in 3.2.2, and we are thus employing a multi-method where we aim for triangulation. This refers to the use of different data collection techniques within one study, which may contribute higher reliability and validity to the findings (Saunders et al., 2009).

3.2.2 Primary and Secondary Data

There are essentially two sources of data, primary and secondary (Saunders et al., 2009). Primary data is obtained directly from the source to answer the problem statement, whereas secondary data has originally been obtained for other purposes (Saunders et al., 2009). The main advantage of primary data is that it is collected for the particular project at hand, which means it is consistent with the research (Ghauri & Grønhaug, 2002). Secondary data provides a comparison instrument with which one can interpret and understand the primary data (Ghauri & Grønhaug, 2002).

In this research, primary data is collected through interviews with companies employing a freemium business model in the B2B enterprise software market. This enables us to gather valid and reliable data that are relevant to answer our research question (Saunders et al., 2009). In-depth interviews also yield an accurate and clear picture of a respondent's position or behavior (Ghauri & Grønhaug, 2002).

As data collection for qualitative research is time consuming (Yin, 2003), we have supplemented the primary data with secondary sources. The use of multiple sources of evidence in case studies allows researchers to address a broader range of historical, attitudinal and behavioral issues (Yin, 2003). Any finding in a case study is likely to be much more convincing and accurate if it is based on several different sources of information. This also helps to triangulate the findings, as a means of further validation (Yin, 2003). For secondary sources, both internal sources from the case company's home pages, as well as external secondary sources are employed. A more thorough discussion about the secondary sources is conducted in paragraph 3.3.3.

3.2.3 Time Horizon

Saunders et al. (2009) distinguish between cross-sectional and longitudinal time horizon in a research. A cross-sectional research takes a snapshot at a particular point in time, whereas a longitudinal study refers to conducting a series of snapshots over a given period. The time horizon is independent of which research strategy is chosen, as well as independent of the choice of method (Saunders et al., 2009). As this research focuses on a particular phenomenon at a particular point in time, this research is cross-sectional. As set research faces time constrains, the case study is based on interviews conducted over a short period of time, which is applicable in a cross-sectional study (Saunders et al., 2009).

3.2.4 Literature

The purpose of the literature review is not to provide a summary of everything that has been written on the specific topic, but to review the most relevant and significant research of the field (Saunders et al., 2009). In the pursuit for relevant literature we started with a broad search across a number of bibliographical databases and journals like CBS Library, Google Scholar, Ebscohost and Business Source Complete. These searches led us to take a business model perspective on value creation. Furthermore, theory on information goods and specifically network theory was found to be useful in enlightening the research question. Lastly, literature on B2B transactions was considered necessary to answer the problem statement. The purpose of the literature review is to identify relevant literature, and develop a framework that we will test using data from the case companies.

3.3 Data Collection

Data collection refers to how data for this research is collected. We thus review the sampling of data (paragraph 3.3.1), the use of semi-structured interviews (paragraph 3.3.2) as well as the use of secondary sources (paragraph 3.3.3).

3.3.1 Sample

When utilizing a qualitative research method, the sample size is more limited than when using a quantitative study (Yin, 2003). Our research question and the nature of this study dictate a non-randomized sample (Saunders et al., 2009). Instead we sampled based on appropriateness, where critical sampling is necessary to select research objects that will yield relevant insights to answer our problem statement (Saunders et al., 2009). As this thesis seeks to find explanations to a phenomenon employed by a variety of businesses, case companies of different size, capital, and industries and in different phases of the life cycle were chosen. The sample is however limited to companies selling to businesses in the enterprise software market, as this reflects the nature of our problem statement. Based on these criteria, a series of companies were identified. We reached out to these companies via e-mail with information regarding the thesis, what resources we would need and how participation could be beneficial for the firm. See appendix A2 for example of an e-mail. In total we e-mailed 11 companies[1] of which four gave positive feedback. These four companies are thus the case companies employed in this thesis: Telenor, Podio, Plytix and Microsoft. The four case companies will be discussed in greater detail in chapter 5.

One interview was conducted with each case company. Our contact person in Microsoft was Ole Roterud, director SMB. Roterud has great knowledge on Microsoft's products and product strategy. From Podio we interviewed CEO and founder Kasper Hulthin. Hulthin has experience with the freemium business model both from Podio and from his newly founded start-up Peakon. Our contact person in Appear.in was the Head of Appear.in in Telenor Digital, Svein Yngvar Willassen. Lastly, in Plytix we had contact with founder and CEO Morten Poulsen, who has experience from working in Google.

[1] Microsoft, Podio, Peakon, Dropbox, Yammer, Box, Google, Telenor, Salesforce, Facebook and Issuu.

3.3.2 Semi-structured Interviews

Semi-structured interviews are non-standardized qualitative research interviews (Saunders et al., 2009). Here, the researcher has a list of themes and questions to be covered, which may vary in the different interviews. The semi-structured interview also provides the opportunity to ask additional and follow-up questions that may be required to explore the research question, and to probe answers where researchers want the interviewees to explain, or build on, their responses (Saunders et al., 2009). Our intention of using semi-structured interviews is based on the desire to cover certain pre-determined topics, but at the same time have the opportunity to be flexible and let the interviewees talk and engage freely in order to gain new insights - which is in line with our inductive approach. Saunders et al. (2009) recommend that when using a semi-structured approach to interviews, researches should employ an interview guide.

The interview guide lists topics we intend to cover in the interviews, as well as initial questions and probes that may be used for follow up questions. The interview guide is also helpful in sorting our data for analysis. The research questions and objectives guided the structure of our interview guide, and together with literature laid the foundation for our interview questions. An example of the interview guide is provided in appendix A3. The interview guide may vary in the different interviews, to fit to the specific companies. The interviews lasted 1-2 hours and were conducted at the office of the interviewees, or via the video conferencing services Skype and Appear.in. In order to provide a more accurate rendition of the interviews, they were all recorded (Yin, 2003). The recordings were used to transcribe the interviews, thus enabling us to go back to the material when needed.

3.3.3 Secondary Sources

The secondary data used in this thesis was collected before, during and after collection of primary sources of information. The gathered data consisted of information from other case companies as well as more extensive information about the case companies interviewed. The data was used to clarify the situation and provide further insights regarding companies utilizing a freemium business model in the B2B market. The secondary data was located using Google Search with search texts like “freemium business model”, “freemium B2B”, “freemium Microsoft” etc. The data was further

filtered by its usefulness and trustworthiness. Most commonly, acknowledged newspapers like Forbes, Business Insider, Wall Street Journal and Washington Post were used as these are recognized as reliable sources of information. However, company blogs and personal blogs are also used to gather expert insights.

3.4 Data Analysis

Data analysis is the process of bringing order, structure and meaning to the mass of collected data (Ghauri & Grønberg, 2002). In this section we review the possibilities of individual vs. cross-case analysis (paragraph 3.4.1) as well as specific methods for structuring data (paragraph 3.4.2).

3.4.1 Individual or Cross-Case Analysis

Patton (1980) differs between individual case analysis and cross-case analysis. *Individual case analysis* is beneficial when the primary focus is to find variations between the cases studied. On the contrary, when the focus is on finding patterns or commonalities between the cases, a *cross-case analysis* is more appropriate (Patton, 1980). As our thesis studies how a freemium business model creates value in different B2B contexts, it aims to find commonalities or patterns between the case companies studied - as such we perform a cross-case data analysis.

Taking a mixed research approach furthermore implies that this research analyses data using both deductive and inductive methods. The conceptual framework offered by Amit & Zott (2001) is employed to help organize and direct the data analysis. Based on supplementing literature we deductively propose modifications to the framework in paragraph 4.6, which helps guide the data analysis. However, as our research is aimed at a specific concept, we recognized a need to also inductively categorize data within the framework's identified sources of value creation. Even though we incorporate an inductive approach, starting our analysis from a theoretical perspective enables us to link the research to an existing body of knowledge (Saunders et al., 2009).

3.4.2 Data Structure

In order to structure data stemming from a research one might use data displays to organize and assemble data in diagrammatic or visual displays (Saunders et al., 2009). In analyzing our data we made use of the computer aided qualitative data analysis software program NVivo. This increased the transparency and methodological rigorousness of the data analysis, and enabled us to structure, store and code the data in a consistent manner (Saunders et al., 2009).

Saunders et al. (2009) differ between three main types of processes when grouping data; summarizing, categorizing and structuring of meanings. All of these can be used on their own, or in combination. As our analysis starts with a pre-defined conceptual framework, we use this framework as categories in which we code the data collected (Appendix A4). Thus, we attach relevant 'chunks' of our primary and secondary data to the appropriate category. As these categories are rather comprehensive, we recognized a need to develop sub-categories (Appendix A5). These subcategories were formed through an iterative process where some emerged from existing theory, and others emerged inductively through patterns and relationships as we gained new insights, and where we were not able to attach these to the pre-defined categories. Yin (2003) refers to this process as 'explanation building' where researchers aim to test theoretical propositions, but in an iterative manner. Explanation building is a useful data analysis method in explanatory case studies (Yin, 2003), as the approach allows for the theoretically based propositions to be initially revised through an iterative process (Saunders et. al, 2009). The proposed categories and subcategories were further tested by seeking alternative explanations and negative examples, in order to draw a conclusion and find explanations (Saunders et al., 2009).

3.5 Evaluation of Credibility

Credibility refers to the reliability (paragraph 3.5.1), validity (paragraph 3.5.2) and generalizability (paragraph 3.5.3) of a study;

3.5.1 Reliability

Reliability refers to the extent to which the data collection techniques or analysis procedures will yield consistent findings (Yen, 2003). Thus, reliability refers to the stability of the measurement in a research (Ghauri & Grønhaug, 2002). Efforts to increase reliability in this study include case studies in several companies and documentation of collected data, e.g. voice recordings of interviews, secondary sources and other material. The use of the data analysis program NVivo further allows potential readers to track the data analysis. Thus, we have to the best of our abilities tried to make the data as replicable as possible, by being transparent and leaving a trail of evidence for the reader to follow.

3.5.2 Validity

Validity is concerned with whether the findings are really about what they appear to be about (Yen, 2003). There are different types of validity (Ghauri & Grønhaug, 2002). *Internal validity* refers to the extent to which one can infer a causal relationship between two or more variables. A correlation between two variables does as such not indicate that there is a causal relationship, as the correlation coefficient does not tell us anything about the direction or if it is influenced by other factors (Ghauri & Grønhaug, 2002). *External validity* refers to in what extent the findings can be generalized to other settings, e.g. to other industries. Furthermore, *interpretive validity* refers to how good the interpretation of the data is. In our research we are not able to reach external validity, as qualitative data rarely can make generalizations to the total population. We are however aiming at reaching internal validity to see if some drivers of the freemium model have a higher chance of leading to value creation than others. We also aim to reach interpretive validity, making it easy to follow our data interpretation.

3.5.3 Generalizability

As previously stated, generalizability is difficult when using qualitative studies, as there is not enough data points to draw conclusions for an entire population (Yin, 2003). Thus, drawing conclusions that are generalizable to entire populations and universes is not the intention of this

study. One can however try to reach what is called theoretical generalization, referring to a generalization from one case to other cases belonging to the scope of the theory involved (Smaling, 2003). Since very little literature exists about the freemium business model in general and on the application of the model in the market for large enterprises in particular, our aim has been to extend the knowledge in this field and enable further research in the area. Thus, as we try to strengthen, weaken or neutralize data in a particular field of interest (Jacobsen, 2005), we are aiming to achieve theoretical generalization.

3.6 Limitations

This section reviews possible limitations with set study, specifically translation (paragraph 3.6.1), the number of interviews per case company (paragraph 3.6.2) as well as the number of case companies used (paragraph 3.6.3).

3.6.1 Translation

Interviews with Microsoft and Telenor were conducted and transcribed in Norwegian, and then translated when there was a need to use information from the interviews in the analysis or discussion. This translation might have led to misunderstandings or misconceptions. We have however tried to translate appropriately, and taken care to not miss any important points or meanings. Furthermore, we believe our skills in both Norwegian and English are satisfactory to avoid important mistakes in understanding what the interviewees wanted to communicate.

3.6.2 Few Interviews per Company

We only conducted one interview per case company in this research. This can be a limitation, as one or two hours interviews might not be enough to properly understand what the research objectives wanted to communicate about the topic. At the same time we had the possibility of asking follow up questions through email. We also incorporated secondary sources on public information about the companies, so we did not have to spend time on such details in the interviews. Additionally, we have included a range of other sources to add rigorousness to the

research. With the additional use of secondary sources, we believe we have been able to satisfactorily answer the research question.

3.6.3 Case Companies

Four case companies have been studied in this research. The sample is limited, which might pose a threat to the validity, reliability and generalizability of our thesis. However, this study is limited to a time period of four and a half months, and as students with limited resources in terms of contacts and goodwill, four case companies seem appropriate to answer our problem statement.

Chapter 4 - Literature Review

To gain an understanding of value and value creation, this chapter will first cover literature on value creation. Thereafter follows a review of literature on information goods, networks and B2B market characteristics. This literature review will serve as a means to identify a model for value creation which will be used throughout the thesis to explain the mechanisms driving value creation in a freemium model for companies operating in B2B markets. This model is revised from Amit & Zott's (2001) value creation model, which is thoroughly reviewed at the end of this chapter.

4.1 Value Creation

In order to build sustained competitive advantage, organizations must understand what creates value (Lapierre, 2000; Lichtenthal et al., 1997). Traditionally there has been five perspectives on value creation: Porter's Value Chain perspective (paragraph 4.1.1), Schumpeterian Innovation (4.1.2), The Resource Based View (RBV) (Barney, 1991) (paragraph 4.1.3), Transaction Cost Economies (paragraph 4.1.4) and Strategic Network Theory (paragraph 4.1.5). Each of these perspectives emphasizes different means of value creation, which will be covered in the following paragraphs.

4.1.1 Porterian View

Porter (1985) analyzes value creation at the firm level through value chains. A value chain is a set of activities that an organization carries out to create value for its customers (Porter, 1985). These activities can for example be logistics, operations, marketing and sales. The way in which activities within the value chain are linked and performed determines costs and profits, which is helpful in understanding sources of value. Porter (1985, p.38) defines value as "the amount buyers are willing to pay for what a firm provides them". A firm is profitable if the value it creates exceeds the costs involved in creating that value. Firms can thus create value and gain a competitive advantage through differentiating any of its activities and/or the links between them. The value chain perspective can be a helpful tool when examining the freemium model as it explains how firms can differentiate by lowering or completely exclude the costs of marketing and sales, or by outsourcing

certain activities through the use of complementary assets. However, the value chain perspective fails to capture the role of important mechanisms such as networking, trust and brand, in creating value in virtual markets *(*Amit & Zott, 2001).

4.1.2 Schumpeterian Theory

Schumpeterian theory (1934) is concerned with innovation as a driver for economic change. Schumpeter (1934; Schilling, 2013) found that innovation and the emergence of technological discontinuities can overturn an industry structure and create new market leaders and losers. In Schumpeterian theory, innovation is the source of value creation; novel combinations of resources lay the foundation for new products and production methods which following creates value. In virtual markets however, innovation involves more than novel products and production methods; it encompasses new transaction and exchange mechanisms, and span across firm and industry boundaries (Amit & Zott, 2001). Although one certainly must acknowledge the role of innovation in creating value, other sources of value not included in Schumpeterian theory, must also be taken into consideration (Amit & Zott, 2001).

4.1.3 Resource Based View

The Resource Based View (RBV) focuses on the internal resources and capabilities of a firm. The RBV assumes that firms can create value and achieve sustained competitive advantage only when the resources they own are heterogeneous and immobile (Barney, 1991). Thus, competitive advantage is developed when a firm implements a value-creating strategy not simultaneously employed by any current or potential competitor. The resource based framework, 'VRIN,' can be used to characterize and analyze strategic resource's potential to provide a sustained competitive advantage. Barney (1991) argues that a firm's resources must be *v*aluable, *r*are, *i*nimitable and *n*on-substitutable to generate a sustained competitive advantage. While this view is helpful in identifying the internal value creating capabilities of a firm, it does not acknowledge that value creation may span beyond firm boundaries and be embedded in interfirm resources and routines (Dyer & Singh, 1998).

4.1.4 Transaction Cost Economics

Transaction Cost Economics questions why some transactions are internalized in firms rather than in the market (Williamson, 1975), and is concerned with explaining "the choice of the most efficient governance form given a transaction that is embedded in a specific economic context" (Amit & Zott, 2001, p. 499). Transaction costs are those costs incurred in planning, adapting, executing and monitoring task completion (Williamson, 1983). Williamson (1975) identified six factors that affect the choice of transaction; bounded rationality, uncertainty, complexity, asymmetric information, and opportunism (Amit & Zott, 2001). In transaction economics, value is created through efficient transactions as efficiency reduces costs. Value creation can thus derive from reducing uncertainty, complexity, information asymmetry etc. in transactions (Amit & Zott, 2001). Other factors like reputation, trust, brand and experience can further enhance transaction efficiency (Williamson, 1975). The Internet, and especially Web 2.0, has significantly reduced transaction costs for e-businesses (Dyer, 1997; Musser & O'Reilly, 2007). Transaction costs theory can thus help enhance our understanding of how the freemium model creates value through efficient transactions. However, it fails to explain how other sources like innovation and complementary assets serve to create value for a firm (Amit & Zott, 2001).

4.1.5 Strategic Network Theory

Strategic networks are "stable interorganizational ties which are strategically important to participating firms" (Amit & Zott, 2001, p. 489; Gulati, Nohria & Zaheer, 2000, p.203). To gain competitive advantage in changing environments, companies need to build strategic network capabilities which enable the firm to place itself in a particular position to gain valuable partnerships in their networks (Hagedoorn, Roijakkers & Van Kranenburg, 2006). Such networks can for example be strategic alliances, joint ventures and long-term buyer-supplier relationships (ibid). Finding beneficial partnerships is especially important in high-tech industries, as these industries are in fast change and important capabilities do not necessarily reside within firm boundaries (Hagedoorn, et al., 2006). In an era of globalization and specialization, a firm's relational assets have furthermore become a vital source of strategic capability and competitive advantage (Bonner, Kim & Cavusgil, 2005; Gulati et al., 2000; Zaheer & Zaheer, 1997).

Network theorists are concerned with the implications of network structure for value creation. Several factors such as density and centrality (Freeman, 1979), size and heterogeneity of ties (Granovetter, 1973) affects the value of a network. Amit & Zott (2001) further add that strategic networks offer the potential to share risk, generate economies of scale and scope, share knowledge, and facilitate learning. Co-creation of value, where two or more firms combine their resources to create value, is a growing trend in today's business environment (Sarker, Sarker, Sahaym & Bjørn-Andersen, 2012), and has increased the range of possible organizational arrangements for value creation (Amit & Zott, 2001). This has raised questions about whether the degree to which value generation occurs is subject to contextual factors. Sarker et al. (2012) suggest these contextual factors might be different *governance mechanisms* like trust and intellectual property protection, a focus on *power and politics* where there are opposing interests within alliances, as well as *collective strength*, which functions as an enabling factor of value generation in networking alliances.

The network perspective is relevant for understanding wealth creation in e-business because of the importance of networks of firms, customers and other partners in virtual markets (Amit & Zott, 2001; Shapiro & Varian 1999; Prahalad and Ramaswamy, 2000). It may be a particularly useful perspective in enhancing our understanding of value creation in a freemium business model, as this model in its very nature relies on networks of users and complementarities. However, the theory may not fully capture the value creation potential of e-businesses that employ new and novel transaction structures (Amit & Zott, 2001). As an example it may not fully explain how and why zero prices create value in information markets.

4.1.6 A Coherent Perspective on Value Creation

As evident from the above assessment, each of these perspectives on value creation can enhance our understanding of value creation in a freemium business model in different ways. However, employing only one of these views may be incomplete in capturing the full potential of firm value creation in virtual markets. Amit & Zott (2001) and Zott & Amit (2009) suggest taking a holistic view on value creation by combining the above mentioned literature in a business model perspective, emphasizing the need to seek outside resources, attribute innovation and at the same time proactively engage in multiple networks.

4.1.7 Value Creation in the Digital Age

The digital age and especially the rise of the Internet, SaaS and availability of information goods have led to the emergence of new digital markets (cf. chapter 1 and 2). This has raised fundamental questions about how firms create and capture value from delivering new information services (Teece, 2010). Virtual markets have a profound effect on how value-creating economic transactions are structured, like the ease of extending a product range to include complementary products, improved access to assets etc. (Amit and Zott, 2001). The Internet is however not just a source of easy access to digital data; it is also a new channel of distribution and for piracy which clearly makes creating and capturing value from Internet transactions difficult (Teece, 2010).

Lytras, Damiani and Ordónez de Pablos (2009) argue that the shift from web 1.0 to web 2.0 resulted in a shift in the way value creation is defined. Traditionally the value was perceived to be inherent in the product itself, whereas the value today is viewed as a system of relations around an offering. This shift from traditional value chains to value creation systems, entails that removal of one part of the system will seriously damage the value of the total offering (Lytras et al., 2009). Zott & Amit (2009, p.260) state that these developments have enabled firms to rethink the ways they "organize and conduct exchanges and activities across firms and industry boundaries". This has increased the possibility of conducting purposeful networking, where firms structure their destiny "within the context of the value networks within which they exist" (Zott & Amit, 2009, p.260). Thus, considering an exchange in isolation from other exchanges that may complement or facilitate that exchange makes it difficult to assess the full value creation potential (Zott, Amit & Massa, 2011). This holistic view is however captured by the business model perspective (Zott & Amit, 2009). Rather than focusing on the focal firm as one of several players in a network, the business model perspective explores how a particular firm structures itself within a value network (Zott & Amit, 2009).

4.1.8 Four Interconnected Sources of Value Creation

Taking a business model perspective, Amit & Zott (2001) identify four interconnected sources of value creation in virtual markets: Efficiency, lock-in, complementarities and novelty;

Transaction efficiency is found to be one of the primary value drivers for e-businesses and reflects the theory on transaction cost economies as covered in paragraph 4.1.4. Transaction efficiency increases when the cost per transaction decreases, and is influenced by factors such as speed, information symmetry, search costs and simplicity (Amit & Zott, 2001). A greater understanding of how efficiency yields value and its importance in virtual markets will be provided in paragraphs 4.2.1 and 4.2.2.

Drivers of Transaction Efficiency	Information Asymmetry	Search Costs	Selection Range	Simplicity	Speed	Scale Economies
Amit & Zott (2001)	Reducing information asymmetry will enhance transaction efficiency through the supply of "up-to-date and comprehensive information", which the Internet facilitates in an easy and convenient manner.	Through reducing information asymmetry, firms can also reduce customers search costs as greater availability of information naturally leads to a more efficient search.	E-businesses provide greater selection at lower costs which reduces information asymmetry.	E-businesses simplify transactions which reduces the likelihood of mistakes, and thus reduces information asymmetry.	The Internet facilitates speedy transactions, which enhances transaction efficiency.	E-businesses allow individual customers to benefit from scale economies through demand aggregation and bulk purchasing.

Table 1. Transaction Efficiency.

Lock-in refers to the extent to which customers are engaged in repeat transactions, and can be enhanced by creating switching costs. In the digital age, such switching costs often encompass network effects, and the driver is thus largely influenced by theory on strategic networks as covered in paragraph 4.1.5. Amit & Zott (2001) argue that there are numerous ways for companies to achieve lock-in, such as loyalty programs, customization, trust and through network externalities. Lock-in, switching costs and network theory will be covered more thoroughly in paragraph 4.2.4 and 4.3.

Drivers of Lock-In	Direct Network Effects	Indirect Network Effects	Customization	Brand and Trust	Loyalty Programs	Dominant Design
Amit & Zott (2001)	Direct network externalities drives lock-in as the value created for customers increases with the size of the customer base.	Indirect network effects drives lock-in as increased usage of a product or service spawn increases in the availability and value of complementary goods, which eventually leads to an increase in the value of the original offering.	Customization enhances lock-in by enabling customers to customize the offering to their individual needs. Customers then have high incentives to use the system, which creates a positive feedback loop.	Through creating trust and familiarity firms may gain more loyal customers, as customers increase their learning costs and thus also their switching costs.	Loyalty programs rewarding repeat customers with special bonuses may enhance lock-in.	Developing dominant designs proprietary standards will increase switching costs, thus enhancing lock-in.

Table 2. Lock-In.

Complementarities yield value whenever having a bundle of goods together provides more value than the total value of having each of the goods separately. Businesses often leverage this value potential by offering bundles of complementary products and services, as opposed to stand-alone offerings (Amit & Zott, 2001). These complementary offerings may be both vertical and horizontal complementarities, and also complementary goods not directly related to the core transaction. We will return to complementarities and bundling of goods in paragraph 4.2.3.

Drivers of Complementarities	Vertical and Horizontal Complementarities	Complementary Activities	Complementary Technologies	On-line and Off-line Complementarities
Amit & Zott (2001)	Vertical and horizontal complementarities enhance the value of the core offering, as several offerings are bundled together.	E-businesses may create value by capitalizing on complementarities among activities such as supply-chain integration.	E-businesses may create value by capitalizing on complementarities among technologies such as linking the imaging technology of one business with the Internet communication technology of another, thereby unleashing hidden value.	Offline assets complement online offerings; Customers who buy products over the Internet value the possibility of getting after-sales service offered through bricks- and-mortar retail outlets, including the covenience of returning or exchanging merchandise.

Table 3. Complementarities.

Lastly, novelty depicts the value creation of innovation as captured by Schumpeter (1934), and also innovation in the way firms do business and structure transactions (Amit & Zott, 2001). Innovation in transaction structures can create value by connecting previously unconnected parties, eliminating inefficiencies in the decision-making process, or by creating entirely new markets (Amit & Zott, 2001). Novelty-centered business models, where new transaction structures are employed, are found to yield higher performance than efficiency-centered business models which enhance transaction efficiency (Zott & Amit, 2008). Amit & Zott (2001) argue that in virtual markets, the possibilities for innovation are endless. For example, companies can incorporate new complementary offerings in novel ways. There can furthermore be substantial first-mover advantages for e-business innovators, as being the first to market with a novel business model or offering makes it easier to create switching costs by developing brand reputation and awareness (Amit & Zott, 2001).

Drivers of Novelty	New Transactional Content	New Transaction Structure	New Transaction Participants
Amit & Zott (2001)	The value creation potential of innovations has been articulated by Schumpeter (1934). The introduction of new products or services, new methods of production, distribution, marketing or the tapping of new markets are traditional sources of value creation through innovation. The main innovation of some e-businesses resides in their complementary elements.	Innovation in transaction structures create value by connecting previously unconnected parties, eliminating inefficiencies in the buying and selling processes through adopting innovative transaction methods, capturing latent consumer needs, and/or by creating entirely new markets. For example, e-business firms can identify and incorporate valuable new complementary products and services into their bundle of offerings in novel ways.	Another dimension of innovation in e-business refers to the appropriate selection of participating parties. For example, firms can direct and intensify traffic to their web site by initiating affiliate programs with third parties, who are compensated for enabling the execution of transactions from their own web sites.

Table 4. Novelty.

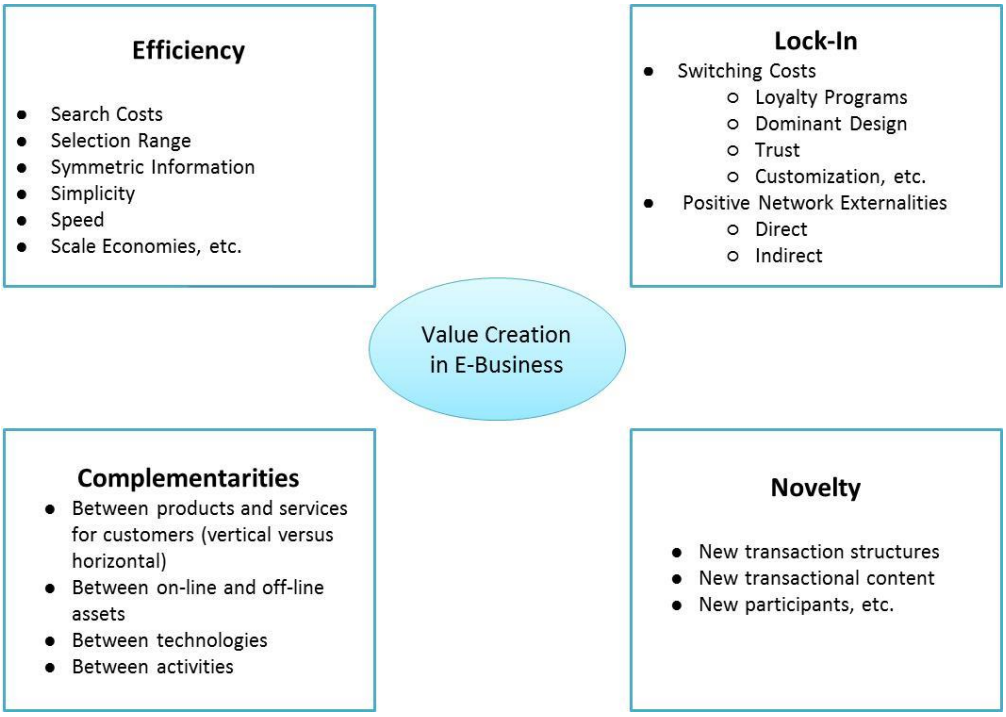

Figure 3. Four Interconnected Sources of Value Creation. Source: Amit & Zott (2001).

Figure 3 provides a complete overview of Amit & Zott's (2001) four interconnected sources of value creation. Amit & Zott's (2001) research suggest that the presence of any of these sources of value creation enhance the value creation potential of a firm's business model in e-businesses. Furthermore, there are important synergies between the four sources of value creation where one can be more valuable when supported by the presence of another. For example, the presence of lock-in can positively influence the value of complementarities as lock-in provides a promise of

potential high-volume repeat business which attracts complementors. Similarly, complementary offerings may serve to attract and retain customers and as such positively influence lock-in (Amit & Zott, 2001). See table 5 for a complete overview of the links between the sources of value creation.

Links between the value drivers	Efficiency	Lock-In	Complementarities	Novelty
Efficiency Enhances		The efficiency features of an e-business may serve to attract and retain customers and partners. The higher the relative benefits offered, the higher their incentives to stick with the network established by the e-business. The increasing return properties inherent to network effects then magnify the relative benefits offered, thus triggering positive feedback dynamics.	Efficiency gains made possible by information technology pave the way for the exploitation of complementarities in e-business. Weaving together the resources and capabilities of distinct firms, a hallmark of e-businesses, is economically compelling when transaction costs, and hence the threat of opportunism, are low.	n.a
Lock-In Enhances	When an e-business creates lock-in, this can also have positive effects on its efficiency. For example, many auction sites enable buyers to rate sellers. This feature increases buyers' trust in the fairness of transactions and therefore fosters stickiness. This feature also provides a strong incentive for repeat sellers to refrain from cheating, which clearly enhances transaction efficiency.		When an e-business creates lock-in, this can also have positive effects on the degree to which it provides complementary offerings. A strong potential for lock-in provides an incentive for high-profile partners to contribute complementary products and services because of the promise of high-volume (repeat) business.	n.a
Complementarities Enhances	Complementarities may lead to increased efficiency, at least from a customer's point of view. When customers have access to products and services that are complementary to the primary product of interest, efficiency may be enhanced, for example, through reduced search costs and improved decision-making.	The complementary product and service offerings of an e-business may serve to attract and retain customers and partners. The higher the relative benefits offered, the higher their incentives to stick with the network established by the e-business. The increasing return properties inherent to network effects then magnify the relative benefits offered, thus triggering positive feedback dynamics.		The main innovation of some e-businesses resides in their complementary elements, such as the resources and capabilities they combine.
Novelty Enhances	Certain efficiency features of e-businesses may be due to novel assets that can be created and exploited in the context of virtual markets. For example, Art-net.com, a European company that enables online art auctions, reduces the asymmetry of information between the buyers and sellers of art (traditionally a source of severe inefficiencies) through maintaining and expanding a data base of transactions (including information on price) that is accessible to its clients.	Novelty affects lock-in, in two important ways: First, e-business innovators have an advantage in attracting and retaining customers, especially in conjunction with a strong brand. Second, being first to market is an essential prerequisite to being successful in markets that are characterized by increasing returns. First movers are in a good position to initiate the positive feedback dynamics that derive from network externalities, and to achieve a critical mass of suppliers and/or customers before others do.	n.a	

Table 5. Links between the Four Sources of Value Creation. Source: Amit & Zott (2001).

In order to identify the mechanisms underlying these sources of value creation in a freemium model, we first aim to develop an understanding of central concepts characterizing virtual markets and information goods. This will contribute to create an understanding of how these interrelated sources create value in information markets or e-business markets, where freemium firms operate. The next section of the thesis will thus commence with literature on Information Goods (paragraph 4.2) before explaining one of its core concepts, "Network Effects", in greater detail in order to understand the powerful impact this has on switching costs and lock-in (paragraph 4.3). Our problem statement further demands a more thorough understanding of B2B transactions (paragraph 4.4). Lastly, the section will conclude with a literature revision and important implications from set literature, which will be used to build the analysis of this thesis (paragraph 4.5 & 4.6).

4.2 Information Goods

This section of the thesis covers important properties of information goods; first, that information goods typically face high cost of production and low cost of reproduction (paragraph 4.2.1); second, that information goods are inherently experience goods (paragraph 4.2.2); and third, discuss the degree to which information goods are public goods (paragraph 4.2.3). Lastly, ways of creating lock-in for firms operating with information goods will be discussed (paragraph 4.2.4).

4.2.1 Pricing Information Goods

An important property of information goods is the high costs of production and low costs of reproduction, implying low marginal costs (Shapiro & Varian, 1999). This cost structure leads to substantial economies of scale, and has implications for competitive pricing strategy. As stated by Bertrand (de Bornier, 1992), in competitive markets, prices will fall to marginal costs, which in the case of information goods, is close to zero. Varian (1997) argues that in this situation, cost-based pricing makes little sense, and that value-based pricing is more appropriate. As different consumers may have different values for information goods, differentiating the goods by pricing can be a sustainable strategy (Varian, 1997).

4.2.1.1 Versioning Information

Versioning means offering information goods in different versions for different market segments (Shapiro & Varian, 1999). The point of versioning information goods is to get customers to sort themselves into categories based on their willingness to pay (Varian, 1997). The fundamental problem in versioning is setting the different categories, in order to make the users who are willing to pay high prices do so. Varian (1997) suggests basing the categories on exogenous characteristics like business customers and private customers, or when this is not possible and/or feasible, base it on endogenous characteristics such as quality, and let users self-select into their desired category.

4.2.2 Information as an Experience Good

Information goods are further characterized as experience goods, meaning users must experience the information upon determining its value (Varian, 2001). This implies that in order to transact in information goods, information suppliers must first demonstrate the potential value. Many media producers overcome this problem through branding and reputation (Shapiro & Varian, 1999). This however requires rather large investments, and is thus not a feasible strategy for all information suppliers. Shapiro & Varian (1999, p.86) argue that information suppliers can prevail by giving away parts of their product for free so that the free products work as advertisements for the proprietary offerings; "the trick is to break your product up into components; some you give away, others you sell". In other words, versioning the offering so one part of the offering is free (Shapiro & Varian, 1999).

4.2.2.1 Zero Pricing

According to standard theoretical perspectives, when faced with a choice of selecting one of several options consumers will choose the option with the highest cost-benefit difference (Shampanier, Mazar, & Ariely, 2007). However, Shampanier et al. (2007) find that decisions about free products differ in that consumers perceive the benefits associated with free products as higher. Their research demonstrates that when consumers are faced with a choice between products where one is free, they 'overreact' to the free product as if zero prices not only imply a low cost, but also increase the benefits of the offering. This implies that versioning products (Shapiro & Varian, 1999), so that one

part of the product is free, will attract significantly more users than if the low-end version of the product had a price that was low, but higher than zero.

4.2.3 Bundling and Complementary Products

As noted thus far, differentiating and versioning a product can be a viable strategy to appropriate rents in the information market. A third property of information goods is however that they, to some extent, are public goods, characterized as non-rival and non-excludable (Varian, 2001). Even though information goods are inherently non-rival, they can in theory be protected by formal intellectual property mechanisms, thus making them excludable. However, with the development of Web 2.0, open source software has increased in popularity and piracy has become rather common (Anderson, 2009). Information goods may therefore be less prone to intellectual property now than before, which makes differentiation a more difficult strategy.

Due to this and the fact that information goods are non-rival, 'bundling' and 'complementarities' is a much used deployment strategy in information markets. Bundling enables a new product or service to piggyback on the success of another good that already enjoys an installed base (Schilling, 2013). Such bundling arrangements have proven to be central to the success of many companies such as Microsoft and Apple. Complements are defined as "products or services that enhance the usefulness or desirability of a good" (Schilling, 2013, p.113). A complementarity is thus a product or service that is compatible with another good. Traditionally, many firms have made their software incompatible with that of others in order to protect their rents (Schilling, 2013). More recently, there has however been a trend to open up and complement each other's goods, especially through the use of APIs (Lyman, 2012).

4.2.3.1 Application Programming Interface (API)

An Application Programming Interface (API) lets programmers use the functionality of a piece of software and develop new innovations and services on top of it, without having access to the source code (Anderson, 2012). An API can as such facilitate integration of new features, and enable data-sharing. In essence, APIs make it easy for developers to connect networks, individual devices, and information streams (Simon, 2014). Thus, APIs enable different pieces of software to interact with each other, allowing different information suppliers to complement each other's goods and services.

4.2.4 Lock-In and Switching Costs

When buyers must bear costs to switch from one information system to another, they are facing switching costs (Shapiro & Varian, 1999). Switching costs occur when the users have invested time and money in learning about an offering and its complementary assets. Users then face significant costs when switching to a new, incompatible system. When these switching costs are substantial the user's face lock-in, meaning that they are locked into a system or brand. Shapiro & Varian (1999) argue that suppliers of information goods should deal with lock-in by utilizing three key principles: Invest in an installed base (paragraph 4.2.4.1), encourage customer entrenchment (paragraph 4.2.4.2) and leverage the installed base (paragraph 4.2.4.3);

4.2.4.1 Investing in an Installed Base

Suppliers of information goods must first invest to build an installed base of customers, using tactics accumulating the lowest possible cost. Such tactics can for example be promotions, discounts, trials or free versions (Shapiro & Varian, 1999). The higher a buyer's switching costs, the more it is worth to spend money on locking in the buyer. In order to attract customers locked into competitor's offerings, firms may need to heavily subsidize their switching costs. Suppliers must accordingly evaluate the profitability of each type of prospective customer in order to determine how much to invest in attracting new customers (Shapiro & Varian, 1999).

4.2.4.2 Encouraging Customer Entrenchment

Following these investments, suppliers must entrench customers so they continue to invest in the good, and become more committed to the supplier over time (Shapiro & Varian, 1999). One way of entrenching customers is to over time offer them more value-added features, like customer service, consulting services and/or new innovative features (Shapiro & Varian, 1999).

4.2.4.3 Leveraging Your Installed Base

Suppliers should further leverage their installed base by selling complementary products, and selling access to these customers to other suppliers (Shapiro & Varian, 1999). Offering complementary products and services may increase customers switching costs, and at the same time encourage customer entrenchment. If the supplier does not have any complementary products of its

own, it can sell access to its customers to other suppliers, for example through the use of APIs as covered in 4.2.3.1.

4.3 Network Effects

A network externality exists when the benefit of a product or service increase with the number of users consuming the same good (Haruvy & Prasad, 1998; Katz & Shapiro, 1985). Positive network effects raise the value received by consumers as markets get larger, and have an impact similar to conventional firm level economies of scale (Margolis, n.d.). With network effects, potential buyers take into account both their own reservation price and the total number of users enjoying the good (Easley & Kleinberg, 2010). To market a product subject to network effects entails there is a need to convince a large initial user group to adopt the product before others will be willing to buy it. One possibility is to set an initial low price, perhaps also for free, in order to increase the conversion rate (Easley & Kleinberg, 2010).

Shapiro & Varian (1999) differ between 'real' and 'virtual' networks; in real networks linkages between nodes are physical connections such as telephone wires, whereas in virtual networks the linkages between the nodes are invisible. In the subsequent, the focus will be on the latter. The Internet is a prime example of a market dominated by virtual network effects as communication between users is easy and transparent (Anderson, 2009). Web 2.0 furthermore enables networks to combine the powerful characteristic of digital economies, low marginal costs, with the opportunity of exponential network growth (Shuen, 2008). Shuen (2008) argues that managing network effects is the key to competitive success in Web 2.0, as online networks form faster and more frequently, and expands more rapidly across market borders. In the following, we will emphasize three types of network effects and their influence in the information goods market; direct network effects (paragraph 4.3.1), indirect network effects (paragraph 4.3.2) and platforms (paragraph 4.3.3). Following, switching costs in markets characterized by network effects will be studied (paragraph 4.3.4).

4.3.1 Direct Network Effects

In the simplest form, direct network effects occur when an increase in the number of users leads to an increase in the value for each user (Katz & Shapiro, 1994). An example of such network effects is Facebook; the utility a user derives from having a Facebook account, clearly depends on the number of other users with an account. Such positive network effects give rise to positive feedback: when a user creates an account on Facebook, the value of another user's Facebook account is advanced since they are now able to communicate (Shapiro & Varian, 1999). Thus, if one user utilizes a good with network effects, other users will be more inclined to utilize the same good - and the consumer utilizing a good with network effects will enjoy increased benefits of that good if other customers utilize it. This creates a positive feedback loop where users will have strong motivation to 'push' the adoption of network goods (Silverman, 2011), resulting in adoption effects (Katz & Shapiro, 1994). Silverman (2011) refer to this as the "Viral Feedback Loop", where 'word of mouth' is a central enabler.

4.3.1.1 Word of Mouth

Word of mouth is defined as "the exchange of information about a product or a service among people who are independent of the producer" (Silverman, 2011, p. 51). Word of mouth is the most powerful force in the marketplace, as customers are thousands of times more likely to act on recommendations from friends, colleagues, or trusted advisors than they are to commercial communication (Silverman, 2011). With word of mouth the user gains indirect experience, which significantly eases the decision making process and decrease the perceived risk associated with making a purchase (Silverman, 2011). This is especially true for information goods which inherently are experience goods (Shapiro & Varian, 1999). Silverman (2011) further finds that word of mouth seems to have a higher than exponential growth rate. This is why value creation, and immediate and intuitive perception of such value, is important.

4.3.2 Indirect Network Effects

Indirect network effects occur as increased usage of a product or service spawn increases in the value of complementary goods, which eventually leads to an increase in the value of the original product or service (Katz & Shapiro, 1994). For example, if many consumers utilize a particular type

of hardware, software vendors will have greater incentives to make their software compatible with that hardware - which eventually leads to an increase in the value of the hardware. Compatibility with complementary technologies is considered an important determinant of a product's success, and a change in the complementary technology thus offers an opportunity for new entrants (Brynjolfsson & Kemerer 1996; Farrell and Saloner, 1987). With Web 2.0, indirect network effects are of even greater importance, as value creation has shifted to a system perspective (Lytras, Damiani, & de Pablos, 2008). Thus, firms increasingly organize their business to create exchanges and activities across firm and industry boundaries, which is further simplified through the use of APIs.

4.3.3 Platforms

Platforms, often referred to as two-sided networks, differ from traditional offerings that has one revenue stream. Platforms enjoy revenue streams from two distinct user groups, one on each side of the platform (Eisenmann, Parker & Van Alstyne, 2006). Two-sided network effects occur when the platform's value to any of the user groups largely depends on the number of users on the network's other side (ibid). Platforms thus yield increasing returns as network effects raise the user's valuation of the platform (Gawer & Cusumano, 2008).

4.3.4 Switching Costs in Markets Characterized by Network Effects

Switching costs in networks work in a nonlinear way: "convincing ten people connected in a network to switch to your incompatible network is more than ten times as hard as getting one customer to switch" (Shapiro & Varian, 1999, p.184). Thus, in network economies firms and users face collective switching costs. This is due to the fundamental economic characteristic of networks: the value of connecting to a network depends on the number of people already connected to it. Because users find it challenging to collectively coordinate a switch to an incompatible technology, control over a large installed user base is very powerful (Shapiro & Varian, 1999).

Brynjolfsson & Kemerer (1996) further find that specifically for software, the purchase price reflects only a relatively small portion of the total consumer expenditure; learning and conversion costs account for much of the remainder. Furthermore, positive network effects from an installed base and compatibility with a dominant standard are approximately as important as any of the

intrinsic product features (Brynjolfsson & Kemerer, 1996). Firms seeking to enter software markets with network effects must thus seek to leverage these networks by making their products or services compatible with the incumbent (Schilling, 2013). For the incumbent firms, networks are far more valuable if the company can control the ability of others to interconnect with the technology (Shapiro & Varian, 1999). At the same time, a failure to open up a technology can spell its demise if customers fear lock-in, or if the company is facing strong rivals whose systems offer comparable performance but is non-proprietary (Shapiro and Varian, 1999). Shapiro & Varian (1999) argue that it is more important to build a valuable technology than to take control over it. Opening up can further freely fuel positive feedback and maximize the total value added (Shapiro & Varian, 1999).

Due to the power of collective switching costs, entry into the software market is found to be more successful when there is a major change in platforms (Brynjolfsson & Kemerer, 1996), like the shift from cellphones to smartphones. However, even with a significant change in architecture, users may be willing to abandon previous software (Brynjolfsson, 1996). An example is the introduction of Google Docs, a word processing service built on new technology architecture, namely cloud computing, which has been able to "steal" many users from Microsoft's predominant desktop product, Office suite (Mackie, 2013).

4.4 B2B Transactions

As explained in chapter 1, the majority of companies utilizing a freemium business model have traditionally targeted the B2C market. Our research does however concentrate on value creation in the B2B market, and the problem statement thus demands a more thorough understanding of B2B transactions. Therefore, we first review some general differences between the B2C and B2B market (paragraph 4.4.1). Thereafter we seek to understand the importance of brand and reputation in B2B industries (paragraph 4.4.2), and lastly we discuss the issue of trust in B2B relations (paragraph 4.4.3).

4.4.1 General Differences between B2B and B2C Markets

B2B and B2C markets differ in many ways. *First*, unlike consumer markets where offerings are consumed personally by the people who buy them, the essence of business markets is that

organizations undertake the consumption (Fill & Fill, 2005). Thus, there are more people involved in the buying decision, which leads to higher complexity in the decision-making process (Kotler & Pfoertsch, 2006). *Second*, greater demand diversity drives a need for more specialization and customization in B2B markets (Kotler & Pfoertsch, 2006). As a result of this, bigger technical and economic considerations need to be made, requiring experts on both side of the transaction. *Third*, in B2B markets there are larger volumes of products/services per customer than in B2C markets (Kotler & Pfoertsch, 2006), as well as more money in circulation. *Fourth*, a close, long-lasting relationship is often required with B2B customers, which implies that more involvement from both the customer and the focal firm is needed (Kotler & Pfoertsch, 2006). Thus, transactions in the B2B market are generally more complex and risky than in the B2C market.

4.4.2 Brand and Reputation

The emergence of the Internet as well as further developments in information technology has dramatically increased the amount of choice for industrial buyers, as this global marketplace cuts the costs of searching and comparing products and services close to zero (Kotler & Pfoertsch, 2006). When the number of offerings increases, customers are likely to sustain preference for brands they already know, as this reduces search time and limits risk (Peter De Legge, 2002). Brands are found to speed up consumer's information processing (Gertner & Kotler, 2004), as well as reduce risk and complexity in the buying decision and increase the value received by customer's (Blacket, 1998). A firm enjoying a strong brand thus has the opportunity to charge premium prices for its products and services (Gertner & Kotler, 2004).

A firm's reputation is closely linked to the brand, but is both a result of how competent the firm seems to the consumer as well as how much the customer likes the company (Burke, Martin & Cooper, 2011). Reputation is closely linked to both personal experiences with the firm and word of mouth (Burke et al., 2011). A firm's reputation is important as "the investment community believes that between 30 and 50 percent of a company's value is intangible, based mostly on corporate reputation" (Burke et al., 2011, p.3). Burke et al. (2007) furthermore find that a favorable organizational reputation will influence purchase decisions when there is little difference in price, quality and design. As discussed in paragraph 4.2.1, this seems especially elevating in information good markets, as prices in these competitive markets falls close to zero (de Bornier, 1992).

4.4.2.1 Bundled Offerings

A complicating factor in B2B markets is the elevated use of bundling solutions rather than sale of single products (Kotler & Pfoertsch, 2006). As discussed in paragraph 4.2.3, this trend is enhanced by the emergence of information goods. As these bundled solutions are quite complex, they can be unclear to the consumer. Brands and a strong reputation are therefore found to be valuable assets in reducing risk and complexity in such offerings. This is recognized by the fact that risk reduction is the most important reason why companies use brand management in B2B markets, closely followed by elevated information efficiency (Kotler & Pfoertsch, 2006). Risk reduction and clear benefits are assumed to lead to loyal customers, which is important as loyal customers are found to be worth up to 10 times more than average customers (Anderson & Srinivasan, 2003).

4.4.3 Trust in B2B Markets

As discussed in paragraph 4.4.1, long-term relationships are generally of higher importance for B2B customers than for B2C customers (Kotler & Pfoertsch, 2006). It therefore seems that building trust with customers is a necessary prerequisite for success in B2B markets (DeMaio, 2001). Trust in B2B markets seems especially important in relation to privacy, intellectual property, anonymity and security (DeMaio, 2001).

4.5 Literature Revision

As there is no model aiming to explain exactly what we research, we find Amit and Zott's (2001) model on value creation the most promising in enhancing our understanding of value creation in a freemium business model (cf. paragraph 4.1). Still, we see the need to highlight some limitations with set model, as well as comment on differences in the way Amit & Zott (2001) utilize this model, and how it is used in this study.

Firstly, even though Amit & Zott's (2001) research is highly cited and relevant, it is at present time 15 years old. As changes in the e-business market are rapid, some aspects of the model are likely outdated. Secondly, Amit & Zott's (2001) main purpose is to identify the value creation potential of all e-business. Their inductive case study led to the identification of four sources of value creation.

Within these they list a number of examples of how firms might seek to enhance or realize these sources of value. Our research, on the other hand, takes a specific angle as we purpose to explain the value creation potential of a specific business model, namely the freemium business model.

Third, we aim to identify the specific mechanisms that enhance or realize the four sources of value creation in a freemium business model. We refer to these mechanisms as drivers. Thus, we are using Amit & Zott's (2001) research for something else than its main purpose. This may raise concerns regarding its applicability in our research. However, as previously stated we find this model to be the most promising in helping us answering the problem statement. Still, we recognize there is a need to adapt the model to our specific research;

4.5.1 View on Value

Amit & Zott (2001, p.515) posits that "total value created through a business model equals the sum of the values appropriated by all the participants in a business model, over all transactions that the business model enables." Though agreeing with their definition of value, our research takes a specific view wherein we focus on the firm. This implies that we are concerned with the value created and appropriated by the firm and not by other participants. Even though there might be disadvantages in not getting a clear perception of customers view on value, most of the case companies had already executed customer research on this subject. Taking a firm perspective furthermore enables us to fulfill the research's purpose and identify the specific mechanisms that drive value for the focal firm, within the identified sources of value creation.

4.5.2 Industry Specific

Amit & Zott (2001) utilize a sample of 59 e-businesses in their study. They define an e-business as "one that derives a significant proportion (at least 10%) of its revenues from transactions conducted over the Internet" (Amit & Zott, 2001, p.500). This definition is very broad and comprises most businesses in present time's society. Due to the purpose of this thesis we emphasize a specific industry, namely the enterprise software industry. This can potentially enable us to discover new sources of value creation and/or connections between them.

4.6 Literature Modifications

In this part of the thesis we propose modifications to Amit & Zott's (2001) model of value creation, based on the literature reviewed in this chapter, and the implications from non-academic articles on freemium models covered in chapter 2. More specifically we add new insights based on information goods, network theory and the freemium model (paragraph 4.6.1). Furthermore, we propose modifications to fit the model to a B2B context (paragraph 4.6.2). We believe these modifications will make the model more applicable to the purpose of this research.

4.6.1 Today's E-Business Market

Based on insights gained in the literary review and with the specific characteristics of a freemium model in mind, we propose some modifications to the value creation model of Amit & Zott (2001) to create a better fit with today's reality of virtual markets. We specifically propose versioning and zero prices as a driver of transaction efficiency (paragraph 4.6.1.1), brand as a driver of transaction efficiency (paragraph 4.6.1.2), word of mouth as a driver of efficiency and lock-in (paragraph 4.6.1.3), platforms as a driver of lock-in (paragraph 4.6.1.4), versioning as a driver of lock-in (paragraph 4.6.1.5), APIs as a driver of complementarities (paragraph 4.6.1.6) and versioning as a driver of complementarities and novelty (paragraph 4.6.1.7).

4.6.1.1 Versioning and Zero Prices as a Driver of Transaction Efficiency

Amit and Zott (2001) find that efficiency can be gained through lowering the cost of transactions. As explained in paragraph 4.2.2.1, versioning information goods may drive adoption as users pay the value they perceive (Shapiro & Varian, 1999). Shampanier et al. (2007) furthermore find that consumers overreact to free products, as if zero prices not only imply a low cost, but also increase its benefits. This implies that versioning a product (Shapiro & Varian, 1999) so that one part is free, will attract significantly more users than if the low-end version of the product had a price that was low, but higher than zero. Zero pricing drives physical transaction costs close to zero, and additionally contributes to higher information asymmetry as users get to experience the good upon determining its value. This is furthermore proposed to reduce search costs as customer's immediately gain greater information about the good. Thus, it seems zero price drives transaction efficiency, which in turn yields value for the focal firm.

4.6.1.2 Brand as a Driver of Transaction Efficiency

Amit and Zott (2001) find that trust and brand has the ability to drive lock-in, as knowledge of and trust in a company make customers more likely to be locked into their offerings. However, the authors do not comment on how trust and brand might drive transaction efficiency. Gertner & Kotler (2004) explain that brands have the possibility of speeding up consumer's information processing, if customers are familiar with a brand and its connotations. We therefore propose that a strong brand might enhance transaction efficiency.

4.6.1.3 Word of Mouth as a Driver of Efficiency and Lock-in

Word of mouth is, as discussed in paragraph 4.3.1 and 4.3.1.1, a central enabler of direct network effects in Web 2.0. With word of mouth, potential users gain a trusted recommendation and indirect experience with an offering (Silverman, 2011). This likely reduces search costs and information asymmetry, and thus enhances transaction efficiency. As more users join the network and network effects play in, lock-in likely also increase in value. Thus, we assume that word of mouth has the possibility of driving efficiency as well as lock-in. However, as covered in paragraph 2.4.2.2, some industry experts argue full featured productivity suites in the B2B market are less likely to go viral than single purpose consumer apps in the B2C market, as simplicity is key to virality. We thus expect there might be some issues in achieving a viral spread in B2B markets.

4.6.1.4 Platforms as a Driver of Lock-In

As explained in paragraph 4.3.3, platforms yield increasing returns as network effects raise the user's value of the platform (Gawer & Cusumano, 2008). As the two distinct user groups attract and retain each other, it will be challenging for one of the parties to leave the service. We therefore propose that a firm who has attracted two distinct and dependent user groups to its platform has the opportunity to create higher switching cost than a firm offering a single service.

4.6.1.5 Versioning as a Driver of Lock-In

As covered in paragraph 2.3.2 freemium firms version their offerings into free and premium versions. As freemium firms do not reap any monetary benefits from free customers it is important to continually convert users to a premium version (Yogev, 2011). We expect that this leads to a higher focus on lock-in. To be able to continually convert users it is important to think about what

features to put in the different versions of the offering (Kumar, 2014). Lankton & Stafford (2013) find that a digital approach lets companies improve sales and marketing by gathering data on potential customers. Furthermore, as seen in paragraph 2.4.1.1, conversion analytics might be a helpful tool in deciding what features to put in a premium version, and what customers to target for conversion. Thus, we propose freemium firms will use conversion analytics as a tool in converting users from a free to a premium version, and thus lock them into the offering.

Furthermore, Kumar (2014) explains that in order to lock customers into a premium offering it is important to have simple and transparent offerings; "If customers don't clearly grasp what they would gain by upgrading, you will monetize fewer of them than you otherwise might" (cf. paragraph 2.4.1.2). Thus, to be able to keep and/or convert customers, we propose that firms utilizing a freemium model need to show value in a simple, quick and transparent manner.

4.6.1.6 APIs as a Driver of Complementarities

As covered in paragraph 4.2.3.1, APIs enable integration and data-sharing. APIs as such make it easy for developers to connect their applications and networks with those of others, and thus complement each other's offerings (Simon, 2014). The extensive use of APIs in today's digital economy entail that it is easier to connect to complementary offerings. This has generated success for large companies like Google, Facebook and Twitter (Simon, 2014), and also smaller companies that can piggyback on the success of others (Proffitt, 2013). APIs are therefore assumed to be a prime driver of complementarities.

4.6.1.7 Versioning as a Driver of Complementarities and Novelty

Amit & Zott (2001) explain that complementary assets create value whenever having a bundle of goods together provides more value than the total value of having each separate (Amit & Zott, 2001). The authors further suggest that complementarities are linked with novelty, as the main innovation of some e-businesses resides in their complementary elements. Versioning is a mechanism that might enhance the importance of complementarities as well as novelty. With the need to convert users from a free to a premium version, and effectively lock in customers, it becomes important to continue innovating and collaborating with other firms in order to add value. We expect this to lead to a higher focus on novelty, especially through the use of complementarities, in order to create indirect network effects and lock in customers.

As a freemium firm gives some versions of their offerings away for free, it is important to add novelty to the offering in order to give customers incentives to pay for the premium version. When firms do not add novel content to the premium offering, incentives for customers to upgrade or stay in the premium version decreases (Kumar, 2014). To retain customers, it is therefore important to continue enhancing the value of the offering over time. We thus assume the need for continued innovation will enhance novelty, and that complementarities will be important in doing so.

4.6.2 B2B Specific

Amit & Zott's (2001) research is conducted in the B2C market, while our research centers on the B2B market. Thus, there may be some limitations regarding its applicability in this context. However, Amit & Zott (2001) put forward that their research should hold true also in a B2B context, though with different weighting on the identified sources of value creation. Based on the literature covered in chapter 4.4 and the suggestions based on freemium literature in paragraph 2.4.3, we wish to extend the model to better fit with B2B contexts, by proposing some modifications to the original model. We specifically propose higher complexity in transactions for B2B companies (paragraph 4.6.2.1), higher switching costs in B2B markets (paragraph 4.6.2.2) and an elevated position of trust and brand in B2B markets (paragraph 4.6.2.3);

4.6.2.1 B2B Complexity in Transaction Efficiency

In paragraph 2.4.3, we suggested the use of a freemium model might be more challenging in B2B markets than B2C markets. As more people are involved in the buying decisions for B2B offerings, B2B transactions have higher complexity than B2C transactions (Kotler & Pfoertsch, 2006). We therefore propose that B2B complexity challenges the process of enhancing transaction efficiency. Furthermore, in relation to paragraph 2.4.2.1, offerings in the B2B market are suggested to meet more stringent requirements where firms and/or firm gatekeepers may prohibit the use of non-approved free services in order to secure proprietary data. This may add to the challenge of achieving transaction efficiency.

4.6.2.2 Switching Costs in B2B Markets

In B2B markets it is the organizations, as opposed to a single consumer, which undertakes the consumption (Fill & Fill, 2005). This may complicate the process of effectively locking in consumers, as the whole organization must be locked in. However, if a company first gets locked in, it will likely lead to higher switching costs than in a B2C context as the company will face collective switching costs both internally, and perhaps also with external businesses in their supply chain.

4.6.2.3 Trust and Brand in B2B Markets

When a consumer has invested time and money in learning about a brand, they increase their learning costs, which create higher barriers for switching to a new and incompatible system (Shapiro & Varian, 1999). As there is a need for long-lasting relationships in B2B markets with time investments for both consumer and firm (Kotler & Pfoertsch, 2007), trust and brand might enjoy elevated positions in B2B markets. We further propose that strong and trusted brands have the ability to positively affect lock-in, as the customer has invested time in getting to know the brand.

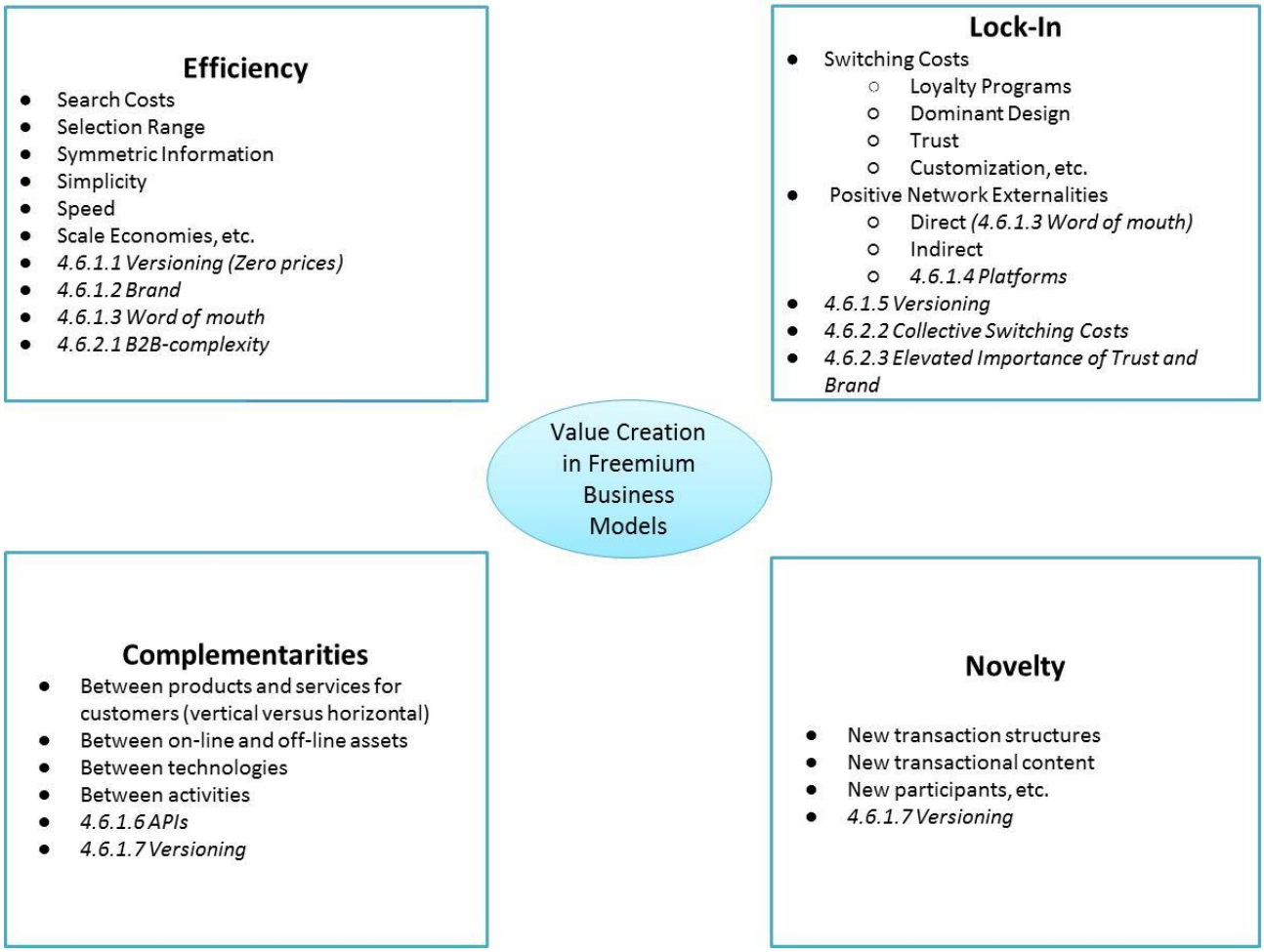

Figure 4. Proposed Literature Modifications.

Chapter 5 - Case Companies

This chapter examines the four case companies interviewed in this research, focusing upon the company's history and freemium offerings. Specifically, we cover Podio (paragraph 5.1), Plytix (paragraph 5.2), Appear.in (paragraph 5.3) and Microsoft (paragraph 5.4). A complete overview of the case companies is provided in paragraph 5.5, table 6.

5.1 Podio

Podio ™ was founded in 2009 (Podio, n.d.), and is a social collaborative work platform. The product works as a tool for small and medium enterprises (SMEs) to organize and track projects and everyday work (Citrix Annual Report, 2013). Podio allows integration and customization with a number of services. The Podio workspace for example enables teams to structure their work with Podio Apps that they build themselves (Citrix Annual Report, 2013). Important elements in Podio's business model is to offer secure, simple and cost efficient solutions for remote and on premise access (Citrix Annual Report, 2013). The firm states in its annual report that factors such as technical expertise, knowledge and innovative skills, relationships and name recognition are significant elements in maintaining a competitive position in the rapidly changing technological industry (Citrix Annual Report, 2013).

5.1.1 Freemium Offerings

Podio is offered free of charge for up to five collaborating users. If additional employees within an organization want to join, different premium versions are available, ranging from $9 to $24 per month (Choose Your Plan, n.d.). These versions are again differentiated based on the number and advancement of the features included. Thus, Podio offers a mixture of a seat-limited and a feature-limited freemium model (cf. paragraph 2.3.3.2 and 2.3.3.3). In 2012 the company was acquired by Citrix Online (Podio, n.d.). Citrix is a large incumbent in the software industry, and the acquisition has significantly raised Podio's adoption rate. Podio is today used by thousands of organizations in more than 170 countries (Goodman, 2012).

5.2 Plytix

Plytix is a Danish start-up company established in 2014 (Akbar, 2015). In 2015 the company received 3.7 million DKK from a variety of investors (Akbar, 2015), and the CEO Morten Poulsen has already branched the company to Spain (About Plytix, 2016). The core of Plytix's service is a cloud-based platform which enables brands to monitor their performance from various third-party web shops (Akbar, 2015). The firm targets the growing issue of sharing product images, which enables their consumers to customize user experiences (About Plytix, 2016). The start-up works with big data (Akbar 2015) and is according to their own webpage, "The #1 online product image bank and the #1 product based analytics platform in the world" (About Plytix, 2016). Being the only cross-site product analytics tool and centralized image sharing platform, the company argues their offering has the ability to "save money, increase productivity and boost sales" (About Plytix, 2016).

5.2.1 Freemium Offerings

At present time the company offers free access to their service for both brands and retailers (Akbar, 2015). They are however looking to add a premium version, which will include analytical and support functions (Akbar, 2015). The basic plan includes offerings like unlimited storage, sharing, and unlimited use of Plytix Product Information Management (PIM) and Digital Asset Management (DAM) systems (About Plytix, 2016). The Premium plan will include all basic features as well as features like dedicated account management, market benchmarking tools and access to Plytix's APIs (About Plytix, 2016). Plytix will thus offer a feature-limited freemium model (cf. paragraph 2.3.3.2).

5.3 Appear.in (Telenor)

Telenor is one of the world's leading suppliers of telecommunication services (About Telenor, n.d.), with a leading position in the Nordic countries. Telenor was founded in 1855 and has over 128 billion NOK in yearly revenues (About Telenor, n.d.). In August 2013, Telenor launched the video conferencing service Appear.in, which is the fastest growing global service in Telenor with users in 196 countries (Telenor Annual Report, 2014). The Appear.in team currently consists of 14 people, and operates as an independent start-up within Telenor Digital (Appear.in, n.d.).

5.3.1 Freemium Offerings

The first launched version of Appear.in was focused on B2C customers, and was offered for free (Boll, 2015). The product is still free today, but has also attracted many B2B customers. The offering is built on WebRTC-technology and consists of a free video chat without any required download or registration. Appear.in states that this is a huge competitive advantage, as it is a lot easier for customers when the product does not require a download (Willassen, 2016). However, in order to claim ownership of a "room" in Appear.in users must register. It is furthermore possible to use Appear.in across multiple platforms and devices, as it operates on any WebRTC enabled browser (Boll, 2015). In 2016, Appear.in will be released as a premium version with more advanced features, built for businesses and professional users (Appear.in Premium, n.d.). Appear.in thus sets out to offer a feature-limited freemium model (cf. paragraph 2.3.3.2).

5.4 Microsoft

Microsoft Corporation was founded in 1975, and is according to the company website "the worldwide leader in software, services, devices and solutions that help people and businesses realize their full potential" (Facts about Microsoft, 2015.). The company offers a range of products and services like Windows, Office, Surface, Bing, Skype and MSN (Facts about Microsoft, 2015). Our focus in this paper will be on their freemium offerings within enterprise software.

5.4.1 Freemium Offerings

Historically Microsoft has focused on their three biggest product lines - Office, Windows client and Windows Server (Capossela, 2015). The firm was for a long time successful in protecting their intellectual assets and offering SaaP in a traditional manner to drive revenues and profits. However, as the world has changed to more mobile and cloud service-oriented solutions, Microsoft has seen the need to change strategy (Capossela, 2015). In an appearance at Convergence 2015, Chief Marketing Officer (CMO) Capossela for example explained that Microsoft's new strategy for Office is to use a freemium business model in which the most common features are given away for free in order to drive adoption of the evolving platform (Capossela, 2015). The Office freemium business model versions based on screen size. This implies that Office is offered as a free service for smaller

devices, but once a user wants to consume the service on larger devices (over 10.2 inch) they will need to pay a premium subscription fee (Capossela, 2015). Office also offers more advanced features for a premium price. This type of freemium model is thus characterized as a mix of a device limited (cf. paragraph 2.3.3.3) and a feature limited freemium model (cf. paragraph 2.3.3.2).

Other products within the Office portfolio like OneNote and OneDrive is offered for free on all devices. The free version of OneNote is however limited to home or school use, while business users must pay a premium subscription fee - thus making it a customer type limited freemium model (cf. paragraph 2.3.3.4) (Keizer, 2014). OneDrive is similar to its competitor Dropbox offered through a capacity limited freemium model (cf. paragraph 2.3.3.3) (Keizer, 2015). Traditionally, Microsoft has also offered Skype for free for audio and video calls over the Internet, whereas calls to landlines and mobile phones can be made for a premium fee. Roterud (2016) adds that Microsoft is in the process of developing and implementing a 'Skype for Business' premium offering with richer features, thus making it a feature limited model. In 2012 Microsoft acquired Yammer, a social networking service for companies (Lardinois, 2012). Yammer´s freemium model consists of a feature limited offering, with a free option as well as the possibility for added features in a paying subscription, thus making it a feature limited freemium model (cf. paragraph 2.3.3.2) (McLaughlin, 2013).

5.5 Information about the Case Companies

Information	Podio	Plytix	Appear.in	Microsoft
Size	Medio	Small	Small	Big
Maturity	Growth	Startup	Startup within mature organization	Mature
Brand name	Medio	Weak	Weak	Strong
Equity	Medio	Weak	Strong	Strong
Industry	Project Management Software	Product Analytics	Telecom	Enterprise Software
Product examined	Podio	Plytix	Appear.in	Office 365, Yammer

Table 6. Information about the Case Companies.

5.5.1 Explanations to Table 6

Size refers to the number of employees in each case company. Microsoft has 112,689 employees worldwide (Facts about Microsoft, 2015), and is thus characterized as a big company. Podio, with its 69 employees (About Podio, n.d.), is characterized as a medium sized firm. Appear.in with its 14 employees (About us, 2015) and Plytix with its 11 (Plytix, 2016) employees, are in this research referred to as small firms.

Maturity refers to the year the case companies were founded. Microsoft was founded in 1975, and is thus characterized as a mature company. Podio was founded in 2009, and is thus viewed as a company facing a growth stage. Appear.in was founded in 2013 and Plytix in 2014, thus we characterize these firms as start-up companies.

Brand name does in this situation refer to how well known a brand is. As this is challenging to measure, we make some assumptions about this issue. Microsoft is by Forbes found to inhibit a very

strong and known brand (America´s Best Employers, 2015). Podio has been in the market for seven years, and the acquisition by Citrix raised the firm's adoption rate. Thus, we assume the firm´s brand to be medium known in relation to the other case companies in this research. Plytix and Appear.in are both start-ups, and we thus assume they both have weak brand names at current time.

Equity does in this table refer to total shareholder equity in 2015. Microsoft, with total shareholder equity of $80,083,000, is characterized as inhibiting a strong equity position. Citrix is revealed to have $1,973,446 in total shareholder equity (Citrix Systems, n.d.), and we thus characterize them as enjoying medium equity strength. Appear.in reaps the financial benefits of its mother company Telenor, and thus relies on $6,669,100 in shareholder equity (Telenor Shareholders, n.d.), attributing them a strong equity position. For the start-up company Plytix we assume they at current time face a weak equity position.

Chapter 6 – Analysis

This chapter covers findings from the case studies, and aims to relate these to the literature covered in chapter 4. The analysis will especially go into depth of the four sources of value creation to explain how they create value in the case companies, thus identifying the drivers of each source of value creation; efficiency (paragraph 6.1), lock-in (paragraph 6.2), complementarities (paragraph 6.3) and novelty (paragraph 6.4).

Each section will commence with an overview of Amit & Zott's (2001) findings followed by a discussion about our relative findings on how a freemium model creates value. Thereafter we will cover the modifications proposed in the literary review under paragraph 4.6, and conclude whether or not we find evidence for these in our research. Lastly we will discuss other empirical findings.

The analysis on each of the sources of value creation will be illustrated in a table, where the color white confirms that our findings are similar to those of Amit & Zott (2001). The color red illustrates that we do not find evidence for said driver. The color green is used where we can confirm the proposed modifications (cf. paragraph 4.6). New empirical findings are also illustrated in green.

White	Our findings are similar to those of Amit & Zott (2001)
Red	We do not find evidence for set driver in our findings
Green	Our findings confirm the proposed modifications (cf.paragraph 4.6) and/or the study reveals new empirical findings

Table 7. Explanations to Color Scheme in Analysis.

In paragraph 6.5 we summarize the identified links and interconnections between the sources of value. Lastly, this chapter will conclude with a summary of how the four sources of value creation are found to create value in a freemium business model in the B2B enterprise software market (paragraph 6.6).

6.1 Efficiency

Transaction efficiency is enhanced when the cost per transaction decreases (Amit & Zott, 2001). Transaction costs are broadly defined, and involve all direct and indirect costs related to a transaction. Amit & Zott (2001) find that reducing information asymmetry and search costs, as well as enhancing selection range, simplicity, speed and scale economies makes it possible for firms to increase transaction efficiency. In the following we will cover how the freemium model is found to enhance transaction efficiency in this research. Specifically, we find that information asymmetry (paragraph 6.1.1), search costs (paragraph 6.1.2), versioning and zero price (paragraph 6.1.4.1), brand (paragraph 6.1.4.2) and word of mouth (paragraph 6.1.4.3) enhance transaction efficiency. We further find that selection range, simplicity, speed and scale economies (paragraph 6.1.3) as well as the proposed B2B complexity (paragraph 6.1.4.4) does not affect transaction efficiency in a freemium model in B2B markets. For a complete overview of the findings regarding transaction efficiency, consult table 8.

6.1.1 Information Asymmetry

One way of enhancing transaction efficiency is by reducing information asymmetry (Amit & Zott, 2001). Information asymmetry occurs when one of the transacting parties has more or superior information relative to the other. With experience goods, like information goods, it can be challenging to achieve information symmetry as users must experience the good upon determining its value (Varian, 2001). Amit & Zott (2001, p.503) suggest that information suppliers can reduce information asymmetry through the supply of "up-to-date and comprehensive information", which the Internet facilitates in an easy and convenient manner. Our findings implicate that even though this reduces information asymmetry, firm's experience that it is often not sufficient in order to achieve transaction efficiency.

Our findings suggest that the freemium model in itself can be efficient in reducing information asymmetry as the customer gets to experience the good, and can determine its value after experimentation.

Drivers of Transaction Efficiency	Amit & Zott (2001)	Proposed Modifications (Paragraph 4.6)	Case Study Findings
Information Asymmetry	Reducing information asymmetry will enhance transaction efficiency through the supply of "up-to-date and comprehensive information", which the Internet facilitates in an easy and convenient manner.	4.6.1.1 & 4.6.1.3: Versioning and zero pricing, and word of mouth will likely reduce information asymmetry.	6.1.1: Reducing information asymmetry enhances transaction efficiency. The freemium model in itself reduces information asymmetry as customers get to experience the good upon determining its value. Zero pricing and word of mouth contributes to this.
Search Costs	Through reducing information asymmetry, firms can also reduce customers search costs as greater availability of information naturally leads to a more efficient search.	4.6.1.1 & 4.6.1.3: Versioning and zero pricing, and word of mouth will likely reduce search costs.	6.1.2: The relationship between information asymmetry and search costs is reciprocal. In a freemium model search costs can be reduced by employing complementarities. Search costs might also be reduced through word of mouth and versioning with zero prices.
Selection Range	E-businesses provide greater selection at lower costs which reduces information asymmetry.	n.a	6.1.3: Greater selection at a lower cost is not found to yield value for the case companies. This finding might be relevant in other industries, or when looking at e-businesses as a whole compared to offline businesses.
Simplicity	E-businesses simplifies transactions which reduces the likelihood of mistakes, and thus reduces information asymmetry.	n.a	6.1.3: For the case companies, simplicity is not primarily a result of e-business, but a result of zero prices which allow customers to experience the offering.
Speed	The Internet facilitates speedy transactions, which enhances transaction efficiency.	n.a	6.1.3: Speed is not found to create particular value for the case companies. This may be due to the commonality of speedy transactions in today's e-business market. The finding is likely more relevant when looking at e-businesses as a whole, compared to offline businesses.
Scale economies	E-businesses allow individual customers to benefit from scale economies through demand aggregation and bulk purchasing.	n.a	6.1.3: Scale economies in terms of demand aggregation and bulk purchasing are not found to be of particular value for the case companies. This is likely more relevant when comparing online and offline businesses, as online businesses are able to offer a larger range of products than offline businesses.
Proposed Modifications			
Versioning and Zero Prices	n.a	4.6.1.1: Versioning information goods may drive adoption as users pay the value they perceive (Shapiro & Varian 1999). Versioning so that one part of the product is free is likely to be even more efficient (Shampanier et.al, 2007). Zero pricing drives transaction costs close to zero, and additionally reduce information asymmetry and search costs as users get to experience the good.	6.1.4.1: Our findings reveal that versioning and zero prices serve to attract a large user base as it reduces both physical and psychological transaction costs. Zero prices is thus effective in enhancing transaction efficiency, and serve to reduce information asymmetry and search costs, and might even incentivize word of mouth.
Brand	n.a	4.6.1.2: Brands have the possibility of speeding up consumers information processing if customers are familiar with a brand and its connotations (Kotler & Gertner, 2004). Brand can thus enhance transaction efficiency.	6.1.4.2: Brands enhance transaction efficiency as users are familiar with a brand and its connotations, which serves to reduce information asymmetry and search costs.
Word of mouth	n.a	4.6.1.3: With word of mouth, potential users gain a trusted recommendation and indirect experience (Silverman, 2012). This likely reduces search costs and information asymmetry, and eventually results in increased transaction efficiency.	6.1.4.3: We find that word of mouth is effective in increasing transaction efficiency as it decrease the perceived risk and reduce information asymmetry and search costs. Freemium models are furthermore beneficial in leveraging the potential of word of mouth.
Complexity in B2B markets	n.a	4.6.2.1: B2B transactions are characterized by higher complexity as more people are involved in the buying decision (Kotler & Pfoertch, 2008). This may make it more challenging to achieve efficient transactions.	6.1.4.4: The case companies solve B2B complexity by targeting end-user employees. Utilizing a bottom-up freemium approach the companies are able to target individual employees as opposed to the whole business. Thus, it is the individual employee who is the decision maker, which naturally decreases complexity in the buying decision.

Table 8. Transaction Efficiency (Analysis).

Poulsen (2016) for example explains that employing a freemium model is important for Plytix's success in order to reduce information asymmetry as the market needs to be educated about their offering. Willassen (2016) further adds that the reason why Appear.in employ´s a freemium model is because their service differentiate from others on "how easy it is to use", something which is very difficult to explain verbally - "customers must experience it themselves (...) it is a necessary prerequisite for customer acquisition". The freemium model thus seems to be an efficient tool in reducing information asymmetry.

6.1.2 Search Costs

Search costs are costs incurred by potential customers in searching for an offering. Amit & Zott (2001) comment that by reducing information asymmetry, firms can also reduce customers search costs as greater availability of information naturally leads to a more efficient search.

The case study findings suggest there is a relationship between search costs and information asymmetry, but also that this relationship is reciprocal. Thus, by reducing search costs firms can reduce information asymmetry. An example of this, as explained in greater detail in paragraph 6.3.1.1, is the relationship between complementarities and search costs; when customers have access to complementarities and compatible offerings, their search costs might be reduced as they can be introduced to the offering via a complementarity. Furthermore, this might serve to reduce information asymmetry; as the customer has knowledge of a complementarity, s/he may also have greater information about the focal offering in terms of in which contexts it is useful, or which brands it is associated with. Our study also suggests that search costs can be reduced by zero prices and positive word of mouth as explained in paragraph 6.1.4.1 and 6.1.4.3 respectively.

6.1.3 Selection Range, Simplicity, Speed and Scale Economies

Amit & Zott (2001) suggest that e-businesses in comparison with traditional business transactions provide greater selection range at lower cost. This is done by reducing distribution costs, streamline inventory management and by allowing individual customers to benefit from scale economies through demand aggregation and bulk purchasing. Even though a greater selection at a lower cost might be important when looking at e-business as a whole, it is not found to yield value for the case

companies in this research. Furthermore, simplicity is for the case companies not a primary result of being an e-business, but a result of zero prices which allows customers to experience the offering. Speed is at the same time not found to create particular value for the firms, which might be due to the commonality of speedy transactions in today's e-business markets. Lastly, scale economies in terms of demand aggregation and bulk purchasing are not found to be of particular value for the case companies. This issue is probably more relevant when comparing online and offline businesses as Amit & Zott (2001) did, as online businesses are able to offer a larger range of products than offline businesses.

6.1.4 Proposed Modifications

6.1.4.1 Versioning and Zero Prices as a Driver of Transaction Efficiency

In paragraph 4.6.1.1 we proposed that versioning and zero prices may drive adoption of a good by reducing information asymmetry and search costs, as the customers gets to experience the good upon determining its value. This seems to hold true for the case companies as well;

Hulthin (2016) explains that the reason why Podio chose to use a freemium model was to quickly scale a large customer base. Poulsen (2016) further explains that Plytix is offered for free to achieve rapid growth; “that is why we offer our service for free, because we don’t want any roadblock” and the zero price enables users to “play around with it [the service] and feel comfortable with it”. Roterud (2016) also explains that Microsoft attracts a significantly larger amount of users after implementing freemium offerings; “We’ve seen that the user’s costs or associated risk of using it [freemium offerings] is a lot lower”.

Our findings further imply that when targeting B2B customers, zero prices might be especially beneficial. Roterud (2016) explains that the way firms operating in a B2B market target consumers have changed in the last years; traditionally companies like Microsoft targeted the IT-department, whereas today it is just as much an end user with a need for an IT solution that starts using one of their products. An employee rarely has a budget to spend on IT-solutions, thus zero prices are very effective in enhancing transaction efficiency for business adoption (Roterud, 2016). In this way, the use of a freemium model reduces physical costs related to the transaction. At the same time, it

seems to contribute to the reduction of psychological costs, and might even as suggested by Shampanier et al. (2007) add a psychological gain. Our findings imply that removal of physical costs reduce information asymmetry as it is easy to experience the goods value. It can furthermore serve to reduce search costs as customer's both immediately gain greater information about the good, and do not have to incur costs in leaving their credit card information when searching for the good's value. The potential psychological gain might lead customers to spread positive word of mouth, which again beneficially affects information asymmetry and search costs, as will be further explained in paragraph 6.1.4.3.

6.1.4.2 Brand as a Driver of Transaction Efficiency

In paragraph 4.6.1.2 we proposed that brand and trust might enhance efficiency, as it has the possibility of speeding up consumer´s information processing (Gertner & Kotler, 2004). Our findings suggest that this holds true for the case companies. Poulsen (2016) e.g. states that it was easier to engage with customers when he was working in Google as opposed to Plytix, because people knew and trusted the Google brand. He explains that as long as he showed his business card, "everyone" was interested in having a meeting. Hulthin (2016) further explains that customers felt more at ease with Podio after they were acquired by the strongly branded company Citrix. Freemium start-ups commonly have little capital to spend on branding; however our research reveals that they use other mechanisms to build a trusted brand, like word of mouth (cf. paragraph 6.1.4.3) and complementarities (cf. paragraph 6.3.4.1).

6.1.4.3 Word of Mouth as a Driver of Efficiency

As proposed in paragraph 4.6.1.3 we expected word of mouth to enhance transaction efficiency, and reduce information asymmetry as well as search costs. For the case companies, it seems like word of mouth is central in driving user adoption, thus enabling efficient transactions.

Roterud (2016) explains that it was important for Microsoft to utilize a freemium model for offerings containing a social component, like Yammer and Skype, to reap the benefits of word of mouth. "We look at the services where it is natural to drive social adoption so that we can drive more people inside an organization to use it and gain an exponential growth as a result of a viral spread". This seems to hold true for several companies in this research. Hulthin (2016) e.g. explains that customers who are satisfied with Podio often attract other users, as this yields higher value for

the individual consumer. Poulsen (2016) furthermore explains that when resellers using Plytix see the value of the service, they want the retailers to use it as well, as this increases the value of the offering. Lastly, Willassen (2016) adds that almost all of Appear.in's growth has been realized through word of mouth.

Word of mouth is thus in the case companies found to positively impact transaction efficiency and reduce both information asymmetry and search costs as potential users gain indirect experience with the product through a trusted referral. The freemium model might also be understood as an enabler of effective word of mouth, as the referred customers have low barriers for trying the offering. This is in contrast to for example Murphy (2010) who argues that it is difficult to achieve a viral spread for B2B offerings.

6.1.4.4 B2B complexity

In paragraph 4.6.2.1 we suggested that as more people are involved in the buying decision for B2B offerings, B2B transactions have higher complexity than B2C transactions (Kotler & Pfoertsch), which will complicate the process of enhancing transaction efficiency in a B2B market. Though not disagreeing with this proposition, our findings suggest that freemium companies solve this complexity by targeting end-user employees via a bottom-up approach to customer acquisition. This is furthermore exemplified by the fact that Plytix, the only company that does not utilize a bottom-up approach, is the only firm that expresses issues with customer acquisition. Thus, by utilizing a bottom-up approach, freemium companies are able to target individual employees as opposed to the whole business. Transaction with a single employee as a decision maker naturally decreases the complexity of the transaction. Furthermore, we did not find that any of the case companies experienced issues with more stringent requirements.

6.2 Lock-in

Lock-in refers to the extent to which customers are engaged in repeat transactions, and can be enhanced through creating switching costs for consumers (Amit & Zott, 2001). Lock-in creates value as it tends to increase transaction volume, and the extent to which strategic partners and complementors have incentives to maintain and improve their associations (Amit & Zott, 2001). Our research finds that lock-in creates value through network effects (paragraph 6.2.1),

opportunities for customization (paragraph 6.2.2), brand and trust (paragraph 6.2.3), word of mouth (paragraph 6.2.5.1), platforms (paragraph 6.2.5.2), versioning (paragraph 6.2.5.3) and collective switching costs (paragraph 6.2.5.4).Our research does not find that loyalty programs and dominant design (paragraph 6.2.4) are relevant for the case companies. For a complete overview of the findings regarding lock-in consult table 9.

6.2.1 Network Effects

Network effects exist when the benefit of a product or service increases with the number of users consuming the same good or service (Haruvy & Prasad, 1998; Katz & Shapiro, 1985). Positive network effects raise the value of the offering as markets get larger (Margolis, n.d.). Network effects are underlined as important by Amit & Zott (2001) for companies operating in virtual markets. This research reveals that network effects also play a major role in freemium business models, to create lock-in.

6.2.1.1 Direct Network Effects

Direct network effects occur when an increase in the number of users leads to an increase in the value for each user (Katz & Shapiro, 1994). Direct network effects are found to both increase the value of the offering itself, and contribute to create higher switching costs as users face collective switching costs (Amit & Zott, 2001).

This seems to hold true also in this research, Roterud (2016) does for example explain that user adoption in Microsoft has changed in that instead of a top-down approach where the firm targets the IT-division, user adoption follows a bottom-up approach where individual employees starts using the product/service. These individual employees attract other employees through word of mouth, in order to advance the value of the offering. This creates network effects, and eventually lock-in, as the user's gain higher collective switching costs. Several freemium model firms have created viable business models based on a bottom-up approach. In fact all case companies except Plytix relies on this approach. Microsoft owned Yammer has e.g. created a successful business where user adoption starts with employees using Yammer as their internal social network, creating network effects as they invite coworkers to join. The employing organization has no control over the network, but is offered the opportunity to acquire administrative control for a premium price (Yoskovitz, 2009).

Drivers of Lock-In	Amit & Zott (2001)	Proposed Modifications, chapter 4.6	Case Study Findings
Direct Network Effects	Direct network externalities drives lock-in as the value created for customers increase with the size of the customer base.	n.a	6.2.1.1: Freemium models naturally create network effects, especially when utilizing a bottom-up approach. This creates high value in terms of lock-in as the value of the offering increases with the number of users. This again leads to collective switching costs, which makes it difficult for customers to churn.
Indirect Network Effects	Indirect network effects drive lock-in as increased usage of a product or service spawn increases in the availability and value of complementary goods, which eventually leads to an increase in the value of the original offering.	n.a	6.2.1.2: Indirect network effects enhance the value of the offerings. As users engage in a network of offerings, their learning costs and thus switching costs multiplies as it is harder for any firm to imitate such a network. Employing a freemium business model seems to be an efficient tool in enabling and leveraging this value as it has the potential of attracting a large user base in which complementors will want to add offerings.
Customization	Customization enhances lock-in by enabling customers to customize the offering to their individual needs. Customers then have high incentives to use the system, which creates a positive feedback loop.	n.a	6.2.2: Customization enhances lock-in in a freemium model. The case companies most commonly rely on complementarities to customize offerings. They furthermore offer opportunity for customization in their premium offerings, which may serve as a way of attracting users to convert.
Brand and Trust	Through creating trust and familiarity firms may gain more loyal customers, as customers increase their learning costs and thus also their switching costs.	4.6.2.3: Trust and a strong brand name is more important in a B2B setting than in a B2C setting.	6.2.3: Brand and trust are important enablers of lock-in, also in a freemium model. Our research suggest that freemium firms might experience that the top management are skeptical of freemium models, which affects the degree to which the business will take part in repeat transaction or convert to a premium price plan.
Loyalty Programs	Loyalty programs rewarding repeat customers with special bonuses may enhance lock-in.	n.a	6.2.4: None of the case companies employ loyalty programs. Loyalty programs is thus not relevant in set study. Our findings suggest that loyalty programs are more relevant in B2C-markets.
Dominant Design	Developing dominant designs proprietary standards will increase switching costs, thus enhancing lock-in.	n.a	6.2.4: None of the case companies rely on proprietary standards for their freemium offerings. It seems relevant to assume that this is due to the fact that their offerings are difficult to formally protect from imitation, which may be the reason why they are giving it away for free.
Proposed Modifications			
Word of mouth as a driver of lock-in	n.a	4.6.1.3: Word of mouth enhances transaction efficiency, and thus attracts more potential users. As more users join the network and network effects play in, lock-in likely also increases in value.	6.2.5.1: Our findings imply that word of mouth is central to network effects in a freemium model, as employees invite their co-workers to join. This leads to higher collective switching costs, which in turn creates lock-in.
Platforms	n.a	4.6.1.4: Platforms, where a firm attracts two distinct user groups, yield increasing returns as network effects raise the user's value of the platform (Gawer & Cusumano, 2008). Platforms will therefore likely create high switching cost.	6.2.5.2: The research indicates that platforms create two-sided network effects, which serve to create high switching costs, and thus positively affects lock-in.
Versioning	n.a	4.6.1.5: Versioning will impact lock-in as there is a need to continually convert users to a premium version. Freemium firms will likely present simple and transparent offerings, and use conversion analytics in converting users.	6.2.5.3: The case companies emphasize a need to convert users to a premium version. In achieving conversion, they employ different analytic tools, which enables them to target users with more personalized and customized offerings.The companies also highlight that in order to retain customers their offering must demonstrate immediate value. Simple and transparent offerings thus enhance lock-in.
Switching costs in B2B markets	n.a	4.6.2.2: In B2B markets it is the organizations, as opposed to a single consumer, that undertake consumption (Fill & Fill, 2005), which may make it more challenging to achieve lock-in. However, if a company gets locked in, it will likely lead to higher switching costs as the company faces collective switching costs.	6.2.5.4: Due to the complexity of B2B transactions, we find that this process can be challenging when gatekeepers lack trust in freemium firms. However, the freemium model is efficient in creating and leveraging network effects, which is found to result in high collective switching costs. Thus, the company might be locked in even before the decision reaches gatekeepers.
Trust and Brand in B2B markets	n.a	4.6.2.3: As there is a need for long-lasting relationships in B2B markets, (Kotler & Pfoertsch, 2007), trust and brand might enjoy elevated positions in B2B markets. This may positively affect lock-in, as the customer has invested time in getting to know the brand.	6.2.5.5: Although some case companies experienced issues with trust, none expressed that brand and trust enjoy an elevated position in B2B markets. Our findings can thus not conclude whether this is more important in B2B markets than in B2C markets.

Table 9. Lock-In (Analysis).

The freemium model seems to be especially beneficial in creating network effects in B2B markets, particularly when leveraging a bottom-up approach that attracts a large user base and stimulates word of mouth (cf. paragraph 6.1.4.1 and 6.1.4.3). This is confirmed by the fact that all the case companies underlined that the rationale for employing a freemium model was to be able to quickly scale a customer base and as such extract network effects. Network effects thus create lock-in, both through advancing the value that each user gains from the offering, and through collective switching costs that increases the barriers of churning.

6.2.1.2 Indirect Network Effects

Katz & Shapiro (1994) find that indirect network effects occur as increased usage of a product or service spawn increases in the availability and value of complementary goods, which eventually leads to an increase in the value of the original offering. Amit & Zott (2001) further find that indirect network effects drives lock-in, as consumers gain higher value and thus higher switching costs.

We find, in conjunction with Amit & Zott (2001), that indirect network effects drives lock-in for the case companies, especially through complementarities. As explained in depth in paragraph 6.3, complementarities enable freemium firms to enhance their offering. This effectively raises the value of the offering, which in turn yields higher switching costs. The increased use of APIs further implies an increased exchange of data and connectivity between different apps and services. This enables firms to make their goods excludable (Varian, 2001) which in turn protects the companies from imitation and eventually leads to higher switching costs and lock-in.

Hulthin (2016) exemplifies by explaining that the real lock-in in today's business environment is when a product or service ties to other offerings. However, in order to attract complementarities, the product or service must have a significant amount of users. Hulthin (2016) explains "no one will build [complementary offerings] on a tool that has 50 users". As explained in paragraph 6.1.3, a freemium business model has the potential of quickly scaling a large user base. Thus, a freemium business model can be useful in leveraging indirect network effects. Roterud (2016) further explains that a big differentiator for Microsoft is the elaborate ecosystem of partners who adds complementarities and customizations on top of their offerings. Microsoft additionally exploits these indirect network effects on internal complementarities. They spend a lot of resources on

making their products work seamlessly together, so that when a customer uses one of their products, this will then naturally lead to consumption of other offerings (Capossela, 2015). As users engage in a network of offerings, their learning costs and thus switching costs multiplies, and it gets harder for any firm to imitate such a network. For B2B customers this effect is of great significance as the business customers must arrange a collective switch (Shapiro & Varian, 1999).

6.2.2 Customization

Customization enhances lock-in by letting consumers customize offerings to their individual needs (Amit & Zott, 2001). Consumers then have higher incentives to use the system, which creates a positive feedback loop. This seems to hold true also in set research, however the case companies rely more on complementarities and connectivity than protectable customized offerings.

In a freemium business model, the supplying firm most commonly relies upon standardized offerings as a way of reducing marginal costs (Baden-Fuller & Haefliger, 2013). Thus, there is little or no room for customized offerings. The general lack of capital for freemium start-ups might thus be an inhibitor in this sense. Firms like Microsoft with greater capital than most freemium firms seem to be more capable of investing in customized offerings. The start-up firms in this research seem to a certain degree to overcome the problem of limited capital by offering integration with complementary services. This way, customers can choose which apps they want integration with. These offerings are however not proprietary as it is most commonly done through open APIs. In that sense there are no barriers to prevent competitors from offering the same through the use of complementarities. However, it seems the interface and connectivity between the offering and the range of complementarities can partially protect the focal firm from imitators. Additionally, the companies offer more customized offerings in their premium versions, like personal service and support. This may thus be a way of trying to force the users to convert to a paying subscription model. Here, the use of conversion analytics (cf. paragraph 6.2.5.3) might be a helpful tool, as it enables the firms to identify the needs of the individual companies. Thus, customization is found to enhance lock-in for freemium offerings by enabling users to customize the offering to their individual needs, which is in conjunction with the findings from Amit & Zott (2001).

6.2.3 Brand and Trust

Amit & Zott (2001) suggest that through creating a trusted and familiar brand, firms may gain more loyal customers. When a consumer has invested time and money in learning about a brand, they increase their learning costs, which in turn create higher barriers for switching to a new and incompatible system (Shapiro & Varian, 1999).

All of the case companies enhance that it is important to establish a trusted and known brand. Even though the case companies focus on the attraction of the end user instead of whole departments, it is still the top management or IT-department who decides whether the company should upgrade to a premium version or not. Our findings suggest that these gatekeepers might be more skeptical of locking the firm into a freemium model. Roterud (2016) does for example explain that for bigger organizations it can be an issue for the IT-department that individual employees start using different software independent of the leadership and organization. Therefore, using a freemium business model might lower the chances of lock-in if the central gatekeepers are not interested in this way of transacting.

However, Roterud (2016) also states that the role of the IT-leader has changed a lot during the last couple of years, and that the focus of "modern" IT-leaders is on value creation for the end user. As he explains; "for some it's probably a pain in the ass, and for others it is very positive as it opens up opportunities to roll out several technologies in the organization that they want to get out there". Thus, even though this way of enrolling technologies yields challenges for many IT-leaders, it has the opportunity of making the interaction between the firm and the end user a lot more efficient.

6.2.4 Loyalty Programs and Dominant Design

Amit & Zott (2001; Varian, 1999) find that loyalty programs rewarding repeat customers with bonuses, can create lock-in. Loyalty programs did not come up in this research as specifically effective in creating lock-in. On the contrary, Hulthin (2016) actually states that Podio unsuccessfully employed a loyalty program. He explains that these sorts of incentives work better in B2C markets than in B2B markets.

Amit & Zott (2001) further explain that a way to achieve customer retention is to develop dominant design proprietary standards. As familiarity with interface design creates customer switching cost, users will likely be loyal to the dominant design. This is however not found to be relevant in this study. This may be due to the fact that set study scrutinizes freemium models; if a company is able to create a dominant design and lock customers in, it would probably not use a freemium model, as a traditional, closed business model would reap higher returns. None of the case companies are found to rely on proprietary standards for their freemium offerings. It seems relevant to assume that this is due to the fact that their offerings are difficult to formally protect from imitation, which may furthermore be the reason why they are giving it away for free.

6.2.5 Proposed Modifications

6.2.5.1 Word of Mouth as a Driver of Lock-in

With word of mouth, potential users gain a trusted recommendation and indirect experience with an offering (Silverman, 2011). Thus, we proposed in paragraph 4.6.1.3 that word of mouth is likely to attract more users to the network, which enables lock-in as customers receive higher value with more participants joining. With word of mouth, employees create incentives for co-workers and other partners to take part in the offering. This leads to higher collective switching costs (cf. paragraph 6.2.5.4), and thus word of mouth seems to drive lock-in.

6.2.5.2 Platforms

As proposed in paragraph 4.6.1.4 platforms yield increasing returns as network effects raise the user's valuation of an offering (Gawer & Cusumano, 2008). Amit & Zott (2001) do not include a discussion about network effects in two-sided platforms, most likely because two-sided platforms have risen to economic and business prominence after their paper was published (Hagiu, 2006). Two of the case companies in this research do however mention platforms as important for lock-in.

Both Plytix and Microsoft aim to make their services a platform. For Plytix, the more retailers use the service, the more brands will naturally be interested in joining, and vice versa. Poulsen (2016) explains that if a brand wants to stop using Plytix, it will face collective switching costs as all their resellers will have to find a new way of sharing pictures with them, something which most likely

will incur huge expenses. Microsoft is engaged in creating a ‘one stop shop’ where customers can access everything they need in one platform (IT Show Case, 2016), offering software such as Office 365, SharePoint, Delve and Lync Online (Anderson, n.d.). The company is also making their products integrated with competing hardware devices such as iPad and iPhone (Clover, 2015). When the firm opens up for third party developers, customers get incentives to start using their offerings because of the added value. At the same time, the platform becomes more attractive for third-party developers as more users join. Thus, it seems platforms are effective in creating lock-in.

6.2.5.3 Versioning

As proposed in paragraph 4.6.1.5 versioning might drive a focus on lock-in as firms need to convert users from a free to a premium version in order to reap monetary returns. Based on existing literature on freemium firms we suggested that firms make use of conversion analytics as a tool to convert users.

A need to convert users from a free to a premium version is mentioned by some case companies as a challenge, and conversion analytics seems to be an important driver in solving this challenge and achieve lock-in. As freemium models generally attract many users who will not convert to a paid subscription, Hulthin (2016) explains that companies need to know which customers to focus their resources on. In fact, all the interviewed case companies employ some sort of analytics tool to identify users, as well as which customers that is likely to convert. Poulsen (2016) e.g. explains that Plytix track “everything” on their service and have different parameters for people who are likely to convert from a freemium to a premium model. Hulthin (2016) adds to the discussion by explaining that Podio use sophisticated data modelling and lead scoring to determine which customers are likely to become premium users. When someone falls into one of these parameters the firm spends more resources to try and convert them. As the companies know who to target, they have the opportunity to create switching costs by creating customized offering.

We further proposed in paragraph 4.6.1.5 that firms utilizing a freemium model need to show value in a simple, quick and transparent manner in order to lock customers effectively into the offering. Hulthin (2016) underlines this issue by explaining that showing the value of an offering is paramount to be able to attract and retain users when using a freemium model. Customers are

expecting to have a “friction free” interaction with the firm, and immediately understand the value of the offering (Hulthin, 2016).

6.2.5.4 Switching Costs in B2B Markets

In B2B markets it is the organizations, as opposed to a single consumer, that undertakes consumption (Fill & Fill, 2005). Therefore, as proposed in paragraph 4.6.2.2, this may complicate the process of effective lock-in. However, if a company first gets locked in, we proposed that this will likely lead to higher switching costs than in a B2C context, as the company will face collective switching costs both internally, and perhaps also with external businesses and complementarities. In paragraph 6.2.1 we found that in B2B markets network effects are very effective in creating lock-in as consumers face collective switching costs when more employees join, and the amount of complementarities increases. In relation to this, Poulsen (2016) explains that it will be difficult for a brand using Plytix to change to another offering, as they will have to switch not only employees in their own organization, but also every reseller; “that would be very expensive for any organization”. Freemium firms utilizing a bottom-up approach are furthermore very efficient in leveraging this effect. This implies that the company might be locked in even before the decision reaches central gatekeepers. Thus, if a company operating in a B2B market has chosen to buy an offering, it seems the firm faces a higher lock-in effect than a single consumer, due to collective switching costs.

6.2.5.5 Trust and Brand in B2B markets

In paragraph 4.6.2.3 we proposed that trust and brand enjoy an elevated position in B2B markets as opposed to B2C markets. We further proposed that strong and trusted brands have the ability to positively affect lock-in, as the customer has invested time in getting to know the brand, which increase their learning costs and creates higher barriers to switching to a new and incompatible system. As discussed in paragraph 6.2.3, all case companies in this research express that it is important to establish a trusted brand in order to effectively locking in their customers. However, neither of the case companies expressed that brand has an elevated position in the B2B market as opposed to the B2C market. Hulthin (2016) explains that some issues relating to trust, like confidential data handling, is probably more important in a B2B setting. He however emphasizes that brand and trust is important no matter what market the focal firm operates in. Thus, this assumption does not seem to hold true.

6.3 Complementarities

Amit & Zott (2001) find that complementary offerings create value whenever having a bundle of goods together yields more value than selling the goods separately. Complementary offerings are found to create value for all case companies in this research, and the findings suggest that complementarities are even more important for success when using a freemium model than more traditional business models. In our research, complementarities are found to create value through enhancing the focal firm's offering (paragraph 6.3.1), reducing search costs (paragraph 6.3.1.1), APIs (paragraph 6.3.3.1), versioning (paragraph 6.3.3.2) and by enhancing brand and trust (paragraph 6.3.4.1). Offline assets complementing online assets are not found to be relevant in this specific research (paragraph 6.3.2). For a complete overview of the findings regarding complementarities, consult table 10.

6.3.1 Enhanced Offering

Amit & Zott (2001) suggest that complementarities can enhance an offering. They differ between vertical complementarities like after-sale services, horizontal complementarities like hardware and software, complementary technologies and complementary activities. In our research we see no reason to differ between these categories. We find that complementary technologies and complementary activities can be both horizontal and vertical complementarities. What these have in common are that they enhance the value of the core offering, and we therefore find it more appropriate to present it as one driver. We thus refer to this new category as "enhanced offering".

Complementarities are in this research revealed to enhance the value of the offerings of all case companies. Podio e.g. makes use of a range of horizontal complementarities like Dropbox, Box, Sharefile, GoToMeeting and ZenDesk (Out of the box, n.d.). Hulthin (2016) explains that this enables the company to add value and continue to charge premium prices, or even add new pricing plans; "Say we had a 25 people company using Podio and everyone was on board - then we would have nothing more to sell them (....) By adding complementarities, you add a new product that costs more and can build on it". In a similar vein, both Appear.in and Plytix aim to attract developers to build complementary apps and services for their offerings in order to add value and differentiate between free and premium price plans. Plytix is additionally dependent on vertical complementary

technologies, more specifically PIM and DAM systems that can store images; “The PIM and DAM system providers are much better than we are in storing (...) so we actually want them to keep doing that, and then we just make them synchronize with Plytix” (Poulsen, 2016). Microsoft, unlike the other case companies, mostly rely on internal complementary offerings and bundle their basic tier freemium services with more advanced services that are offered for a premium subscription fee. Thus, both vertical and horizontal complementarities contribute to create value for the case companies.

Drivers of Complementarities	Amit & Zott (2001)	Proposed Modifications (Paragraph 4.6)	Case Study Findings
Vertical and horizontal complementarities	Vertical and horizontal complementarities enhance the value of the core offering, as several offerings are bundled together.	n.a	6.3.1: In our research we find no reason to differ between vertical and horizontal complementarities, complementary technologies and complementary activities. We find that complementary technologies and complementary activities can be both horizontal and vertical complementarities. What these have in common is that they enhance the value of the core offering. We therefore refer to this new category as 'Enhanced Offering'. Our findings reveal that freemium firms make use of complementarities to enhance the value of their core offering, as it creates more value to offer it as a bundle rather than separately. This enables the firms to charge higher prices for the added value. Complementarities may furthermore reduce search costs and thus enhance transaction efficiency.
Complementary activities	E-businesses may create value by capitalizing on complementarities among activities such as supply-chain integration.	n.a	
Complementary technologies	E-businesses may create value by capitalizing on complementarities among technologies such as linking the imaging technology of one business with the Internet communication technology of another, thereby unleashing hidden value.	n.a	
On-line and off-line complementarities	Offline assets complement online offerings; Customers who buy products over the Internet value the possibility of getting after-sales service offered through bricks- and-mortar retail outlets, including the covenience of returning or exchanging merchandise.	n.a	6.3.2: None of the case companies leverage offline complementary offerings on their freemium products. This may be due to a need of keeping costs low, but we also assume that this was more relevant in 2001 when users were not that accustomed to online offerings.
Proposed Modifications			
APIs as a driver of complementarities	n.a	4.6.1.6: APIs make it easy for developers to connect their applications and networks with those of others, and thus complement each other's offerings (Simon, 2014). The extensive use of APIs in today’s digital economy entail that it is easier to connect to complementary offerings.	6.3.3.1: Our findings conclude that APIs drive the use of complementarities as APIs enable firms to make use of complementary assets and leverage the connectivity between different offerings in a more efficient manner.
Versioning as a driver of complementarities	n.a	4.6.1.7: Versioning is a mechanism that might enhance the importance of complementarities. With the need to convert users from a free to a premium version, and effectively lock customers in, it becomes important to continue innovating and collaborating with platforms in order to add value.	6.3.3.2: In our research it seems that freemium firms drive adoption on their premium offerings by adding more sophisticated features, often enabled through the use of complementary offerings. Having different versions of a product thus seems to enhance the use of complementarities.
Empirical Findings			
Enhanced Brand and Trust	n.a	n.a	6.3.4.1: Our findings suggest that complementarities may enhance brand and trust. Customers may know the complementarities of an offering, and have trust in these complementarities, which serves to enhance the credibility of the focal offering.

Table 10. Complementarities (Analysis).

As complementarities are used by the case companies as a way of adding value and creating novelty in their offerings, the value creation of complementarities and novelty are highly connected. Complementarities may furthermore enhance lock-in as customers have the opportunity to engage in a larger system of offerings.

6.3.1.1 Reduced Search Costs

Amit & Zott (2001) further comment that vertical and horizontal complementarities can enhance efficiency through reduced search costs. This seems to hold true also in this research. Microsoft e.g. significantly reduces search costs for it's more advanced and costly offerings by bundling and introducing them via freemium products. Lately, Microsoft has also realized the importance of compatibility and has for example created Office Apps for both Apple and Android smartphones (Skillicorn, 2014b), which enables their software to run efficiently on competing platforms. Roterud (2016) explains that integration has been a major focus in Microsoft, especially in the last two years. "It's a recognition about the fact that we are a lot more logical in relation to where we go, so when Apple is the most established, homogenic platform for smartphones, we start there". These integrations serve to reduce search costs for customers that are not using Microsoft hardware. In a similar vein, Appear.in highly values their integration with Slack due to its ability to function as a marketing channel where Appear.in can acquire customers. On the other hand, Appear.in might experience some difficulties with the compatibility of their service, as they are currently not compatible with Internet Explorer, which is the preferred web-browser for many enterprise customers (Willassen, 2016).

6.3.2 Offline Assets Complement Online Assets

Amit & Zott (2001) find that customers who buy products over the Internet value the possibility of getting certain services, like after-sales services, offered through offline assets. In this research, none of the case companies leverage offline complementary offerings for their freemium products/services. This may be a result of a need to keep costs low. We also assume that complementary offline assets were more relevant when e-business was first introduced, as users were not accustomed to all online offerings.

6.3.3 Proposed Modifications

6.3.3.1 APIs as a Driver of Complementarities

As suggested in paragraph 4.6.1.6, APIs might drive the use of complementarities. We find that complementarities are indeed most commonly enabled through the use of APIs in the case companies, and that the use of APIs has led to an increased connectivity between different offerings, thus enhancing the use of complementarities. APIs enable firms to make use of complementarities in a more efficient manner. Hulthin (2016) explains that it was a deliberate, strategic decision to make all of Podio's products compatible with open APIs so everyone could connect to the service. He continues by stating that "everything is API-based nowadays" and that a lot of companies make buying decisions based on what complementarities are offered. Poulsen (2016) further explains that Plytix use open APIs to attract complementary products and services. This extended use and ease of offering complementarities creates value for companies utilizing a freemium business model, as they usually have little capital to invest in building these complementary offerings themselves.

6.3.3.2 Versioning as a Driver of Complementarities

Versioning was in paragraph 4.6.1.7 assumed to enhance the importance of complementarities, as they may nudge users to move from a freemium to a premium subscription plan. Complementarities were furthermore in paragraph 6.3.1 found to enhance the value of a firm's offerings, and can thus be used to attract users to a premium version. Podio, Plytix, Appear.in and Microsoft owned Yammer are all driving adoption to their premium offerings by adding more sophisticated features, often enabled through complementary offerings. Hulthin (2016) states that complementarities are one of the most important features for Podio´s customers, as he explains; "no one wants to enter a locked system". Thus, it seems having different versions of a product drives an increased use of complementary offerings.

6.3.4 Empirical Findings

6.3.4.1 Enhanced Brand and Trust

Our findings suggest that complementarities not only reduce search costs, but also have the potential of creating value through building strong and trusted brands. Podio did for example use complementarities as a means to gain credibility. Hulthin (2016) for example explains that it was highly beneficial for Podio to integrate with large, successful companies like Box, as it functioned as a way of telling potential customers that "these guys are OK". Complementarities are in this way used to enhance brand and trust, something that leads to greater transaction efficiency.

6.4 Novelty

Amit & Zott (2001) find that in virtual markets novelty creates value both through novel offerings and services as first portrayed by Schumpeter (1934), and also from novel ways of doing business and structuring transactions. Our research finds that the case companies create value through new transactional content (paragraph 6.4.1), new transaction structures (paragraph 6.4.2) and new transaction participants (paragraph 6.4.2). We also find that freemium firms create novelty through versioning (paragraph 6.4.3.1) and crowdsourcing (paragraph 6.4.4.1). A full overview of the ways in which freemium firms create value through novelty is provided in table 11.

6.4.1 New Transactional Content

Amit & Zott (2001) find that Schumpeterian innovation through new transaction content creates value through combining resources that lays the foundation for new products and services. The introduction of new products or services, new methods of production, distribution, marketing, or tapping of new markets is traditional sources of value creation through innovation (Amit & Zott, 2001). Innovative content raises the value of the offering which serves to both attract and retain consumers. In conjunction with these findings, our research shows that new transactional content creates value for the case companies.

Drivers of Novelty	Amit & Zott (2001)	Proposed Modifications (Paragraph 4.6)	Case Study Findings
New Transactional Content	The value creation potential of innovations has been articulated by Schumpeter (1934). The introduction of new products or services, new methods of production, distribution, marketing or the tapping of new markets are traditional sources of value creation through innovation. The main innovation of some e-businesses resides in their complementary elements.	n.a	6.4.1: New transactional content creates value as it serves to attract new users. Adding innovative content furthermore raises the value of the offering, attract consumers to convert to a premium version and prevent customers from churning. Our findings suggest that freemium firms often add novel content through complementarities and crowdsourcing.
New Transaction Structure	Innovation in transaction structures create value by connecting previously unconnected parties, eliminating inefficiencies in the buying and selling processes through adopting innovative transaction methods, capturing latent consumer needs, and/or by creating entirely new markets. For example, e-business firms can identify and incorporate valuable new complementary products and services into their bundle of offerings in novel ways.	n.a	6.4.2: In our research we find no reason to differ between new transaction structure and new transaction participants, as we find that new participants can be a way of creating new transaction structures. We therefore refer to this category as *New Transaction Structure*. Our research implies that freemium firms add value by creating new transactions structures in which they integrate complementarities, add new transaction participants or change the linkages between them. This serves to enhance both transaction efficiency and lock-in.
New Transaction Participants	Another dimension of innovation in e-business refers to the appropriate selection of participating parties. For example, firms can direct and intensify traffic to their web site by initiating affiliate programs with third parties, who are compensated for enabling the execution of transactions from their own web sites.	n.a	
Proposed Modifications			
Versioning as a driver of novelty	n.a	4.6.1.7: With the need to convert users from a free to a premium version, it becomes important to continue innovating and collaborating with platforms in order to add value. We expect this to lead to a higher focus on novelty, especially through the use of complementarities in order to create indirect network effects and lock customers in.	6.4.3.1: Versioning seems to drive a focus on novelty as the firm's need to retain users, and incentivize users to convert to a premium version. Our research suggests that firms often employ complementarities in adding novelty.
Empirical Findings			
Crowdsourcing	n.a	n.a	6.4.4.1: The research suggests that crowdsourcing is commonly used as a source of innovation in freemium firms. Crowdsourcing as a way of innovating adds value as it is asset light and allows firms to focus on customer's needs and wants.

Table 11. Novelty (Analysis).

In a freemium model, innovative offerings serve to attract new users, and enable firms to charge the same price over time, or/and add new price plans. Adding value to an offering through innovation furthermore attracts consumers to convert to the premium version, as well as preventing customers from churning to competitive offerings. Traditionally, firms add novel content through advancements in R&D. Microsoft has for example been able to create new transaction content through heavy R&D spending. To appropriate rents on these investments, the firm has traditionally relied heavily on traditional protection mechanisms like patents and trademarks, successfully protecting their offerings (Microsoft 10-K, 2011). However, in light of the increased competition in the industry, Microsoft has been moving towards a more open appropriability strategy, relying on APIs (Skillicorn, 2014a) and bundles of offerings to informally protect their goods (Roterud, 2016).

Firms that only offer freemium offerings generally suffer from rather low capital, thus it is challenging for these companies to invest in R&D for creation of novel content. At the same time these firms usually offer software services which are difficult to formally protect, as discussed in paragraph 6.2.2. To overcome this issue, many companies utilizing a freemium model open up their offerings, using APIs and open source coding to be able to create novelty through complementary offerings. As discussed in paragraph 6.3.1, the case companies make use of complementary assets in order to add novelty and value to their offerings. Effectively it is the connectivity between the focal offering and the range of complementarities that makes the content novel. This is exemplified by Hulthin (2016) "The real value is that it [Podio] ties to other things". This furthermore enables the firms to add customized offerings (cf. paragraph 6.2.2), which raise switching costs and locks customers into the offering. As more consumers are locked in, the company has more capital to invest in adding novelty in their services. Adding novel features can furthermore drive users to convert to more expensive price plans. The use of complementarities in adding novel features seems to blur the lines between novel transaction content and transactional structures.

6.4.2 New Transaction Structure

Amit & Zott (2001) suggest that e-businesses also innovate in the way they do business, meaning the structuring of transactions, and the selection of participating parties involved in a transaction. We see no reason to differ between the two as we find that the involvement of new transaction participants is a type of new transaction structure. Thus, we refer to this category as new transaction structure. Zott & Amit (2008) furthermore propose that a novelty-centered business model yields higher performance than an efficiency-centered business model. This emphasizes the importance of novel transaction structures in value creation and appropriation.

The emergence of SaaS enabled a range of new transaction structures, and made it possible to utilize new business models like the freemium model. When the first companies started to employ freemium models in the late 2000s, they could probably extract first-mover advantages stemming from the new transaction structure, and "steal" customers from incumbent firms. However, as freemium models are quite common for SaaS offerings today, it might be more challenging to create value through new transaction structures using only this model (Hulthin, 2016). The competition between firms utilizing set model in the enterprise software market is fiercer, which

makes it more challenging to compete. Firms must thus find new ways of structuring transactions within their freemium models in order to create value.

A freemium company that successfully created value by employing a new transaction structure, is Dropbox. In 2011, the company faced great competition, both from incumbents like Google, Microsoft and Amazon as well as smaller start-ups (Barret, 2011). Drew Houston, the CEO of Dropbox, chose to protect the company by signing a deal with HTC, making Dropbox the default cloud storage option on every one of its Android phones. Deals with other phone companies were also expected in the near future. By integrating with new transaction participants, Dropbox created a new transaction structure and managed to face the rising competition. New transaction structures may in this way also serve to enhance transaction efficiency as integration with complementarities reduces search costs.

In a similar manner as Dropbox, several of the case companies have utilized new transaction structures through the use of complementarities. Podio has for example made it possible to relate and connect different apps (Podio Help Center, n.d.), making Podio´s core strength its ability to "keep everything connected" (Froda, 2012). Integration with third party developers has further enhanced the possibilities of new transactional content (Podio extensions, n.d.). Three of the case companies also leverage a bottom-up approach in which they target the end user, as opposed to the buying customer. This transaction structure, which includes new transaction participants, enables the firms to create network effects in a novel and efficient manner.

Lastly, as noted in paragraph 6.2.5.3 firms can enhance lock-in by quickly revealing the value of their offerings to the consumer. This is done by employing simple and transparent products or services, so the customers quickly perceive the value and also what they will gain by converting to a premium version. New transaction structures can be an efficient tool in realizing this. As an example, Microsoft versions their Office offerings based on the size of the device. This is a novel transaction structure that is relatively easy to understand; users get acquainted with and accustomed to the offerings on smaller devices, and when they want to utilize it on larger devices they need to upgrade to a premium account (Capossela, 2015). By quickly attracting and retaining the customer by efficiently showing the offerings value, new transaction structures may enable firms to enhance lock-in.

6.4.3 Proposed Modifications

6.4.3.1 Versioning as a Driver of Novelty

Versioning means offering information goods in different versions for different market segments (Shapiro & Varian, 1999). The purpose of versioning information goods is to get customers to sort themselves into categories based on their willingness to pay (Varian, 1997). In a freemium model it is paramount to convert users from a free to a premium version, as well as effectively lock the customers in. Thus, it becomes important to focus on continued innovation. In paragraph 4.6.1.7 we therefore proposed that versioning may drive novelty in freemium offerings, as novel content or transaction structures has the potential of raising the value of the offering, and thus attract users to convert to a premium offering. We especially expected that the case companies would make use of complementarities in enhancing the value of the offering.

Hulthin (2016) explains that customers paying for premium subscription models in Podio highly value APIs and other possibilities for complementary offerings. He states that customers will not enter a "locked system", and that complementarities therefore are expected in the market. The firm actually found that many customers convert to the premium version even before they reach the freemium limit. This implies that complementarities and novel offerings in the premium version are perceived as worth paying for. Poulsen (2016) adds to the discussion by explaining that Plytix aim to attract users to the premium version by enabling access to more advanced features like benchmarking and product tracking technology.

The case companies further explain that in order to advance revenues and retain customers over time, it is important to continually add novelty to the offering (Hulthin, 2016; Willassen, 2016). In deciding what features to add the firms commonly rely upon their users (cf. paragraph 6.4.4.1).

Versioning thus seems to drive a focus on innovation and novelty, as the firms need to attract and retain users to a premium price plan. Furthermore, complementarities seem to be an important aspect in this value creation, thus linking the two value drivers closely together.

6.4.4 Empirical Findings

6.4.4.1 Crowdsourcing as a Source of Novelty

As seen in paragraph 6.4.1, freemium firms make use of alternative methods to add novelty to their offerings. Our findings imply that in addition to employing complementarities (cf. paragraph 6.4.1), freemium firms also employ customers in the innovation process through 'crowdsourcing'. Hulthin (2016) explains that when adding new features to Podio, the company listened to their customers as well as the market to identify needs and wants. Appear.in also relies on crowdsourcing in deciding what features to develop, and whether to add specific features to the free or premium version. This is done through direct contact and piloting with their customers, and also queries on their webpage (Willassen, 2016). Appear.in further emphasizes the 'YAGNI' (You Aren't Gonna Need It) principle, in which they only add features when deemed absolutely necessary (How we work, 2015). This implies that the firm does not add features they *think* their customers might appreciate. Crowdsourcing as a way of innovating thus seems to add value as it is asset light and allows firms to focus on customer´s needs, which prevents spending's on unnecessary features.

6.5 Links between the Four Sources of Value Creation

Amit & Zott (2001) find that there are important synergies between the four sources of value creation where one can be more valuable when supported by the presence of another. Throughout the analysis we have touched upon how the sources of value creation are linked together, for example how complementarities are linked with novelty (cf. paragraph 6.3.1, 6.4.1 and 6.4.2). In this section of the thesis we will clarify the linkages revealed in the case study and compare them with the findings of Amit & Zott (2001). Table 12 depicts the different linkages suggested by Amit & Zott (2001), as well as new and refined linkages revealed in this study. We will therefore in this paragraph only briefly go through the linkages revealed to connect the four sources of value creation;

Links between the value drivers	Efficiency	Lock-In	Complementarities	Novelty
Efficiency Enhances		**Amit & Zott (2001)**The efficiency features of an e-business may serve to attract and retain customers and partners. The higher the relative benefits offered, the higher their incentives to stick with the network established by the e-business. The increasing return properties inherent to network effects then magnify the relative benefits offered, thus triggering positive feedback dynamics. **Holthe & Langø (2016)** As freemium models enhance transaction efficiency, firms utlizing this model can potentially attract large user bases. This serves to enhance lock-in as freemium models leverage this user base through network effects.	**Amit & Zott (2001)** Efficiency gains made possible by information technology pave the way for the exploitation of complementarities in e-business. Weaving together the resources and capabilities of distinct firms, a hallmark of e-businesses, is economically compelling when transaction costs, and hence the threat of opportunism, are low. **Holthe & Langø (2016)** Efficiency gains in terms of APIs facilitates efficienct explotation of complementarities. Furthermore, other efficiency gains in a freemium model make firms capable of attracting large user bases, which serves to attract suppliers of complementary offerings and gives rise to indirect network effects.	x
Lock-In Enhances	**Amit & Zott (2001)** When an e-business creates lock-in, this can also have positive effects on its efficiency. For example, many auction sites enable buyers to rate sellers. This feature increases buyers' trust in the fairness of transactions and therefore fosters stickiness. This feature also provides a strong incentive for repeat sellers to refrain from cheating, which clearly enhances transaction efficiency. **Holthe & Langø (2016)** An installed base that is locked into the offering raises the value of the offering when network effects are present. This incetivizes word-of-mouth, enhance the brand and reduce search costs which again enhances transaction efficiency and attracts users.		**Amit & Zott (2001)** When an e-business creates lock-in, this can also have positive effects on the degree to which it provides complementary offerings. A strong potential for lock-in provides an incentive for high-profile partners to contribute complementary products and services because of the promise of high-volume (repeat) business. **Holthe & Langø (2016)** When a business has locked in a large user base, complementors are keen to add new products and services because of the promise of repeat business.	**Amit & Zott (2001)**: x **Holthe & Langø (2016)** When a firm utilizing a freemium model has an installed base that is locked into the offering it has more capital to invest in creating novel transactional content. A focus on lock-in furthermore drives a focus on novelty as firms need to add novel features in order convert customers to premium price plans.
Complementarities Enhances	**Amit & Zott (2001)** Complementarities may lead to increased efficiency, at least from a customer's point of view. When customers have access to products and services that are complementary to the primary product of interest, efficiency may be enhanced, for example, through reduced search costs and improved decision-making. **Holthe & Langø (2016)** Complementarities may lead to enhanced transaction efficiency as complementarities serves to reduce search costs, and enhance trust and brand.	**Amit & Zott (2001)** The complementary product and service offerings of an e-business may serve to attract and retain customers and partners. The higher the relative benefits offered, the higher their incentives to stick with the network established by the e-business. The increasing return properties inherent to network effects then magnify the relative benefits offered, thus triggering positive feedback dynamics. **Holthe & Langø (2016)** In close relation to Amit & Zott's (2001) findings we find that complementors add products and services on top of the focal offering, which raises the value of the offering. This raises switching costs and creates lock-in through indirect network effects.		**Amit & Zott (2001)** The main innovation of some e-businesses resides in their complementary elements, such as the resources and capabilities they combine. **Holthe & Langø (2016)** Complementary offerings enhance the novelty of the offering. The case companies leverage complementarities in adding value for their customers, which again enables them to keep charging premium prices and/or add new price plans.
Novelty Enhances	**Amit & Zott (2001)** Certain efficiency features of e-businesses may be due to novel assets that can be created and exploited in the context of virtual markets. For example, Art-net.com, a European company that enables online art auctions, reduces the asymmetry of information between the buyers and sellers of art (traditionally a source of severe inefficiencies) through maintaining and expanding a data base of transactions (including information on price) that is accessible to its clients. **Holthe & Langø (2016)** The case companies employ novel transaction structures that may serve to enhance efficiency. For example, Appear.in enables online video conferencing without a need to sign up for an account. This reduces search costs and may enhance information symmetry.	**Amit & Zott (2001)** Novelty affects lock-in, in two important ways: First, e-business innovators have an advantage in attracting and retaining customers, especially in conjunction with a strong brand. Second, being first to market is an essential prerequisite to being successful in markets that are characterized by increasing returns. First movers are in a good position to initiate the positive feedback dynamics that derive from network externalities, and to achieve a critical mass of suppliers and/or customers before others do. **Holthe & Langø (2016)** Our findings suggests that in order to create sustained switching costs abd entice conversions, firm's must be able to provide value in terms of novelty. Second, our research implies that first movers in transaction structures, yields positive outcomes in terms of network effects. However, we do not find any evidence for whether or not there are any first-mover advantages in relation to novel content.	x	

Table 12. Links between the Four Sources of Value Creation (Analysis).

Transaction efficiency is in freemium business models found to enhance lock-in through facilitating low entry barriers, which serves to attract large user bases. This gives rise to network effects, which increases switching costs and thus enhances lock-in (cf. paragraph 6.2.1.1). Furthermore, the attraction of large customer masses serves to entice suppliers of complementary offerings, and thus give rise to indirect network effects (cf. paragraph 6.2.1.2). The development of API-technology further simplifies this process, and makes it easy and efficient to leverage complementary offerings (cf. paragraph 6.3.3.1). Efficiency is neither in this study nor Amit & Zott's (2001) research found to enhance novelty. The presence of transaction efficiency thus enhances the value creation potential inherent in lock-in and complementarities.

The study reveals that lock-in enhances transaction efficiency as an installed base that is locked into an offering raise the value of the offering when network effects are present (cf. paragraph 6.2.1.1). This increased value incentivizes word of mouth, enhances the brand value and reduces search costs.

Furthermore, with an installed base locked into the offering, complementors are more interested in adding new products and services on top of the offering, as there is promise of repeat business - which again adds value to the focal offering (cf. paragraph 6.2.1.2). Lastly, with a locked-in installed base, the freemium firm likely enjoys higher revenues, which may facilitate higher investments in the creation of novel offerings. To keep customers locked in, the firms have incentives to innovate through new transactional content and transaction structures. The value creation potential of transaction efficiency, complementarities and novelty may thus be advanced when the firm has a customer base locked into its offering.

Complementarities are in this study found to be able to enhance transaction efficiency as the presence of complementary offerings may reduce search cost and enhance trust and brand (cf. paragraph 6.3.1.1 & 6.3.4.1) When a freemium firm is associated with a respectable brand or a complementary offering that the customer is familiar with, this has the ability to reduce search costs and thus enhance transaction efficiency. Complementary offerings furthermore enhance the value of the service or product and can be used as a way of enabling customization, which creates higher switching costs and thus lock-in (cf. paragraph 6.2.1.2, 6.2.2 & 6.3.1). Complementarities also enhance the value potential of lock-in through indirect network effects. Lastly, the use of

complementarities is found to add novelty to the offering (cf. paragraph 6.3.1, 6.4.1, 6.4.2 & 6.4.3.1). Thus, the presence of complementary offerings seems to enhance the value of transaction efficiency, lock-in and novelty.

Novelty is found to enhance efficiency through novel transaction structures (cf. paragraph 6.4.2). For example, zero prices and the bottom-up transaction structures serve as ways of reducing entry barriers and thus enhance transaction efficiency (cf. paragraph 6.1.4.1 & 6.4.2). Novelty also has a big impact on lock-in, as freemium firms must create novel offerings to entice users to convert to a premium version, and prevent customers from churning (cf. paragraph 6.2.5.3, 6.4.1, 6.4.2 & 6.4.3.1). The research reveals that first movers in transaction structures, in terms of being the first to employ a freemium model, receive positive feedback in terms of network effects, as these firms can steal customers from incumbent firms and generate lock-in (cf. paragraph 6.4.2). There are however no evidence as to whether or not there are any first mover advantages in relation to novel offerings. We suggest this is due to the fact that none of the case companies offer truly novel products or services. Novelty, both novel transactional content and transaction structures, thus enhances the value creation potential of transaction efficiency, lock-in and complementarities.

6.6 Four interconnected Sources of Value Creation

The findings identified throughout this analysis bear important insights as to how the four sources of value creation create value in a freemium business model, and how they are connected. In the following we will summarize how each source of value creation spawn increases in value, which is furthermore illustrated in figure 5. New findings added to the model are illustrated in italic.

6.6.1 Transaction Efficiency

In conjunction with Amit & Zott (2001), our findings suggest that transaction efficiency in a freemium model is enhanced by reducing information asymmetry and search costs. Additionally, the research implies that versioning and zero prices serves to drive efficiency as it enables consumers to experience the offering upon determining its value. Network effects is found to be an enabler of transaction efficiency, as direct network effects raise the value of the offering, and word

of mouth serves to reduce search costs and information asymmetry. Brands are furthermore found to enhance efficiency.

6.6.2 Lock-in

This study reveals that network effects are essential to lock-in in a freemium business model. Direct network effects raise the value of the offering, which leads to higher and collective switching costs. Indirect network effects, hereunder a link with complementarities, furthermore raise the value of the offering. Indirect network effects also raise switching costs as the consumers face a web of connected offerings. Lock-in is additionally enhanced by customization, as well as brand and trust. Lastly, versioning seems to drive a focus on lock-in as firms need to convert users from free to premium offerings in which time-to-value and conversion analytics are of significance.

6.6.3 Complementarities

Complementarities are in this study found to enhance the value of an offering through adding novel features and linkages, which eventually raise switching costs and increase transaction efficiency through reduced search costs and enhanced brand and trust. A focus on complementary offerings are furthermore driven by a focus on creating and leveraging indirect network effects, and versioning the offerings through customization, which in turn enhance lock-in. Efficiency gains made possible by new technology (APIs) furthermore enhance the value of complementarities.

6.6.4 Novelty

In this study firms are found to enhance the value creation potential of novelty through new transactional content and transaction structures. In doing so, the firms most commonly rely upon complementarities. The need to create and leverage network effects in a freemium model furthermore drives novelty, as firms for example employ new transaction structures like the bottom-up approach in order to create network effects. Lastly, a focus on versioning offerings drives a focus on novelty, as firms add new transactional content and structures in order to attract users to a premium version.

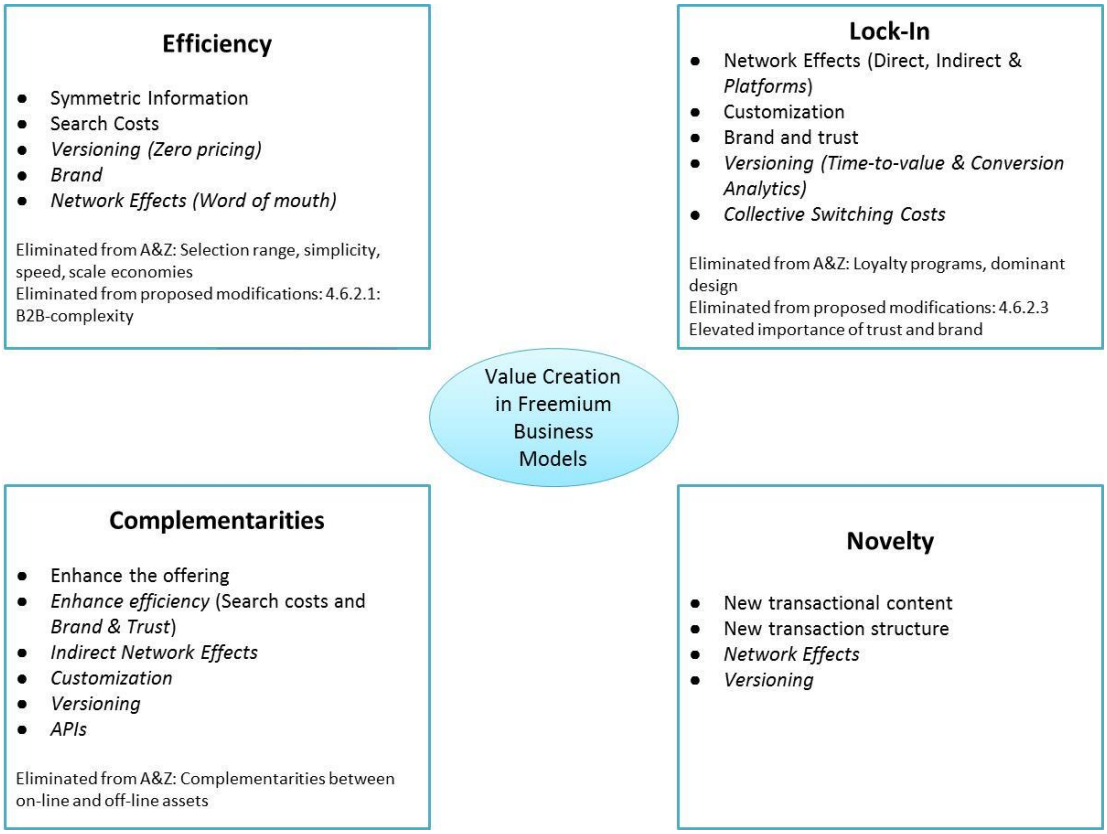

Figure 5. Four Interconnected Sources of Value Creation (Analysis).

Chapter 7 – Discussion

This chapter will discuss the contribution of our findings. The chapter commences with a review of how this research may contribute to the literature on value creation in a freemium business model (paragraph 7.1). Thereafter, we scrutinize how the findings may affect managerial decision making in firms utilizing a freemium business model (paragraph 7.2). Perspectivation and possible limitations associated with the research is discussed in paragraph 7.3. Lastly, this chapter provides suggestions for future research (paragraph 7.4).

7.1 Implications for Literature

This study set out to explain how a freemium business model creates value in B2B markets, a concept that is not well understood by existing literature. In seeking to understand the value creation potential of said business model we found that the existing literature only partially advanced our knowledge of the field. As covered in paragraph 4.5, Amit & Zott's (2001) research purposed to identify the value creation potential of all e-business relative to offline businesses. Our research purpose is on the other hand to use this model to explain the value creation potential of a specific business model. This study therefore bears important implications for how the freemium business model creates value in the B2B enterprise market, and identifies differences in value creation between small and large firms. The research is thus interesting from an academic point of view as it advances our understanding of how the freemium business model creates value in B2B contexts, as well as adds to the existing body of research on value creation in virtual markets.

Two implications seem most prevailing in this research, extending the knowledge of academia in our focus area; Freemium value creation in B2B markets (paragraph 7.1.1), and the impact of firm size on value creation (paragraph 7.1.2).

7.1.1 Freemium Value Creation in B2B Markets

Our findings imply that the freemium business model creates value in the B2B enterprise software market. This is contrasting many non-academic articles reviewed in paragraph 2.4.2, who argues that the freemium business model is of limited value in B2B markets. For example, it was suggested that free offerings do not work well in B2B markets as it is difficult to achieve a viral spread in these markets and that B2B clients have more stringent requirements, and are more skeptical towards free offerings. Literature covered in paragraph 4.4 furthermore suggested there would be noteworthy differences between B2B and B2C markets in terms of greater complexity in B2B markets, as well as an elevated focus on trust and brand which we proposed would complicate the value creation potential of the case companies. Our findings do however interestingly imply that the freemium business model creates value in B2B companies, and that it creates this value in a similar manner to value creation in B2C markets.

Prior non-academic articles and industry experts seem to have failed to understand that companies operating in the B2B enterprise market are changing the game of user acquisition. The foremost important finding in this research is the fact that freemium firms operating in a B2B context most commonly utilize a bottom-up approach to user acquisition in which they target the end user, and not a traditional sales strategy targeting the IT-division or top management. Thus, the case companies move away from the classical way of customer acquisition in B2B markets, and in this manner manage to circumvent B2B transaction complexity and issues related to trust and brand. This is furthermore supported by the fact that Plytix, the only case company that did not utilize a bottom-up approach, was the only firm that expressed issues with transaction efficiency or an elevated focus on trust and brand when using a freemium business model (cf. paragraph 6.1.4.4).

Contrary to prior knowledge on the topic, our research interestingly implies that the freemium business model is a very helpful tool in overcoming issues with B2B complexity. Existing literature suggested that B2B firms utilizing a freemium business model would experience difficulties with gatekeepers in relation to trust, brand and more stringent requirements. Our research however reveals that the use of a freemium business model actually enables firms to employ a bottom-up approach to user acquisition as individual employees can start using the offering for free. This serves to dramatically reduce transaction costs and increase transaction efficiency. Our findings

furthermore imply that the user adoption following this approach creates tremendous network effects as users engage in word of mouth, which serve to lock the customers in even before the gatekeepers are involved.

To be clear, we do not suggest that the proposed B2B characteristics are wrong, but that freemium firms utilizing a bottom-up approach manage to circumvent these challenges - thus making them less relevant in our case study. This way of structuring transactions thus makes B2B transactions more similar to B2C transactions. However, there are certain characteristics that are of high importance to achieve value creation in B2B markets with a freemium bottom-up approach, namely versioning (paragraph 7.1.1.1) and network effects (paragraph 7.1.1.2);

7.1.1.1 Versioning

The bottom-up approach to user acquisition is effective in circumventing gatekeepers as zero prices allow individual employees to start using the offering for free. As suggested by Shapiro & Varian (1999), versioning offerings is found to drive adoption, and versioning so that one part of the product is free is found to attract even more users (Shampanier et al., 2007). Thus, as expected in paragraph 4.6.1.1, versioning offerings and zero prices dramatically reduce adoption barriers in B2B markets; As individual employees rarely has a budget to spend on enterprise software, zero prices is very effective in reducing transaction costs. Versioning is thus a central mechanism in employing a bottom-up approach to customer acquisition. Our study furthermore implies that versioning enhances lock-in, as the attraction of a large user base gives rise to positive network externalities which will be discussed in paragraph 7.1.1.2.

Versioning is furthermore a central mechanism in overcoming the challenge with less virality in B2B markets as a result of complexity in the offerings. As versioning allows the case companies to keep the free version of the offering simple and transparent, they are able to achieve a viral spread similar to that of single purpose consumer apps in the B2C market. Versioning furthermore results in an enhanced focus on conversion. To able to convert business users to the premium version, the case companies are found to add novel content and transaction structures. Thus, versioning enables value creation through a bottom-up approach by lowering the complexity of the offerings and attracting a large user base with zero prices.

7.1.1.2 Network Effects

Contrary to the belief of non-academic papers, our research implies that the freemium model is especially prominent in extracting network effects also in B2B markets. With a bottom-up approach to customer acquisition, network effects are central to value creation, as it is key to spread adoption of the offering and attract complementary assets. The elevated importance of network effects differentiates from the quite narrow focus Amit & Zott (2001) yield this phenomenon. This is most likely due to the fact that the possibility for purposeful networking, and thus the potential of leveraging network effects, has been advanced with the development of Web 2.0 (cf. paragraph 2.2). In addition to having a greater impact on lock-in than formulated by Amit & Zott (2001), the research reveals that network effects also enhance the value creation potential inherent in transaction efficiency, complementarities and novelty.

Network effects are found to enhance transaction efficiency as direct network effects lead consumers to attract more users and hence increase the value of an offering. Word of mouth is revealed as an important driver in this situation, as word of mouth in accordance with Silverman (2011) is found to yield a viral spread among consumers. This viral spread serves to reduce search costs and enhance the brand of a firm, thus increasing transaction efficiency.

Network effects are furthermore found to be the main driver of lock-in in a freemium model, through both direct and indirect network effects. The impact of network effects on lock-in is actually found to be so great that it alone can create lock-in when utilizing a bottom-up approach to customer acquisition. Familiar examples of this are Podio and Yammer, where the consuming organizations had so many of their employees networked in the offering that the gatekeepers had no other choice than to acquire a premium subscription in order to gain control of the network.

Indirect network effects where the value of the offering is enhanced through integration with complementary offerings are also found to enhance value creation in freemium models. The use of complementarities serves to entrench customers (Shapiro & Varian, 1999) through adding value and novelty, and can potentially also protect the focal offering from imitation. Complementarities are thus a prime driver of 'added value' in terms of novelty, customization and brand and trust. In B2B markets, complementarities seem to enjoy an elevated position in value creation as there is a strong

focus on compatibility and connectivity, where value stems from the novel interconnections between different offerings.

Network effects are thus revealed to be extremely important in value creation in freemium business models in B2B markets, enabling all value drivers to co-evolve and create spiraling effects.

7.1.2 The Impact of Firm Size on Value Creation

As stated in chapter 1 and 2, the freemium model has commonly been employed by disruptive start-up firms with relatively cheap and uncomplex products. The simplicity of the offering has been essential to effectively drive adoption and gain traction for their offerings. However, in later years more established industry players have started utilizing the freemium business model for complex and traditionally costly offerings like Microsoft's enterprise software. These established industry players commonly have large installed bases of users, which raises the question about whether the freemium business model creates value in a similar manner for start-ups and incumbent firms.

To be able to form a differentiated view on value creation in freemium firms, we chose companies of different size and maturity. The study has revealed that the use of a freemium business model in fact creates value in different ways for different firms. In adherence with existing literature, start-up companies, which most frequently has been the firms utilizing a freemium model (Kim, 2015), is found to use the model as a means to attract a large customer base, and thus be able to show a value potential for complementary offerings as well as investors. For start-ups, customer acquisition seems to be the most important aspect for value creation in a freemium model, while the revenue model, what features to put in the different versions of the offering as well as continued innovation are of second priority.

For the bigger companies in the study, like Microsoft and partly Telenor (owner of Appear.in) and Citrix (owner of Podio), the freemium model seems to be employed in order to create value for, and attract users to, other more advanced and proprietary offerings. Established companies thus make a small and simple part of their offering available for free, while more advanced features are offered at a premium price. In addition they bundle freemium offerings with other products and services in-house, in order to drive adoption for these offerings. Microsoft for example aims at building

products that work 'beautifully' together, so that consumption of one of their products naturally leads to consumption of another (Capossela, 2015). For the more established industry players it seems that they do not have to spend a lot of resources on user adoption, as they can utilize their strong brand to attract users to the platform. Microsoft's CMO Capossela for example explains "The reality is we have no issue with acquisition, zero. Tons of people know the brand. I can do a press release, and all of a sudden, "Oh my gosh, Office is on the iPhone, let me go download Word and Excel and PowerPoint" (Capossela, 2015). Established firms thus seem to focus their resources on the revenue model, continued innovation and feature differentiation.

This implies that the freemium business model can create value and be leveraged in different ways by different firms. While the freemium model primarily create value through lowering adoption barriers and attracting large user bases in start-ups, it seems to create higher value when combined with other business models and strategies to drive engagement for more advanced and proprietary offerings in established firms.

7.2 Managerial Implications

In this paragraph we first propose some managerial implications stemming from the findings in chapter 6. We specifically propose three managerial implications; the use of a bottom-up approach (paragraph 7.2.1), differentiation between free and premium offerings (paragraph 7.2.2), and protection of freemium offerings (paragraph 7.2.3). Thereafter, we shortly discuss challenges with utilizing a freemium model (paragraph 7.2.4).

7.2.1 Bottom-up Approach

An important managerial implication stemming from this research is the use of a bottom-up approach to customer acquisition. Microsoft, Podio and Appear.in all employ this approach, in which they reach out to the individual employee in contrast to selling to the IT-department. The bottom-up approach has the opportunity to attract many users who join a network and then get dependent on the offering, which again creates high collective switching costs. This furthermore implies that a bottom-up approach is best utilized in combination with an offering that has a social component. A social component gives the customers a natural incentive to share and include other

users. Yammer is a classic example of a product that naturally drives sharing, as users gain much higher value with more people in the network.

Several of the case companies expressed that firms likely would be unwilling to try freemium offerings through traditional sales methods. This is why it might be challenging for Plytix to gain huge traction for their service, as the firm employs a more traditional sales strategy. This strategy might not work with a freemium model, something that can be exemplified by the fact that Plytix got more meetings with department leaders when they on a trial basis added a price to their offering. Thus, it seems it might be challenging to use a freemium model in conjunction with traditional customer acquisition models. This implies that managers must wisely assess which products to deliver with a freemium model, as those inhabiting a social component seem more successful.

7.2.2 Differentiating Free and Premium Offerings

As suggested in chapter 2 (cf. paragraph 2.3.4.1.1), the research further implies that it is important to balance the features offered in the free and premium version of a freemium offering. This is in coherence with Kumar (2014), who finds that the freemium version needs to show enough value to attract a huge customer base, but not so much that no consumer sees the value of upgrading to the premium version. The offerings furthermore need to exhibit simplicity and transparency both in regards to the offering and in relation to what users gain by upgrading to a premium version. Time-to-value is therefore important for value creation in a freemium model, as the density of freemium offerings in the market makes it paramount to clearly differentiate offerings. The issue of which features to put in each offering is highlighted by many of the case companies, with no 'one' right answer, as this naturally depends on the service. However, all of the case companies emphasize that conversion analytics are important in order to reach this conclusion. Thus, it seems important to study how customers use the service in order to determine which features to put in the different versions of the offering.

The research furthermore implies that it is important to continue innovating in order to increase the value of the premium services and accelerate upgrades over time (cf. paragraph 6.4.3.1). Attention to continued innovation might also create switching costs and entrench customers to continue investing in a good and thus become committed to the focal firm (Shapiro & Varian, 1999) (c.f.

4.2.4.2). The same seems true for transaction structures; in order to yield high performance, firms utilizing a freemium business model need to find new ways of structuring transactions in order to create value, for example by integrating with complementarities or linking transaction participants in novel ways.

7.2.3 Protection of Freemium Offerings

Lastly, as explained in chapter 6, the supplying firm in a freemium business model most commonly does not rely on proprietary protection mechanisms. The case companies are found to overcome this issue by offering integration with complementary services. However, these bundled offerings are not proprietary as it is most commonly done through the use of open APIs. In that sense there are no barriers to prevent competitors from offering the same through the use of complementarities. Thus, if a firm has a novel offering and the opportunity to formally protect it, it might not be beneficial to use a freemium business model. Managers should in this situation consider whether they should use a more traditional business model to create the highest possible value.

7.2.4 Challenges with Freemium Models

This research finds that the use of a freemium model in the enterprise software market has the potential of creating value in B2B contexts. However, both literature and the case study findings in accordance with prior literature suggest that there might be some challenges with value creation when using a freemium business model. Specifically, we address two prevailing issues; Negative associations to the word 'free' (paragraph 7.2.4.1), and deferred revenues (7.2.4.2);

7.2.4.1 Negative Associations to the Word 'free'

Anderson (2009) suggests customers might face negative associations to the word 'free', as companies might want to pay for a product or service to lower the risk of it not being what they want. Prices come with guarantees, while free typically does not (Anderson, 2009). This issue can be seen in relation to Plytix, as the firm found it was easier to get a meeting with divisional leaders when they on a trial basis added a price to the offering. Shampanier et al. (2007) further explain that consumers might associate zero prices with low quality. Poulsen (2016) underlines this issue for Plytix; "It is a big challenge right now to go out there and offer a very good service for free,

because people become skeptical, that it's too good to be true. I don't really know what the solution is".

7.2.4.2 Deferred Revenues

For big enterprises using the freemium business model, it is possible to sustain overall revenues even though the freemium offering does not reap high revenues in the early stages. For start-ups however, using a freemium model involves high costs from supporting the service to a mass of non-paying customers, which pose challenges to the viability of the model (Niculescu & Wu, 2011). The start-ups in this research have both relied on funding, which will be a prerequisite to be able to launch a product or service with a freemium model. Thus, it might be more challenging for start-up firms to succeed when utilizing a freemium model, as they are more reliant on direct income.

7.3 Perspectivation and Limitations

The Internet has made transactions between firms and customers significantly more efficient. In later years, improvements in the infrastructure of the Internet have furthermore changed the way consumers and firms interact, and created possibilities beyond efficient transactions. The Internet has effectively changed the rules of the game for all firms operating in today's e-business market, and enabled a range of new ways to structure offerings, the links between participants, communication with customers, distribution of products and services, etc. The freemium business model can be viewed as one example of a structure made possible by the improvements in set technology.

As explained in this thesis, prior literature deemed the freemium model as not applicable in B2B markets, as these markets are more complex, and business customers were assumed to face inertia in relation to free offerings. We have in this thesis however revealed that this does not fit with the reality of the case companies. We do not believe that the industry experts and others who stated set business model would be of limited value in a B2B market were wrong per say, but that they overlooked or failed to understand how the improved technologies of the Internet transform relationships between firms and customers. Through a traditional sales strategy where firms transact with the IT-department or other gatekeepers, the proposed complexity would likely fit the reality. However, the case companies are found to overcome this issue by utilizing a bottom-up approach

for customer acquisition, thus utilizing new relational structures to attract individual employees instead of whole departments or organizations. This way of transacting has in the bigger picture been enabled by the Internet and SaaS cloud infrastructure, which enables firms to deliver offerings to individual's independent of their location and organizational affiliation, and without the need for on premise software. Thus, this research is in a greater sense an example of how the Internet has transformed the rules of the game for all companies operating over the Internet, which today inhibits most firms.

One should not conclude that one has seen everything made possible by the Internet. The availability of pervasive networking along powerful computing communications will probably enable a range of new transaction structures, ways of communication and distribution methods in the future. It will therefore be interesting to see how the evolution of the Internet will enable firms to exploit new sources of value creation in the coming years.

Even though we find our study highly relevant and useful in explaining how the freemium model can create value for B2B firms operating in the enterprise software market, we see a need to discuss the choices made in the case study, and how these choices might have limited set research. These limitations inhibit drawbacks regarding choice of literature in relation to the value creation model of Amit & Zott (2001) (paragraph 7.3.1.1) and in relation to the use of a business model perspective (paragraph 7.3.1.2). Further, we highlight a possible lack of focus on value appropriation in set research (paragraph 7.3.2), as well as possible validity threats (paragraph 7.3.3).

7.3.1 Choice of Literature

7.3.1.1 Value Creation Model

Amit & Zott´s (2001) model for value creation has been employed to answer the research question. Even though the model is found to yield great help in reviewing and structuring the findings, we recognized that some drivers proposed by Amit & Zott (2001) were not relevant in enhancing our understanding of the freemium models value creation potential. These drivers consist of selection range, simplicity and scale economics for efficiency (cf. paragraph 6.1.3), loyalty programs and dominant design for lock-in (cf. paragraph 6.2.4) and offline assets complementing online assets for

complementarities (cf. paragraph 6.3.2). At the same time, some value drivers not mentioned by Zott, Amit & Massa (2011) were revealed to be highly relevant in this research. One might therefore suggest other theoretical lenses could have been more applicable in understanding and answering the research question.

As networks are found to be of much higher importance in this study than in the research of Amit & Zott (2001), network theory could have provided a better understanding and basis for answering the research question. For example, as strategic network theory is concerned with the implications of network structure for value creation, this could have explained the importance of complementarities and co-creation of value in a more efficient manner. However, this view focuses on the network as a whole and not a specific firm's position in set network (cf. paragraph 4.1.5). Furthermore, strategic network theory would probably not enable us to explain the value creation potential of new transaction structures and the direct efficiency gains made possible by versioning and zero price. The value creation model proposed by Amit & Zott (2001) centers on a particular firm and questions how the firm structures itself within the network in which it exists. Thus, we believe the network perspective would have limited our understanding of the holistic firm value creation. Even though the value creation model proposed by Amit & Zott (2001) needed adjustments to fit this specific research, we consider this model as the best possible way to conduct a holistic view on the value creation potential of freemium firms in a B2B context.

7.3.1.2 Business Model View on Value Creation

Employing Amit & Zott's (2001) value creation model in answering the research question implies taking a business model perspective on value creation. This view on value creation is not as known and well cited as the other views covered in paragraph 4.1, like the Resource Based View. Thus, in employing this view we limited our opportunity to search for related studies that has employed the view in seeking an explanation to a similar concept. The fact that we are employing the business model view in a new way and seek explanation for the value creation potential of a specific business model (cf. paragraph 4.5), further underlines this limitation. As we are not dealing with a highly investigated stream of literature, misunderstandings or misconceptions might arise. However, as explained in chapter 4 (cf. paragraph 4.1 & 4.5), employing only one of the other views on value creation would probably have been insufficient in capturing the full value creation potential of a freemium business model. For example, as this research takes a firm perspective, one

might suggest the Porterian view or Resource Based View could have been employed to answer set research question. Even though this could have advanced our understanding of how the freemium model creates value through differentiating activities and the links between them, or the internal resources and capabilities of a freemium firm, these views fail to capture the role of networking (cf. paragraph 4.1.1 & 4.1.3).

As a business model perspective accommodates, to our opinion, a more holistic view of value creation and integrates the other views mentioned in chapter 4, this model became the preferred choice. In order to circumvent the limited available literature on the business model view, we have additionally studied and incorporated literature that Amit & Zott (2001) builds their arguments and reasoning on, as well as other supporting or complementary literature as covered in paragraph 4.2, 4.3 and 4.4.

7.3.2 Appropriation

As our research focuses upon value creation, other aspects of a business model, like value appropriation, is given less focus. The value creation model used to explain this research question mostly focuses on creation and not on appropriation of value (Zott & Amit, 2008). In later years, Amit & Zott's attention has shifted to value appropriation by linking some of the sources of value creation, namely efficiency and novelty, to focal firm performance. Zott & Amit (2008) highlight that value creation and appropriation needs to be simultaneously considered in order to assess firm performance.

Even though the employment of Amit & Zott's (2001) value creation model yields little explicit focus to value appropriation, this research has given a lot of attention to how freemium firms create value through lock-in, hereunder how firms convert free customers to paying ones. This indirectly yields insights into how firms might appropriate rents. The focus on conversion rates and other ways to lock in customers thus bear implications for how freemium firms might reap monetary value. However, as we on several occasions have touched upon, the case companies employ a variety of complementary offerings, and our research yields little insight as to how this may affect the companies' ability to appropriate rents.

7.3.3 Validity Threats

This research has studied the use of a freemium business model in four case companies, and is thus limited to the scope of these firms. This may cause validity threats in regards to the generalizability of the research. At the same time, two of the case companies have only just started utilizing a freemium business model, making it challenging to know whether or not they will create value in the future. Furthermore, other findings could have been revealed if other case companies had been studied. One might for example vision interviewing firms in B2B markets which has not succeeded with the freemium business model, in order to reveal aspects that might challenge the value creation potential of set business model. Our informants in the case companies also reside in Norway (Microsoft and Telenor) and Denmark (Plytix and Podio), which makes the research geographically restricted. At the same time, as this is a master thesis with limited time and resources available, more than four case studies might not have been possible. As our case companies vary in maturity, size and resources, we further feel the research presents a realistic insight into how different firms create value with a freemium business model.

7.4 Suggestions for Future Research

The primary concern for this study has been to identify how the use of the freemium business model creates value for companies operating in the B2B enterprise software market, using Amit & Zott's value creation model (2001) to structure the research. Even though the study has increased the knowledge in set area, there are still some gaps that would benefit from further research. Thus, we propose four suggestions for future research; a longitudinal research (paragraph 7.4.1), customer point of view (paragraph 7.4.2) and merging factors between the B2C and B2B market (paragraph 7.4.3);

7.4.1 Longitudinal Research

As this thesis faced a time restriction of four and a half month, it would be interesting to test how the freemium business model yields value creation over time in the B2B enterprise market. A longitudinal study might yield better insights about how different firms utilize the freemium business model, and reveal new important drivers for value creation. A longitudinal study could

also focus on both value creation and retention, and thus possibly increase the knowledge about the appropriation part of the business model.

7.4.2 Customer Point of View

As our study focuses on value creation from the firm's point of view, the customer's perception of value creation in a freemium model is not explicitly researched. To enlighten how the freemium business model creates value in B2B markets, a study of value creation from a customer point of view would probably be highly beneficial.

7.4.3 Merging Factors between the B2C and B2B market

An important finding in this thesis is how the freemium business model seems to create value in very similar ways in the B2C and B2B market. As explained in chapter 2, (cf. paragraph 2.3.4.2) prior knowledge on the freemium model predicted its use in the B2B market would be of limited value. Our study however suggests that the freemium model can create value in a B2B market, and furthermore do so in a very similar way as in a B2C market. Therefore, an in depth study of the merging factors between B2C and B2B markets would be beneficial in order to confirm and/or enhance knowledge in this area.

Chapter 8 – Conclusion

The purpose of this study has been to answer the research question "How does the freemium business model create value for firms operating in the B2B enterprise software market?" The research question has been examined through a qualitative case study using four case companies.

In answering set research question this thesis has taken a business model perspective on value creation. Amit & Zott's (2001) value creation model has laid the foundation for our analysis, wherein we purposed to identify the mechanisms driving the four sources of value creation; transaction efficiency, lock-in, complementarities and novelty. Additionally, we examined and incorporated complementary literature for this specific research topic, namely literature on information goods, network theory and B2B transactions. This enabled us to propose ten modifications to the value creation model (cf. paragraph 4.6). Taking a mixed research approach furthermore enabled us to identify new concepts in our data.

The research revealed that transaction efficiency, lock-in, complementarities and novelty are all important sources of value creation in a freemium business model in the B2B enterprise software market. As for what drives these sources of value creation, we found that some were in adherence with Amit & Zott's (2001) research, while other drivers emerged from new insights. In our analysis we specifically found that symmetric information, search costs, versioning, word of mouth and brand drive enhanced transaction efficiency. The findings further suggest that lock-in is enhanced by network effects, customization, versioning and brand and trust. Complementarities are revealed to enhance the value of the offering aa well as transaction efficiency, through indirect network effects and versioning. APIs are furthermore found to drive an increased use of complementary assets. Lastly, novelty is found to create value through new transactional content as well as new transaction structures, network effects and versioning. Through the case study we thus managed to identify the specific mechanisms driving value creation in a freemium business model. These new findings bear important implications for the literature of the field, as well as for managers. Specifically, our findings imply that the freemium business model creates value for firms operating in the B2B enterprise software market by enabling a bottom-up approach to user acquisition. Two mechanisms are of high importance in this approach, namely versioning and network effects;

Our research implies that in a freemium business model, 'versioning' enhances the value creation potential of all sources of value creation. Versioning and zero prices are especially prominent in enhancing transaction efficiency in B2B markets as it reduces barriers to adoption and allows individual employees to start using the offering for free. This gives rise to network externalities which serves to enhance lock-in. Versioning furthermore drives a focus on converting users from a free to a premium subscription plan, which drives the value creation of complementarities and novelty. For managers in B2B markets this implies that they must carefully assess what features to put in their free and premium offerings, and also focus on continued innovation to increase revenues over time.

We furthermore find that network effects enhance the value potential of all sources of value creation in a freemium business model in the B2B enterprise software market. Contrary to the belief of prior literature, our research implies that the freemium model is especially prominent in extracting network effects, also in B2B markets. Network effects enable all value drivers to co-evolve and create spiraling effects. Network effects are especially important to enhance lock-in, as there seems to be an elevated focus on connectivity and compatibility in B2B markets. For managers in B2B markets this implies they should design their freemium business model with a social component and utilize a bottom-up approach, in order to attract a large user base as well as complementarities, and thus leverage the value creation potential inherent in network effects.

The foremost important implication of this research is that for firms operating in the B2B enterprise software industry, the freemium business model creates value through a bottom-up approach to user acquisition. This is contrasting prior literature that suggested the freemium business model would be of limited value in B2B markets as these markets are complex, more conservative and less prone to network externalities. However, prior research seems to have failed to understand that employing a freemium business model in a B2B market enables firms to utilize a bottom-up approach to customer acquisition, wherein the firms target individual employees. Freemium firms thus greatly eliminate and/or circumvent challenges experienced with more traditional sales strategies, such as B2B complexity and gatekeepers like the IT-department. The bottom-up approach furthermore enables word of mouth and spiraling direct and indirect network effects, which serve to lock organizations in even before the decision reaches top management. Therefore we find that the

differences between B2B and B2C markets are of less relevance when firms utilize a freemium business model in combination with a bottom-up approach to user acquisition. Our research furthermore implies that the freemium business model creates value for start-ups as well as established firms, albeit in different ways. In an enterprise software market characterized by fiercer competition due to the developments of SaaS and Web 2.0, the freemium business model can therefore be a powerful tool in creating value. This study thus contributes to enhance existing literature on value creation in freemium business models and increase the knowledge of value creation in the B2B enterprise software market and in virtual markets.

Through a case study with four case companies, employing a business model perspective on value creation, this research has revealed that the freemium business model creates value for firms operating in the B2B enterprise software market through increasing transaction efficiency, creating lock-in, employing complementarities and adding novelty. In a B2B market this is enabled by a bottom-up approach to user acquisition, where the freemium business model greatly enhances the sources of value creation through versioning and zero prices, as well as network effects. These findings make important contributions to extant literature. However, the research area of freemium business models and value creation in the B2B enterprise software market will benefit from further research in the years to come, as SaaS and other technological developments enable new ways to create value.

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Appendix

A1 - Recurrently Used Terms

A2 - Sample Mail

Dear XXX,

In relation to our master thesis at Copenhagen Business School, my thesis partner and I are conducting a study on the freemium business model. In this study we aim to answer the problem statement: “How does a freemium business model create value in the B2B enterprise software market?”

We are conducting this research as a case study, where we from a theoretical perspective aim to suggest how certain factors are of higher significance in creating value in B2B markets, and how variables like trust, adoption rate, network effects and 'reaction' to free may impact value creation in a freemium model. In the practical perspective we will conduct interviews with firms in different phases and industries in order to approach a conclusion based on the firm’s experience with the freemium model.

In this regard, we would very much like to conduct an interview with XXX. We think it is very interesting that you employ the freemium model on your XXX service, and would very much like to hear more about this and your experiences with the freemium model, and SaaS in general. As it is not conducted much research in this specific area yet, we hope that our thesis can be an interesting contribution in the freemium debate and improve the understanding of how a freemium business model creates value in B2B markets.

We would like to conduct a semi-structured interview in 1-2 hours, depending on how much time you can set aside, in the end of February or beginning of March. If desired, we can sign a confidentiality agreement, as well as keep your, and the company's identity anonymous throughout the thesis.

We hope that you find this interesting, and are looking forward to hear back from you.

Best Regards,
Karen Langø & Gunnhild Holthe

A3 - Interview Guide

General questions

- How many users does xxx have today?
- Who are these users/what customer segment?

General freemium questions

- What is the rationale behind employing a freemium business model?
- How does the free version of xxx create value for the users?
- How many pricing plans do you have?

Efficiency

- How does the freemium business model affect the customer acquisition process?
- How does zero prices affect the customer acquisition process?
- How do customers find your offering?
- How can/does the freemium business model simplify transactions?
- How important is speed in your offering?

Lock-in

- What is your strategy for retaining users?
- Do you gather any customer data to find the conversion potential among your current customers?
- Do you target individual employees or senior managers?
- Do you see any challenges in attracting business users instead of private consumers?
- Have you made your service compatible with the offerings of others in order to attract users?
- Can customers customize the offering to their needs?
- Do you employ a loyalty program?
- How does xx create network effects?
- How do you leverage these network effects?

- How do you create trust?

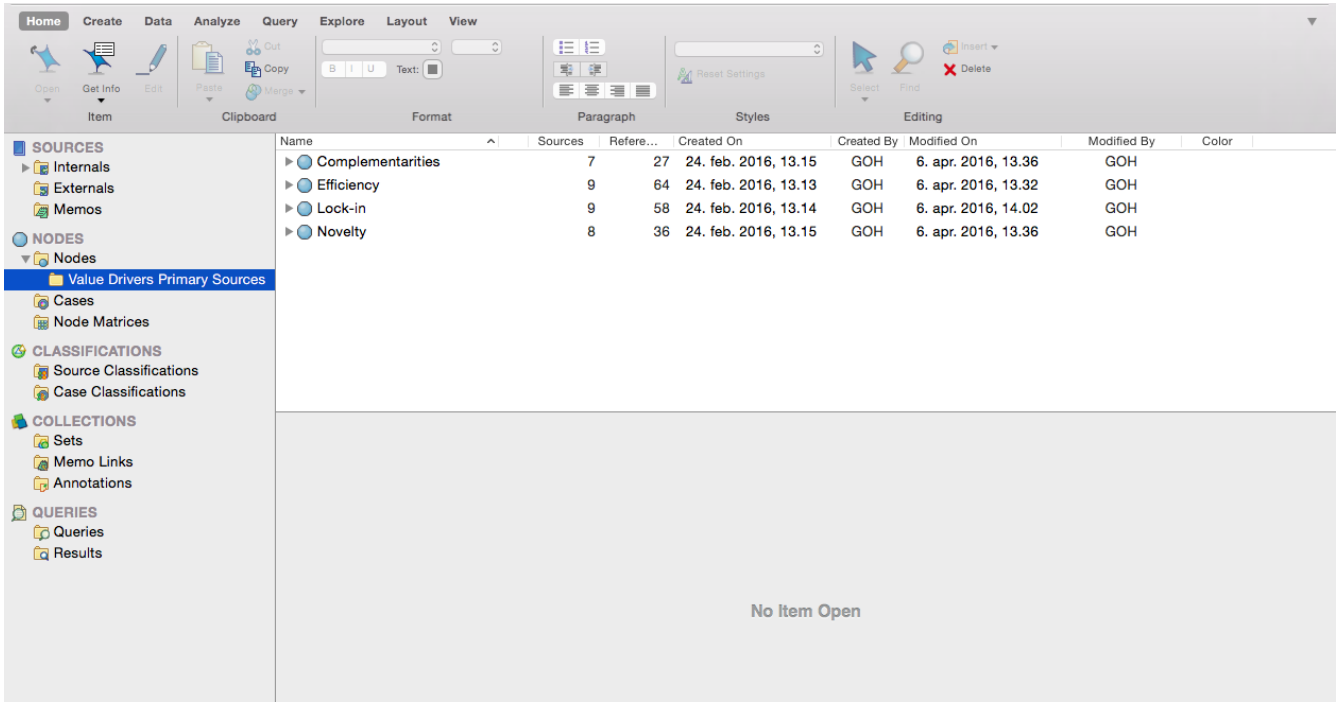
Complementarities

- Do you have complementary offerings on your service?
- How do these complementary offerings create value for your offering?
- Are any of the complementarities perceived as more important than the others?
- How do you attract suppliers of complementary assets?
- Do you function as a complementarity for other services?

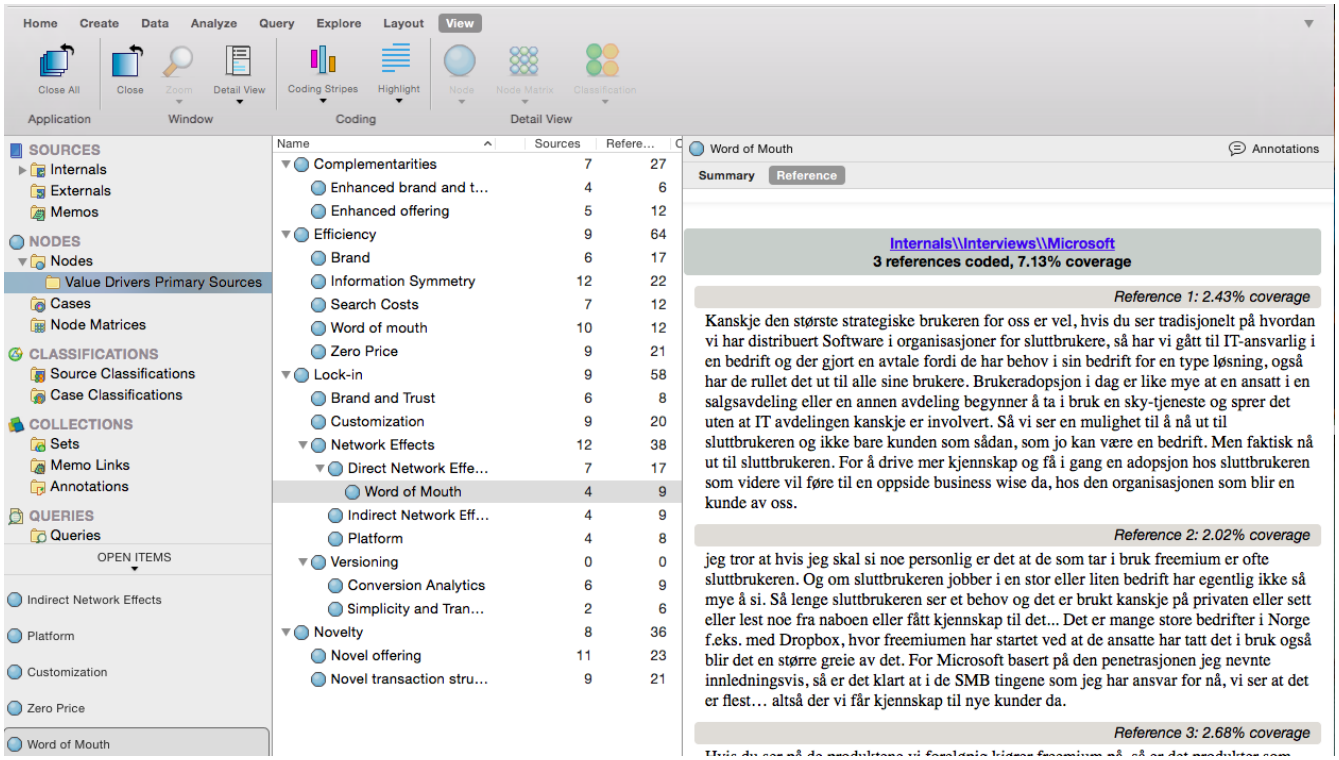
Novelty

- Do you have any competitor's offering a similar product/service?
- How does your offering/business model differ from theirs?
- How and when do you add features to the different versions of the offering?
- How do you determine what features to add to your free and premium offerings?
- How and when do you add features to the different versions of the offering?
- Have you changed your business model over time?

A4 - NVivo 1



A5 – NVivo 2



A6 - Transaction Efficiency (extended version)

Drivers of Transaction Efficiency	Amit & Zott (2001)	Proposed Modifications (Paragraph 4.6)	Appear.In	Microsoft	Plytix	Podio	Case Study Findings
Information Asymmetry	Reducing information asymmetry will enhance transaction efficiency through the supply of "up-to-date and comprehensive information", which the Internet facilitates in an easy and convenient manner.	4.6.1.1 & 4.6.1.3: Versioning and zero pricing, and word of mouth will likely reduce information asymmetry.	Customers must experience how easy Appear.in is to use themselves	The freemium model reduces the barriers to adoption	The market needs to be educated about our product	The freemium model is about time-to-value	6.1.1: Reducing information asymmetry enhances transaction efficiency. The freemium model in itself reduces information asymmetry as customers get to experience the good upon determining its value. Zero pricing and word of mouth contributes to this.
Search Costs	Through reducing information asymmetry, firms can also reduce customers search costs as greater availability of information naturally leads to a more efficient search.	4.6.1.1 & 4.6.1.3: Versioning and zero pricing, and word of mouth will likely reduce search costs.	The integration with Slack serves as a customer acquisition channel, and reduces search costs as customers are introduced to Appear.in via a known complementary	Reduce search costs for its more advanced and costly offerings by bundling and introducing them via its freemium products	Incentivize resellers to tell the retailers about the service, and vice versa - thus reducing search costs	Integration with Box to gain credibility and reduce search costs	6.1.2: The relationship between information asymmetry and search costs is reciprocal. In a freemium model search costs can be reduced by employing complementarities. Search costs might also be reduced through word of mouth and versioning with zero prices.
Selection Range	E-businesses provide greater selection at lower costs which reduces information asymmetry.	n.a	n.a	n.a	n.a	n.a	6.1.3: Greater selection at a lower cost is not found to yield value for the case companies. This finding might be relevant in other industries, or when looking at e-businesses as a whole compared to offline businesses.
Simplicity	E-businesses simplifies transactions which reduces the likelihood of mistakes, and thus reduces information asymmetry.	n.a	n.a	n.a	n.a	n.a	6.1.3: For the case companies, simplicity is not primarily a result of e-business, but a result of zero prices which allow customers to experience the offering.
Speed	The Internet facilitates speedy transactions, which enhances transaction efficiency.	n.a	n.a	n.a	n.a	n.a	6.1.3: Speed is not found to create particular value for the case companies. This may be due to the commonality of speedy transactions in today's e-business market. The finding is likely more relevant when looking at e-businesses as a whole, compared to offline businesses.
Scale economies	E-businesses allow individual customers to benefit from scale economies through demand aggregation and bulk purchasing.	n.a	n.a	n.a	n.a	n.a	6.1.3: Scale economies in terms of demand aggregation and bulk purchasing are not found to be of particular value for the case companies. This is likely more relevant when comparing online and offline businesses, as online businesses are able to offer a larger range of products than offline businesses.
Proposed Modifications							
Versioning and Zero Prices	n.a	4.6.1.1: Versioning information goods may drive adoption as users pay the value they perceive (Shapiro & Varian 1999). Versioning so that one part of the product is free is likely to be even more efficient (Shampanier et.al, 2007). Zero pricing drives transaction costs close to zero, and additionally reduce information asymmetry and search costs as users get to experience the good.	Zero prices is a necessary prerequisite for customer acquisition. "It would not be possible to attract customers it they were not able to try it for free	User's costs or associated risks of using freemium offerings is a lot lower. An end user in an organization rarely has a budget to spend on IT-solutions, and zero prices is thus very effective	We offer our service for free, because we don't want any roadblock. Zero prices enables users to play around and feel comfortable about our offering	Zero prices allows a firm to quickly scale a large customer base	6.1.4.1: Our findings reveal that versioning and zero prices serve to attract a large user base as it reduces both physical and psychological transaction costs. Zero prices is thus effective in enhancing transaction efficiency, and serve to reduce information asymmetry and search costs, and might even incentivize word of mouth.
Brand	n.a	4.6.1.2: Brands have the possibility of speeding up consumers information processing if customers are familiar with a brand and its connotations (Kotler & Gertner, 2004). Brand can thus enhance transaction efficiency.	n.a	"The reality is we have no issue with acquisition, zero. Tons of people know the brand"	"It is easier to engage with customers when they know and trust the brand"	Being associated with a known and trusted brand as Citrix made it easier to attract customers	6.1.4.2: Brands enhance transaction efficiency as users are familiar with a brand and its connotations, which serves to reduce information asymmetry and search costs.
Word of mouth	n.a	4.6.1.3: With word of mouth, potential users gain a trusted recommendation and indirect experience (Silverman, 2012). This likely reduces search costs and information asymmetry, and eventually results in increased transaction efficiency.	Almost all of Appear.in's growth has been realized through word-of-mouth	Incorporate a freemium model on offerings with a social component where they can get an exponential growth as a result of a viral spread	When resellers using Plytix sees the value of the offering, they will want retailers to use it as well	Users who are satisfied with Podio often attract other users	6.1.4.3: We find that word of mouth is effective in increasing transaction efficiency as it decrease the perceived risk and reduce information asymmetry and search costs. Freemium models are furthermore beneficial in leveraging the potential of word of mouth.
Complexity in B2B markets	n.a	4.6.2.1: B2B transactions are characterized by higher complexity as more people are involved in the buying decision (Kotler & Pfoertch, 2008). This may make it more challenging to achieve efficient transactions.	It is not the companies that starts using appear.in. It is employees in the companies.	Targets end users with a need for an IT solution.	Find it difficult to get meetings with top management or IT-leaders.	Bottom up buying decision. "You could see one team using it and then another team and another team - it kind of floats to the top and then someone decides to buy it."	6.1.4.4: The case companies solve B2B complexity by targeting end-user employees. Utilizing a bottom-up freemium approach the companies are able to target individual employees as opposed to the whole business. Thus, it is the individual employee who is the decision maker, which naturally decreases complexity in the buying decision.

A7 - Lock-In (extended version)

Drivers of Lock-In	Amit & Zott (2001)	Proposed Modifications, chapter 4.6	Appear.in	Microsoft	Plytix	Podio	Case Study Findings
Direct Network Effects	Direct network externalities drives lock-in as the value created for customers increase with the size of the customer base.	n.a	Network effects are inherent in the product itself. Users gain more value as more people are using our service.	Emphasize the importance of a bottom-up approach where individual employees attract other employees, in order to advance the value of the offering.	If a retailer decides they will not be using the service anymore, they need to get all their resellers to switch to another service as well.	Individual employees started using Podio and spread it to their co-workers. Many companies then felt dependent upon the service and chose to buy subscriptions.	6.2.1.1: Freemium models naturally create network effects, especially when utilizing a bottom-up approach. This creates high value in terms of lock-in as the value of the offering increases with the number of users. This again leads to collective switching costs, which makes it difficult for customers to churn.
Indirect Network Effects	Indirect network effects drive lock-in as increased usage of a product or service spawn increases in the availability and value of complementary goods, which eventually leads to an increase in the value of the original offering.	n.a	Integration with complementary offerings are mainly used to be able to efficiently drive customer acquisition, creating the biggest possible customer mass.	A big differentiator for Microsoft is the elaborate ecosystem of partners who adds complementarities and customizations on top of Microsoft's offerings.	Plytix enables open source as well as open APIs for complementors to build on their offering. More users make the offering more attractive for complementarities to build on, and with more complementarities, more users are interested in joining.	The real lock-in in today's business environment is that the product or service ties to other offerings.	6.2.1.2: Indirect network effects enhance the value of the offerings. As users engage in a network of offerings, their learning costs and thus switching costs multiplies as it is harder for any firm to imitate such a network. Employing a freemium business model seems to be an efficient tool in enabling and leveraging this value as it has the potential of attracting a large user base in which complementors will want to add offerings.
Customization	Customization enhances lock-in by enabling customers to customize the offering to their individual needs. Customers then have high incentives to use the system, which creates a positive feedback loop.	n.a	Customers can customize what complementarities they want integration with. Will implement opportunity for customization in the premium version.	Opportunity for customization in premium offerings.	Opportunity for customization in premium offerings.	Customers can customize what complementarities they want integrated in the offering. Opportunity for customization in the premium version.	6.2.2: Customization enhances lock-in in a freemium model. The case companies most commonly rely on complementarities to customize offerings. They furthermore offer opportunity for customization in their premium offerings, which may serve as a way of attracting users to convert.
Brand and Trust	Through creating trust and familiarity firms may gain more loyal customers, as customers increase their learning costs and thus also their switching costs.	4.6.2.3: Trust and a strong brand name is more important in a B2B setting than in a B2C setting.	Appear.in has not faced issues in relation to brand, even though they are a startup. Have however been faced with criticizm regarding the product in itself, as video conferencing is not seen as a very reliable service.	It can be an issue for the IT-department when individual employees start using different software independent of the leadership and organization.	It is challenging to be a startup company without a known brand when utilizing a freemium model. The CEO states it was a lot easier getting meetings with customer when he was working in Google.	Got questioned on whether or not Podio would exist the following year, as customers did not want to expose sensitive data to a company that might not be around for a long time.	6.2.3: Brand and trust are important enablers of lock-in, also in a freemium model. Our research suggest that freemium firms might experience that the top management are skeptical of freemium models, which affects the degree to which the business will take part in repeat transaction or convert to a premium price plan.
Loyalty Programs	Loyalty programs rewarding repeat customers with special bonuses may enhance lock-in.	n.a	n.a	n.a	n.a	Tried to employ a loyalty program without success	6.2.4: None of the case companies employ loyalty programs. Loyalty programs is thus not relevant in set study. Our findings suggest that loyalty programs are more relevant in B2C-markets.
Dominant Design	Developing dominant designs proprietary standards will increase switching costs, thus enhancing lock-in.	n.a	n.a	n.a	n.a	n.a	6.2.4: None of the case companies rely on proprietary standards for their freemium offerings. It seems relevant to assume that this is due to the fact that their offerings are difficult to formally protect from imitation, which may be the reason why they are giving it away for free.
Proposed Modifications							
Word of mouth as a driver of lock-in	n.a	4.6.1.3: Word of mouth enhances transaction efficiency, and thus attracts more potential users. As more users join the network and network effects play in, lock-in likely also increases in value.	Almost all user adoption has happened through word of mouth. Finds it challenging to use external marketing to attract users.	User adoption follows a bottom-up approach where individual employees attract other employees through word of mouth, in order to advance the value of the offering.	When the resellers see the value, they wille tell all the brands to use Plytix too, and vice versa. "So the network effect will kick in".	Podio grew as a result of word-of-mouth, which resulted in network effects. Many companies then felt dependent upon the service and chose to buy subscriptions for higher value and control of the network.	6.2.5.1: Our findings imply that word of mouth is central to network effects in a freemium model, as employees invite their co-workers to join. This leads to higher collective switching costs, which in turn creates lock-in.
Platforms	n.a	4.6.1.4: Platforms, where a firm attracts two distinct user groups, yield increasing returns as network effects raise the user's value of the platform (Gawer & Cusumano, 2008). Platforms will therefore likely create high switching cost.	Not using a platform.	Microsoft creates lock-in through a bundle of offerings that work together in one platform. Attraction to one service naturally directs the user to another inside the platform.	It will be difficult for a brand using Plytix to change to another offering, as they would have to switch not only employees in their own organization, but also all resellers.	Not using a platform.	6.2.5.2: The research indicates that platforms create two-sided network effects, which serve to create high switching costs, and thus positively affects lock-in.
Versioning	n.a	4.6.1.5: Versioning will impact lock-in as there is a need to continually convert users to a premium version. Freemium firms will likely present simple and transparent offerings, and use conversion analytics in converting users.	Appear.in`s biggest competitive advantage is its simplicity and quick time to value. The firm asks their customers to gain insights into which features should be in the free and premium version of the offering.	Versioning features in a free and premium version is a balance between giving enough in a freemium model for it to be interesting over time, but at the same time showing a clear upside by upgrading to the premium version.	Plytix use conversion analytics to know which features to but in the premium version in the future. All tests are at the moment in beta mode.	Podio uses convertion analytics to monitor which features are used most by the customers, and therefor know what to put in the premium versions of their offering. Explains the value must be clear and quicly understood by the customer.	6.2.5.3: The case companies emphasize a need to convert users to a premium version. In achieving conversion, they employ different analytic tools, which enable them to target users with more personalized and customized offerings, which in turn can drive lock-in. The companies also highlight that in order to retain customers their offering must demonstrate immediate value. Simple and transparent offerings thus enhance lock-in.
Switching costs in B2B markets	n.a	4.6.2.2: In B2B markets it is the organizations, as opposed to a single consumer, that undertake consumption (Fill & Fill, 2005), which may make it more challenging to achieve lock-in. However, if a company gets locked in, it will likely lead to higher switching costs as the company faces collective switching costs.	As Appear.in does not require any download or payment, collective switching costs are quite low.	Yammer has been highly efficient in attracting individual users to a bottom-up approach, creating a huge user base which face collective switching costs. The firm feels dependent on the network, and thus decides to buy control of it.	When resellers and brands have started using the service, they will both face high collective switching costs as they are part of a platform with two dependent parties.	Many companies felt dependent on the service and chose to buy subscriptions.	6.2.5.4: Due to the complexity of B2B transactions, we find that this process can be challenging when gatekeepers lack trust in freemium firms. However, the freemium model is efficient in creating and leveraging network effects, which is found to result in high collective switching costs. Thus, the company might be locked in even before the decision reaches gatekeepers.
Trust and Brand in B2B markets	n.a	4.6.2.3: As there is a need for long-lasting relationships in B2B markets, (Kotler & Pfoertsch, 2007), trust and brand might enjoy elevated positions in B2B markets. This may positively affect lock-in, as the customer has invested time in getting to know the brand.	n.a	n.a	n.a	n.a	6.2.5.5: Although some case companies experienced issues with trust, none expressed that brand and trust enjoy an elevated position in B2B markets. Our findings can thus not conclude whether this is more important in B2B markets than in B2C markets.

A8 - Complementarities (extended version)

Drivers of Complementarities	Amit & Zott (2001)	Proposed Modifications (Paragraph 4.6)	Appear.in	Microsoft	Plytix	Podio	Case Study Findings
Vertical and horizontal complementarities	Vertical and horizontal complementarities enhance the value of the core offering, as several offerings are bundled together.	n.a	Appear.in aims to attract developers to build complementary apps and services for their offerings in order to add value and differentiate between free and premium price plans.	Microsoft mostly relies on internal complementary offerings and bundle their basic tier freemium services with more advanced services that are offered for a premium subscription fee. Microsoft has furthermore made Office Apps available on Android and Apple smartphones. These integrations serve to reduce search costs for customers that are not using Microsoft hardware.	Plytix is dependent on vertical complementary technologies - more specifically Personal Information Management (PIM) and Digital Asset Management (DAM) systems that can store images.	Podio makes use of a range of horizontal complementarities like Dropbox, Box, Sharefile, GoToMeeting and ZenDesk. This enables the company to add value and continue to charge premium prices, or even add new pricing plans. It furthermore reduces search costs.	6.3.1: In our research we find no reason to differ between vertical and horizontal complementarities, complementary technologies and complementary activities. We find that complementary technologies and complementary activities can be both horizontal and vertical complementarities. What these have in common is that they enhance the value of the core offering. We therefore refer to this new category as 'Enhanced Offering'. Our findings reveal that freemium firms make use of complementarities to enhance the value of their core offering, as it creates more value to offer it as a bundle rather than separately. This enables the firms to charge higher prices for the added value. Complementarities may furthermore reduce search costs and thus enhance transaction efficiency.
Complementary activities	E-businesses may create value by capitalizing on complementarities among activities such as supply-chain integration.	n.a					
Complementary technologies	E-businesses may create value by capitalizing on complementarities among technologies such as linking the imaging technology of one business with the Internet communication technology of another, thereby unleashing hidden value.	n.a					
On-line and off-line complementarities	Offline assets complement online offerings; Customers who buy products over the Internet value the possibility of getting after-sales service offered through bricks- and-mortar retail outlets, including the covenience of returning or exchanging merchandise.	n.a	n.a	n.a	n.a	n.a	6.3.2: None of the case companies leverage offline complementary offerings on their freemium products. This may be due to a need of keeping costs low, but we also assume that this was more relevant in 2001 when users were not that accustomed to online offerings.
Proposed Modifications							
APIs as a driver of complementarities	n.a	4.6.1.6: APIs make it easy for developers to connect their applications and networks with those of others, and thus complement each other's offerings (Simon, 2014). The extensive use of APIs in today's digital economy entail that it is easier to connect to complementary offerings.	Appear.in has an API-integration with Slack, Trello and HipChat	Employs open APIs on top of offerings like Office 365 and Delve, in order to gain compatibility with other offerings.	Plytix use open APIs to attract complementary products and services on top of their offerings.	"It was a deliberate, strategic decision to make all of Podios products compatible with open APIs so everyone could connect to the service."	6.3.3.1: Our findings conclude that APIs drive the use of complementarities as APIs enable firms to make use of complementary assets and leverage the connectivity between different offerings in a more efficient manner.
Versioning as a driver of complementarities	n.a	4.6.1.7: Versioning is a mechanism that might enhance the importance of complementarities. With the need to convert users from a free to a premium version, and effectively lock customers in, it becomes important to continue innovating and collaborating with platforms in order to add value.	Aims to develop more sophisticated features for the premium version, enabled through complementarities.	Drives adoption on freemium offerings, by adding more advanced features in the premium version.	Plans to add more advanced feateures in the premium version like benchmarking and product tracking technology.	Complementarities are one of the most important elements for customers, as "no one wants to enter a locked system"	6.3.3.2: In our research it seems that freemium firms drive adoption on their premium offerings by adding more sophisticated features, often enabled through the use of complementary offerings. Having different versions of a product thus seems to enhance the use of complementarities.
Empirical Findings							
Enhanced Brand and Trust	n.a	n.a	The integration with Slack serves to enhance the brand name of Appear.in, and reduces search costs.	n.a	Plytix aims to gain integration with familiar PIM and DAM-systems and provide their customers with different ways of integrating with their platform in order to increase both compatibility and trust.	Podio use complementarities as a means to gain credibility. Hulthin (2016) explains it was highly beneficial to have integration with large, successful companies like Box, as it functioned as a way of telling potential customers that "these guys are OK".	6.3.4.1: Our findings suggest that complementarities may enhance brand and trust. Customers may know the complementarities of an offering, and have trust in these complementarities, which serves to enhance the credibility of the focal offering.

A9 - Novelty (extended version)

Drivers of Novelty	Amit & Zott (2001)	Proposed Modifications (Paragraph 4.6)	Appear.in	Microsoft	Plytix	Podio	Case Study Findings
New Transactional Content	The value creation potential of innovations has been articulated by Schumpeter (1934). The introduction of new products or services, new methods of production, distribution, or marketing, or the tapping of new markets are traditional sources of value creation through innovation. The main innovation of some e-businesses resides in their complementary elements.	n.a	Add new features through complementarities and crowdsourcing.	Add new features through internal R&D.	New transaction content with opportunities for benchmarking and product-tracking across web-sites.	Add new features through complementarities and crowdsourcing.	6.4.1: New transactional content creates value as it serves to attract new users. Adding innovative content furthermore raises the value of the offering, attract consumers to convert to a premium version and prevent customers from churning. Our findings suggest that freemium firms often add novel content through complementarities and crowdsourcing.
New Transaction Structure	Innovation in transaction structures create value by connecting previously unconnected parties, eliminating inefficiencies in the buying and selling processes through adopting innovative transaction methods, capturing latent consumer needs, and/or by creating entirely new markets. For example, e-business firms can identify and incorporate valuable new complementary products and services into their bundle of offerings in novel ways.	n.a	Employ a bottom-up approach in which they link participants in a novel manner. Appear.in furthermore does not require users to sign up in order to use the service. This novel transaction structure serves to enhance transaction efficiency.	Versions their Office-products based on screen size.	Novel link between resellers and retailers.	Novel link to complementarities.	6.4.2: In our research we find no reason to differ between new transaction structure and new transaction participants, as we find that new participants can be a way of creating new transaction structures. We therefore refer to this category as *New Transaction Structure*. Our research implies that freemium firms add value by creating new transactions structures in which they integrate complementarities, add new transaction participants or change the linkages between them. This serves to enhance both transaction efficiency and lock-in.
New Transaction Participants	Another dimension of innovation in e-business refers to the appropriate selection of participating parties. For example, firms can direct and intensify traffic to their web site by initiating affiliate programs with third parties, who are compensated for enabling the execution of transactions from their own web sites.	n.a					
Proposed Modifications							
Versioning as a driver of novelty	n.a	4.6.1.7: With the need to convert users from a free to a premium version, it becomes important to continue innovating and collaborating with platforms in order to add value. We expect this to lead to a higher focus on novelty, especially through the use of complementarities in order to create indirect network effects and lock customers in.	Aims to develop more sophisticated features for the premium version, enabled through complementarities.	Drives adoption on freemium offerings, by adding more advanced features in the premium version.	Aims to develop more advanced features in the premium version of the offering, like benchmarking and product tracking technology.	Continuously add features to the premium model, and make 'commodity' premium features available in the free version.	6.4.3.1: Versioning seems to drive a focus on novelty as the firm's need to retain users, and incentivize users to convert to a premium version. Our research suggests that firms often employ complementarities in adding novelty.
Empirical Findings							
Crowdsourcing	n.a	n.a	Use crowdsourcing to know what features their customers need, and which features to add to the premium and free version. They do this both through direct contact and piloting with their customers, and also queries on their webpage.	n.a	n.a	When adding new features to Podio, they listen to their customers and the market to identify needs and wants.	6.4.4.1: The research suggests that crowdsourcing is commonly used as a source of innovation in freemium firms. Crowdsourcing as a way of innovating adds value as it is asset light and allows firms to focus on customer's needs and wants.